

BRAID STABILITY AND A FLOER THEORY OF BRAID ISOTOPIES

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ABSTRACT. We build a Floer theory on the space of braids of periodic orbits of a Hamiltonian flow on a closed symplectic surface. The differential and continuation maps are defined by counting Floer isotopies of braids: tuples of Floer cylinders with pairwise disjoint graphs. With this new perspective, we prove a quantitative braid stability result that shows the persistence of braids under possibly large Hamiltonian perturbations. As an application, we show that for every $\alpha \geq 0$ there is a sequence of Hamiltonian diffeomorphisms ϕ_k on the two-torus T^2 such that

$$d_H(\text{Ent}_{\leq \alpha}(T^2, \omega), \phi_k) \rightarrow \infty \quad (k \rightarrow \infty),$$

where $\text{Ent}_{\leq \alpha}(T^2, \omega) \subset \text{Ham}(T^2, \omega)$ denotes the set of Hamiltonian diffeomorphisms with topological entropy at most α and d_H the Hofer metric. We prove this result by studying only contractible periodic orbits, whereas analogous higher-genus statements were previously obtained using non-contractible orbits.

1. INTRODUCTION

Let (S, ω) be a closed symplectic surface of genus $g \geq 0$. For every smooth Hamiltonian $H : S^1 \times S \rightarrow \mathbb{R}$ the equation

$$\omega(X_{H_t}, \cdot) = -dH_t$$

uniquely defines the Hamiltonian vector field X_{H_t} whose flow we denote by ϕ_H^t . Such time-1 maps ϕ_H^1 are called Hamiltonian diffeomorphisms, and their collection forms the group of Hamiltonian diffeomorphisms denoted by $\text{Ham}(S, \omega)$. On $\text{Ham}(S, \omega)$, there is a non-degenerate bi-invariant metric called the *Hofer metric*, which is defined as follows. For a Hamiltonian H set

$$E(H) = \int_0^1 \left(\max_{x \in S} H(t, x) - \min_{x \in S} H(t, x) \right) dt.$$

Then the Hofer metric is defined by

$$d_H(\phi, \psi) := \inf_{H, G} E(H - G),$$

where the infimum runs over all H and G such that $\phi_H^1 = \phi$ and $\phi_G^1 = \psi$. For every smooth loop $x : S^1 \rightarrow S$ and a capping C given by a homotopy from x to a fixed reference loop in its homotopy class, we define

$$\mathcal{A}_H([x, C]) = \int_0^1 H(t, x(t)) dt - \int_C \omega.$$

In accordance with standard convention, we let C be a disk in the case when x is contractible.

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1.1. The Braided Floer Space. We denote the set of one-periodic orbits of the flow ϕ_H^t by $\mathcal{P}(H)$. For $x = \{x_1, \dots, x_k\} \subset \mathcal{P}(H)$ one can define a (*pure closed*) *surface braid*

$$B(H, x) = \{(x_1(t), t), \dots, (x_k(t), t)\}_{t \in S^1},$$

where the loops $(x_i(t), t)$ are called *strands*. The isotopy class of this braid will be denoted by $\mathcal{B}(H, x)$. We define the *braided Floer space in the class β* by

$$\text{BCF}_\beta(H) = \langle \{x_1, \dots, x_k\} \subset \mathcal{P}(H) \mid \mathcal{B}(H, \{x_1, \dots, x_k\}) = \beta \rangle_{\mathbb{Z}_2}.$$

In this paper, we build a Floer theory for this space and study the stability of braids of orbits $B(H, \{x_1, \dots, x_k\})$ under Hamiltonian perturbations $H + h$ of H , where $E(h)$ is possibly large.

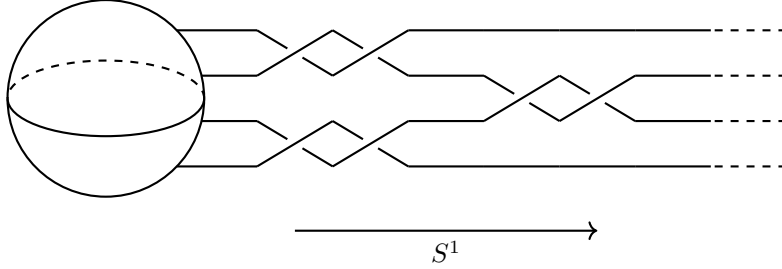


FIGURE 1. A (*pure closed*) surface braid on S^2 with four strands.

1.2. An Application: High-Entropy Braids Stable under Large Perturbations.

From a topological perspective, it is very natural to study flows of vector fields through the braids traced by fixed points $\{z_1, \dots, z_k\}$ of the time-1 map of the flow. There is a push map

$$\text{Push} : \text{PB}_k(S, z_1, \dots, z_k) \rightarrow \text{MCG}(S, z_1, \dots, z_k)$$

from the pure braid group at $z_1, \dots, z_k$ to the mapping class group of the punctured surface $S \setminus \{z_1, \dots, z_k\}$ (see [FM12]). At the level of conjugacy classes, we obtain a map from closed braids to conjugacy classes of mapping classes, where $\mathcal{B}(H, \{z_1, \dots, z_k\})$ is mapped to the conjugacy class of $[\phi_H^1|_{S \setminus \{z_1, \dots, z_k\}}]$. Thus, properties determined by the conjugacy class of $[\phi_H^1|_{S \setminus \{z_1, \dots, z_k\}}]$ can be inferred from the existence of the corresponding braid. One such property is the minimal *topological entropy* in the conjugacy class (see Section 5.1 for a definition), which measures dynamical complexity. To estimate the topological entropy, we use a notion of entropy at the level of braids. For a non-degenerate Hamiltonian H this estimate takes the form

$$h(\mathcal{B}(H, \{x_1, \dots, x_k\})) \leq h_{\text{top}}(\phi_H^1).$$

This motivates the study of braid stability under Hamiltonian perturbations. To the best of the author's knowledge, Alves and Meiwes first proved braid stability in [AM24] for closed surfaces and the disk, and Hutchings generalized it to general symplectic surfaces with boundary [Hut25]. Alves and Meiwes [AM24] proved that under small Hamiltonian perturbations, braids persist, thereby showing that topological entropy is lower-semicontinuous with respect to the Hofer metric. To quantify the perturbations for which braid stability is guaranteed, they had to consider the action of all periodic orbits. The machinery built in this work allows more refined estimates: braid stability is essentially quantified by the energy of Floer isotopies that end at the braid. For action estimates, this sharply restricts

the set of action values we need to consider. As a proof of concept that this is in fact an improvement on the theory, we prove the following theorem by only doing action estimates of contractible orbits.

Theorem A. *For every $\alpha \geq 0$ there is a sequence of Hamiltonian diffeomorphisms $\phi_k \in \text{Ham}(T^2, \omega)$ of the torus such that*

$$d_H(\text{Ent}_{\leq \alpha}(T^2, \omega), \phi_k) \rightarrow \infty \quad (k \rightarrow \infty),$$

where $\text{Ent}_{\leq \alpha}(T^2, \omega)$ denotes the set of Hamiltonian diffeomorphisms with topological entropy at most α .

The braid stability result proven in this work is essential to this theorem; we apply a new type of action estimate that relies on the energy of Floer isotopies rather than on the energy of Floer cylinders alone. Importantly, we prove this result by only considering contractible periodic orbits. For surfaces of genus ≥ 2 , the analogous statement was proved by Chor and Meiwes [CM23]. Their work builds on that of Polterovich and Shelukhin [PS16], who used persistence modules in a Floer theory of non-contractible orbits to show that there are Hamiltonian diffeomorphisms of a closed symplectic surface of genus $g \geq 4$ that lie arbitrarily far, in the Hofer metric, from the set of *autonomous* Hamiltonian diffeomorphisms. Also using non-contractible orbits, Chor [Cho22] extended this result to $g \geq 2$, and Khanevsky [Kha22]. Since autonomous Hamiltonian diffeomorphisms of surfaces have zero entropy, Theorem A extends the last result. That an analogous statement to that in [CM23] holds for the case $g = 1$ is not surprising. The novelty, however, is that we need only consider contractible orbits, making the case $g = 0$, where little is known, more approachable. Next, we outline the machinery that is used in the proof of Theorem A: we prove a quantitative version of braid stability by counting Floer isotopies.

1.3. Quantitative Braid Stability and Floer Isotopies. To study the stability of the space $\text{BCF}_\beta(H)$, we define continuation maps between $\text{BCF}_\beta(H)$ and $\text{BCF}_\beta(G)$ by counting *Floer isotopies*. These Floer isotopies are given by Floer cylinders $u_1, \dots, u_k$ that connect the periodic orbits of the strands. Each cylinder satisfies the Floer equation

$$\partial_s u_i + J_{s,t}(\partial_t u_i - X_{F_{s,t}} \circ u_i) = 0,$$

for a Hamiltonian $F : \mathbb{R} \times S^1 \times S \rightarrow \mathbb{R}$ and a generic family of almost complex structures $(J_{s,t})_{(s,t) \in \mathbb{R} \times S^1}$, and the cylinders additionally satisfy the following: for all $(s,t) \in \mathbb{R} \times S^1$ and $i \neq j$

$$u_i(s,t) \neq u_j(s,t).$$

In contrast to standard Floer theory, where the relevant moduli space of Floer cylinders of index 1 is a union of intervals, the moduli space of isotopies in which each curve has index 1 is a union of hypercubes that arise as a product of moduli spaces of Floer cylinders.

These continuation maps preserve the *Conley–Zehnder index* $\mu = \mu_{CZ}$ of each strand. This index is a symplectic twisting number associated to a periodic orbit $x \in \mathcal{P}(H)$ and takes values in $\mathbb{Z}$ for $g \geq 1$ and in $\mathbb{Z}_4$ for $g = 0$. Since continuation maps preserve the index of every strand, the maps preserve the *framing*

$$f_x : x_i \mapsto \mu(x_i)$$

of the underlying sets $\{x_1, \dots, x_k\} \subset \mathcal{P}(H)$ of the braids of orbits. A *(closed pure) $\mathbb{Z}$ -framed surface braid* or *(closed pure) $\mathbb{Z}_4$ -framed surface braid* is a (closed pure) surface braid b with strands $\{s_1, \dots, s_k\}$ equipped with a map, called *framing*,

$$f : \{s_1, \dots, s_k\} \rightarrow \mathbb{Z} \quad \text{or} \quad f : \{s_1, \dots, s_k\} \rightarrow \mathbb{Z}_4.$$

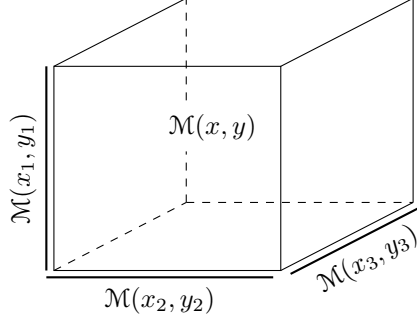


FIGURE 2. A cube as a moduli space.

When it is clear from the context, we drop the prefix $\mathbb{Z}$ or $\mathbb{Z}_4$. The framing-preserving isotopies induce an equivalence relation on framed surface braids; the equivalence classes are called *(closed pure) framed surface braid classes*, denoted by the tuple (β, f) . For every framed braid class (β, f) where the framing is $\mathbb{Z}$ -valued when $g \geq 1$ and $\mathbb{Z}_4$ -valued when $g = 0$ we define

$$\text{BCF}_{\beta, f}(H) = \langle x \subset \mathcal{P}(H) \mid \mathcal{B}(H, x) = \beta, f_x = f \rangle_{\mathbb{Z}_2}.$$

For every non-degenerate Hamiltonian H and a set $x = \{x_1, \dots, x_k\} \subset \mathcal{P}(H)$, we will define the *stability radius* $\mathcal{S}(H, x) \in \mathbb{R} \cup \{\infty\}$ of the braid $\mathcal{B}(H, x)$ (see below for the definition when the framing f_x is injective and Section 4 for the general definition) such that the following theorem holds.

Theorem B (Quantitative Braid Stability). *Let H be a non-degenerate Hamiltonian on a closed symplectic surface (S, ω) and let $x \subset \mathcal{P}(H)$. Let $\beta = \mathcal{B}(H, x)$ and $f = f_x$ be the associated braid class and framing. For every non-degenerate Hamiltonian G such that $E(H - G) < \mathcal{S}(H, x)$ we have*

$$\dim \text{BCF}_{\beta, f}(G) \geq 1.$$

The essential improvement over the results in [AM24] is that we can estimate the stability radius $\mathcal{S}(H, x)$ by the action differences of orbits that appear in β -braids, as we explain next for an injective framing f . For a smooth family of ω -compatible almost complex structures $(J_t)_{t \in S^1}$ we consider Floer isotopies consisting of cylinders $u = (u_1, \dots, u_k)$ that solve the Floer equation

$$\partial_s u_i + J_t(\partial_t u_i - X_{H_t} \circ u_i) = 0.$$

For a smooth cylinder $u : \mathbb{R} \times S^1 \rightarrow S$ the energy is defined by

$$E(u) = \int_{\mathbb{R}} \int_0^1 \omega(\partial_s u, J_t \partial_s u) dt ds.$$

We define the energy of a Floer isotopy by

$$E(u) = \sum_{i=1}^k E(u_i).$$

When $E(u)$ is finite, the cylinders u_i converge to periodic orbits $\lim_{s \rightarrow -\infty} u_i(s, \cdot) = x_i$ and $\lim_{s \rightarrow +\infty} u_i(s, \cdot) = y_i$ of H . In this case, one can define the index $\mu(u_i)$ of u_i by

$\mu([x_i, C_{x_i}]) - \mu([y_i, C_{x_i} \# u_i])$, where C_{x_i} is an auxiliary choice of capping of x_i on which the difference does not depend. The maximal index of a Floer isotopy is defined by

$$\mu_{\max}(u) = \max_i \mu(u_i).$$

The central quantity in defining the stability radius is the minimal energy of non-trivial Floer isotopies with one end in x for which the Conley–Zehnder index drops by at most 1 for each strand

$$\mathcal{E}(H, J, x) = \min\{E(u) \mid u \text{ is a Floer isotopy with one end given by } x \text{ and } \mu_{\max}(u) = 1\}.$$

This generalizes the minimal energy of Floer cylinders connecting single orbits of index difference one used in similar previous work (see for example [Kha22]). Moreover, we define the *interior minimal energy of Floer cylinders of x* by

$$\mathcal{E}^{\text{int}}(H, J, x) = \min\{E(u) \mid u \in \mathcal{M}(x_i, x_j), \mu([x_i, C]) - \mu([x_j, C \# u]) = 1\},$$

where C is any choice of capping of x_i . The stability radius for an injective framing is now defined by

$$(1) \quad \mathcal{S}(H, x) = \frac{1}{k} \sup_J \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x)\},$$

where the supremum runs over all loops of ω -compatible almost complex structures such that (H, J) is regular and $k = |x|$ is the number of strands. When f is not an injective framing, the minimum in the definition of $\mathcal{S}(H, x)$ includes an additional term $\mathcal{E}^2(H, J, x)$ that is sensitive to configurations of two Floer cylinders with ends in x . The general definition is given in Section 4.

1.4. Braided Floer Homology. Although this is not relevant for our approach to proving braid stability, for non-degenerate Hamiltonians and a generic almost complex structure J as auxiliary data, we define a differential ∂ on $\text{BCF}_\beta(H)$ that satisfies $\partial^2 = 0$. This differential counts Floer isotopies given by Floer cylinders $u_1, \dots, u_k$ which all drop the Conley–Zehnder index by 1. We call the associated homology *braided Floer homology in the class β* and denote it by $\text{BHF}_\beta(H, J)$. To avoid Novikov coefficients, in the case $S = T^2$ we assume that all the strands of β are contractible. We note that with the Floer-theoretic setup of $\text{BCF}_\beta(H)$, we recover a refined version of the Arnold conjecture for surfaces, which already follows from the work of Connery-Grigg [CG24]:

The flow of any non-degenerate time-1 periodic Hamiltonian of a closed symplectic surface of genus g has at least $2g + 2$ contractible orbits of period 1 that form the trivial braid.

Indeed, this is an immediate consequence of the following result about braided Floer homology.

Theorem C. *Let (S, ω) be a closed symplectic surface of genus g and let β_0 be the closed braid with $2g + 2$ constant strands. For every non-degenerate H and generic almost complex structure J , we have*

$$\dim \text{BHF}_{\beta_0}(H, J) \geq 1.$$

Importantly, the homology theory constructed in this paper differs from the braid Floer homology defined by van den Berg, Ghrist, Vandervorst, and Wójcik in [vdBGVW14]. In [vdBGVW14], a homology theory is defined that is independent of the Hamiltonian H , and the differential essentially counts braid isotopies where the index drops by 1 in a single orbit. Our homology theory depends on the Hamiltonian H and counts tuples of Floer cylinders pairwise connecting strands in braids of periodic orbits, where the index drops by 1 in every

strand. Next, we outline what we think are the essential new perspectives and ideas in this work.

1.5. Elements of the proofs. All proofs build on standard Hamiltonian Floer theory with additional topological and combinatorial input. In this work, the differential ∂ and continuation maps Φ of braided Floer spaces are defined by counting tuples of Floer cylinders $(u_1, \dots, u_k)$ all of the same index such that

$$(2) \quad u_i(s, t) \neq u_j(s, t)$$

for all $(s, t) \in \mathbb{R} \times S^1$ and $i \neq j$. This property can be expressed by a topological intersection number of the graphs of the cylinders.

1.5.1. The Intersection Number. We consider a purely topological intersection number of cylinders u_i , which is a priori only defined when the cylinders converge to distinct loops in the limit $\lim_{s \rightarrow \pm\infty} u_i(s, t) \neq \lim_{s \rightarrow \pm\infty} u_j(s, t)$ for $i \neq j$. The intersection number $\iota(u_i, u_j)$ is the algebraic count of intersections of the graphs $\hat{u}_i$ and $\hat{u}_j$ of u_i and u_j . Since u_i and u_j are Floer cylinders, their graphs are pseudoholomorphic (for a suitable almost complex structure), and the intersection number is non-negative. This means that (2) holds if and only if $\iota(u_i, u_j) = 0$ for all $i \neq j$. In many proofs, we need to study configurations of two broken trajectories $u_i \# v_i$ for $i = 1, 2$ of Floer cylinders that break at the same orbit y . In

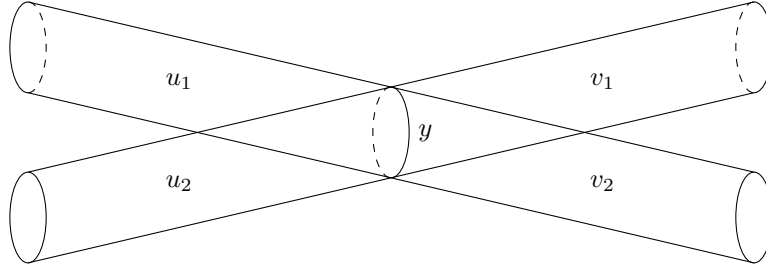


FIGURE 3. Two cylinders breaking at a single orbit y .

this situation, the intersection number equals the count of intersections away from y , plus an asymptotic term $\lambda(u_1 \# v_1, u_2 \# v_2)$ accounting for intersections near y :

$$\iota(u_1 \# v_1, u_2 \# v_2) = |\hat{u}_1 \cap \hat{u}_2| + |\hat{v}_1 \cap \hat{v}_2| + \lambda(u_1 \# v_1, u_2 \# v_2).$$

This asymptotic term is defined purely topologically: near a breaking orbit y we associate to the ends of cylinders $\{u_i\}_{i=1,2}$ and $\{v_i\}_{i=1,2}$ two two-component links L^u and L^v in S^3 for the positive and negative ends of $\{u_i\}_{i=1,2}$ and $\{v_i\}_{i=1,2}$. The asymptotic term is the difference of the linking numbers of these links: $\lambda(u_1 \# v_1, u_2 \# v_2) = \ell(L^v) - \ell(L^u)$. This perspective is essential to the proofs of the main theorems in this paper and is, to the best of the author's knowledge, new. The following question arose in a discussion with Beomjun Sohn.

Question. *How does this topological asymptotic term of the intersection number relate to the more classical geometric intersection number defined by Siefring [Sie08]?*

1.5.2. *The Moduli Spaces.* For a Hamiltonian $H : \mathbb{R} \times S^1 \times S \rightarrow \mathbb{R}$ that is locally constant outside of a compact set, a smooth family of ω -compatible almost complex structures $(J_{s,t})_{(s,t) \in \mathbb{R} \times S^1}$, and a positive integer k , we consider the space of Floer isotopies consisting of k Floer cylinders

$$\mathcal{M}(H, J, k) = \left\{ (u_1, \dots, u_k) \left| \begin{array}{l} \partial_s u_i + J_{s,t}(\partial_t u_i - X_{H_{s,t}} \circ u_i) = 0, \\ E(u_i) < \infty, \iota(u_i, u_j) = 0 \text{ for } i \neq j \end{array} \right. \right\}.$$

Using positivity of intersections, the condition $\iota(u_i, u_j) = 0$ is equivalent to $u_i(s, t) \neq u_j(s, t)$ for all $(s, t) \in \mathbb{R} \times S^1$, while the condition $E(u_i) < \infty$ is a standard assumption and ensures that the Floer cylinders converge to periodic orbits for $s \rightarrow \pm\infty$. The space $\mathcal{M}(H, J, 1)$ is simply the classical space of Floer cylinders, and

$$\mathcal{M}(H, J, k) \subset \mathcal{M}(H, J, 1) \times \dots \times \mathcal{M}(H, J, 1),$$

where $\mathcal{M}(H, J, k)$ is given by the union of the connected components on which $\sum_{i \neq j} \iota(u_i, u_j) = 0$. Here, a path in the product space gives rise to a homotopy of cylinders relative to limiting

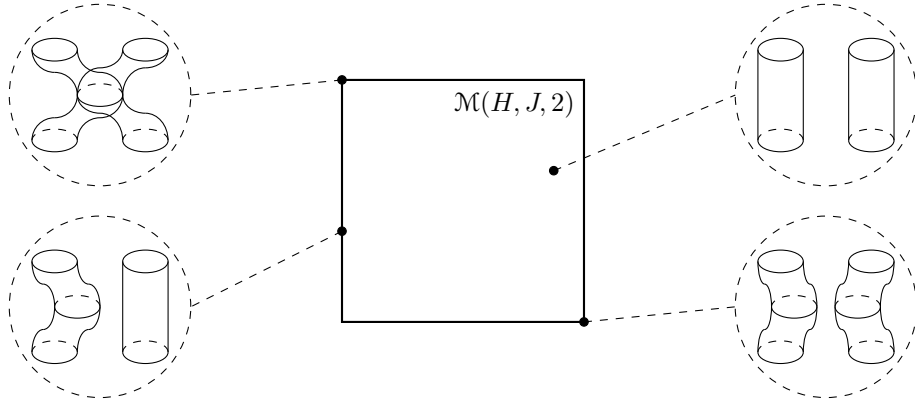


FIGURE 4. Cylinders at different points in the moduli space $\mathcal{M}(H, J, 2)$.

orbits, so the intersection number is locally constant. To define and analyze the differential and continuation maps, we restrict the moduli space to cylinders with the same index. Here, the most relevant part of the moduli space is a product of one-dimensional intervals forming a hypercube. Passing to a face of this hypercube means one cylinder breaks into a broken trajectory, with the extreme case at the corner points, where all cylinders are broken (see Figure 4).

1.5.3. *The Symmetry.* Understanding the composition of continuation maps and the differential is central to Floer theory. These compositions always count broken Floer isotopies that consist of tuples of Floer cylinders that all break at distinct orbits. Such broken Floer isotopies appear as corner points of the moduli space we defined above. So to understand the composition of maps, we need to count these corner points. A simple but deep observation in classical Floer theory is that the number of boundary components of a compact 1-dimensional manifold is even. In our case, this is replaced by the observation that the number of corner points of hypercubes is even. Unfortunately, the moduli space is not sensitive to configurations of breaking orbits at corner points. Still, we want to count only those where all breaking orbits are distinct and hence correspond to honest broken isotopies. We

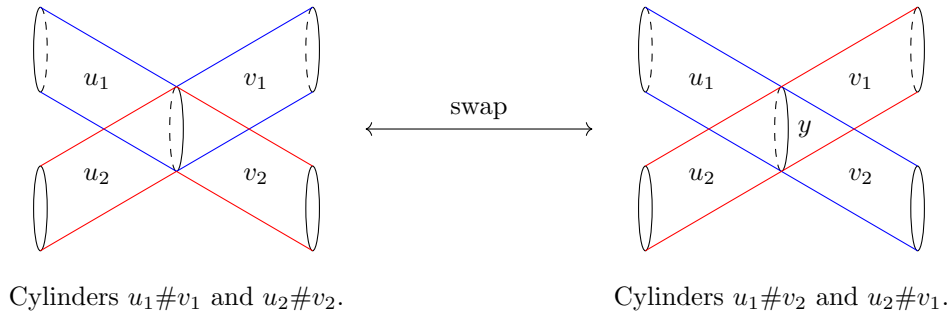


FIGURE 5. The swapping operation.

approach this problem by using the symmetry of broken trajectories with the same breaking orbits and show that they pair up and do not affect the count modulo 2. Given broken trajectories (u_1, v_1) and (u_2, v_2) breaking at an orbit y , we use that (u_1, v_2) and (u_2, v_1) are two other broken trajectories and that this swapping operation does not change the intersection number of the broken trajectory. Hence, they remain in the moduli space. At its core, this argument relies on the asymptotic expression for the intersection number and on the symmetry of the linking number.

Organization of the paper: In Section 2, we define an intersection number for cylinders and study its behavior under Floer-theoretic gluing. In Section 3, we define a differential on $\mathrm{BCF}_\beta(H, J)$ and define continuation maps between braided Floer complexes. We then proceed to prove Theorem C. In Section 4, we prove Theorem B by defining *bounded continuation maps*. In Section 5, we prove Theorem A by constructing eggbeater maps on T^2 with high-entropy braids of periodic contractible orbits and a large stability radius. Finally, in the Appendix we carry out a Maslov index calculation which we use in Section 5.

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2. SOME PREPARATION: INTERSECTIONS

In this section, we define an intersection number of cylinders $u, v : \mathbb{R} \times S^1 \rightarrow S$ that algebraically counts points $(s, t) \in \mathbb{R} \times S^1$ such that $u(s, t) = v(s, t)$. To do this, we pass to graphs of cylinders in a 4-dimensional almost complex manifold. We then make use of the well-known intersection theory of pseudoholomorphic cylinders in such manifolds to conclude that solutions to Floer equations intersect only positively. A priori, the intersection number we use is purely topological and is only defined when the Floer cylinders converge to distinct orbits in the limit. We do, however, discuss the behavior when the limiting orbits agree, specifically when they arise as breaking orbits of broken trajectories (see Figure 6). In

this case, an additional asymptotic term appears, equal to the linking number of the knots given by slices of the Floer cylinders near the breaking orbit (see Lemma 2.1 and Figure 7).

First, we discuss the intersection number from a purely topological standpoint. We study cylinders $u : [-1, 1] \times S^1 \rightarrow S$, where S is a closed surface. For two such cylinders u_1, u_2 , we would like to define an intersection number that counts, algebraically, the number of points (s, t) such that $u_1(s, t) = u_2(s, t)$. To do so, we consider the graphs of u parametrized by

$$\hat{u} : [-1, 1] \times S^1 \rightarrow [-1, 1] \times S^1 \times S, \quad \hat{u}(s, t) = (s, t, u(s, t))$$

and consider the intersection of $\hat{u}_1, \hat{u}_2$ in $M = [-1, 1] \times S^1 \times S$. For simplicity of notation, we denote by $\hat{u}$ both the graph of u and its parameterization. A pair of cylinders u_1, u_2 is called *admissible* if for every $t \in S^1$ we have

$$u_1(-1, t) \neq u_2(-1, t) \quad \text{and} \quad u_1(1, t) \neq u_2(1, t).$$

For admissible u_1 and u_2 such that $\hat{u}_1$ and $\hat{u}_2$ can be put in transverse position via a homotopy that is constant at the boundary, the intersection number is defined by

$$\iota(u_1, u_2) = \sum_{p \in \hat{u}_1 \cap \hat{u}_2} \epsilon_p,$$

where the orientation of the intersection determines $\epsilon_p \in \{\pm 1\}$. This intersection number is symmetric and invariant under homotopies constant on the boundary. We remark that there is a purely algebraic definition of this intersection number using Poincaré–Lefschetz duality

$$D : H_2(M, A) \rightarrow H^2(M, B),$$

where $A \cup B = \partial M$ is a partition of the boundary of M into two manifolds which overlap only at their common boundary $A \cap B = \partial A = \partial B$. For admissible cylinders u_1, u_2 , the intersection number is given as follows. Choose a partition $A \cup B$ of $\partial([-1, 1] \times S^1 \times S) = \{\pm 1\} \times S^1 \times S$ such that $\hat{u}_1(\pm 1, t) \in A$ and $\hat{u}_2(\pm 1, t) \in B$ for all $t \in S^1$. For admissible cylinders u_1 and u_2 we have

$$\iota(u_1, u_2) = D([\hat{u}_1])([\hat{u}_2]).$$

2.1. Intersections of Cylinders in Degenerate Breaking Position. We now discuss the intersection number of cylinders that arise as follows. Let $u_1, u_2 : [-1, 0] \times S^1 \rightarrow S$ and $v_1, v_2 : [0, 1] \times S^1 \rightarrow S$ such that for all $t \in S^1$

$$y(t) = u_1(0, t) = u_2(0, t) = v_1(0, t) = v_2(0, t)$$

and consider

$$u_i \# v_i : [-1, 1] \times S^1 \rightarrow S, \quad (s, t) \mapsto \begin{cases} u_i(s, t), & s \in [-1, 0] \\ v_i(s, t), & s \in [0, 1] \end{cases}.$$

Further assume that $\widehat{u_1 \# v_1}$ and $\widehat{u_2 \# v_2}$ are admissible and that away from $\hat{y}$, $\widehat{u_1 \# v_1}$ and $\widehat{u_2 \# v_2}$ are transverse $\widehat{u_1 \# v_1}$ and intersect only finitely many times. Cylinders that arise in this form are *in degenerate breaking position with breaking orbit y* . That is, two admissible cylinders w_1, w_2 are in degenerate breaking position if their graphs $\hat{w}_1, \hat{w}_2$ intersect only finitely many times away from $0 \times S^1$, where they agree. Now suppose that w_1, w_2 are cylinders in degenerate breaking position with breaking orbit y . We next discuss the local behavior of the cylinders near the breaking orbit. Consider

$$\hat{y} : S^1 \rightarrow [-1, 1] \times S^1 \times S, \quad t \mapsto (0, t, y(t))$$

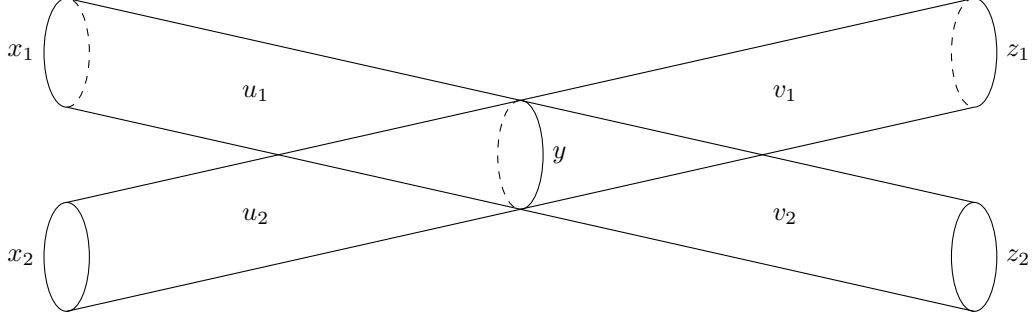


FIGURE 6. Cylinders in degenerate breaking position.

and choose a small tubular neighborhood U of $\hat{y}$ that is homeomorphic to $(-\varepsilon, \varepsilon) \times S^1 \times D^2$ where D^2 is the two-dimensional disk. Since the $\hat{w}_i$ are the graphs of w_i we have that the slice of the cylinder $\hat{w}_i$ at $s = \delta$ for $\delta \in (-\varepsilon, \varepsilon)$ lies in $\{\delta\} \times S^1 \times D^2$, that is

$$\hat{w}_i|_{\{\delta\} \times S^1} \subset \{\delta\} \times S^1 \times D^2.$$

The neighborhood U can be embedded in $(-\varepsilon, \varepsilon) \times S^3$ by embedding the filled torus $S^1 \times D^2$ as the standard filled torus in S^3

$$\iota : (-\varepsilon, \varepsilon) \times S^1 \times D^2 \hookrightarrow (-\varepsilon, \varepsilon) \times S^3.$$

Now, let us assume that $\varepsilon > 0$ is small enough so that $w_1(s, t) \neq w_2(s, t)$ for every $(s, t) \in (-\varepsilon, \varepsilon) \times S^1$ where $s \neq 0$. This is possible because the pair (w_1, w_2) is in degenerate breaking position. In this case, the cylinders $\hat{w}_i|_{(-\varepsilon, \varepsilon) \times S^1}$ define isotopies of knots in S^3 by

$$K_i^\delta : S^1 \rightarrow \{\delta\} \times S^3, \quad K_i^\delta = \iota \circ \hat{w}_i|_{\{\delta\} \times S^1}.$$

For every $\delta \in (-\varepsilon, \varepsilon) \setminus \{0\}$ the knots K_1^δ and K_2^δ are disjoint and the linking number $\ell(K_1^\delta, K_2^\delta)$ is well-defined. Since in $((-\varepsilon, \varepsilon) \setminus \{0\}) \times S^1$ the cylinders $\hat{w}_1$ and $\hat{w}_2$ define an isotopy of $K_1^\delta \dot{\cup} K_2^\delta$ as links, the linking number $\ell(K_1^\delta, K_2^\delta)$ is constant on each interval $(-\varepsilon, 0)$ and $(0, \varepsilon)$. Finally, we find a well-defined quantity that describes the behavior of the cylinders at the breaking orbit; namely, we define

$$(3) \quad \lambda(w_1, w_2) = \ell(K_1^\delta, K_2^\delta) - \ell(K_1^{-\delta}, K_2^{-\delta}),$$

where $\delta > 0$ is small enough. The following question about this asymptotic term arose in a discussion with Beomjun Sohn.

Question. *In the classical work of Siefring [Sie08], an intersection number of pseudoholomorphic curves is defined by counting intersections in the interior of the curves plus an asymptotic term which relies on the winding numbers of the cylinders near the limiting orbits. What is the relation between this geometric asymptotic term and the one defined here, which relies only on topological information?*

Lemma 2.1. *Let w_1, w_2 be a pair of cylinders in degenerate breaking position. Then the intersection number of w_1, w_2 is given by*

$$\iota(w_1, w_2) = \lambda(w_1, w_2) + \sum_{p \in \hat{w}_1 \cap \hat{w}_2 \setminus \hat{y}} \epsilon_p.$$

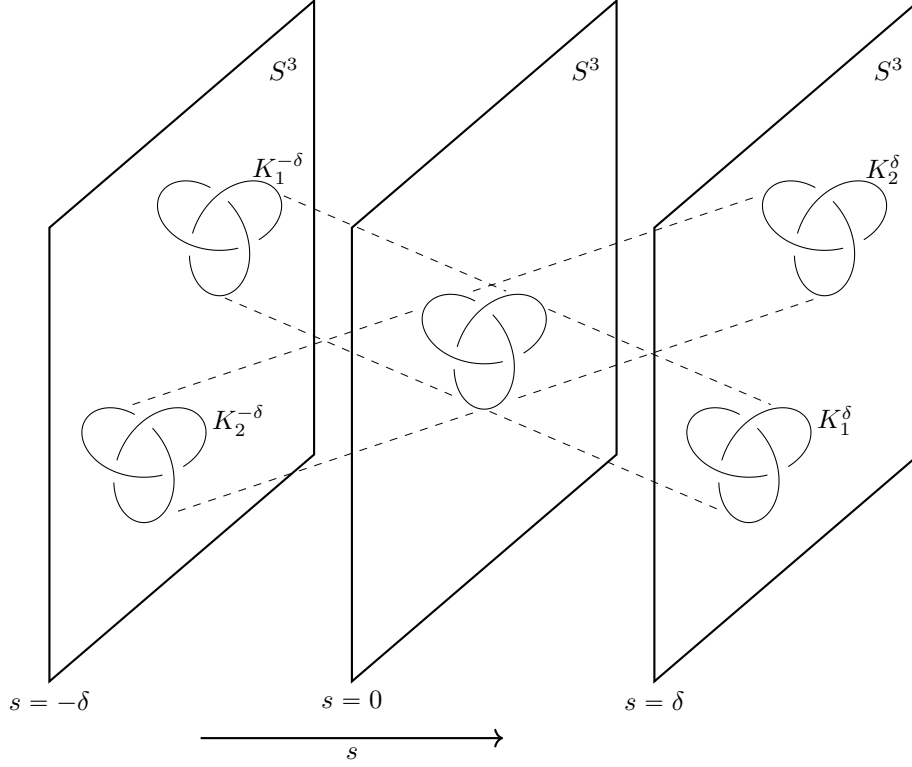


FIGURE 7. Local behavior at breaking orbit.

Proof. We will homotope the cylinders in a neighborhood of the breaking orbit into transverse position and then see that in this neighborhood the number $\lambda(w_1, w_2)$ counts exactly the intersections with sign that have appeared. We use the following formulation of the linking number of two knots in S^3 . Consider S^3 as the boundary of the 4-ball B^4 and let S_1, S_2 be embedded surfaces in B^4 such that $\partial S_i = K_i$. Then the algebraic intersection number of the homotopy classes of S_1, S_2 relative to the boundary equals the linking number $\ell(K_1, K_2)$. Let $\{-\delta\} \times S^3$ bound B^4 for $\delta \in (0, \varepsilon)$ and choose surfaces S_1, S_2 that bound $K_1^{-\delta}$ and $K_2^{-\delta}$. Homotope $\hat{w}_1$ and $\hat{w}_2$ in a small tubular neighborhood U of $\hat{y}$ such that $U \cong (-\frac{\delta}{2}, \frac{\delta}{2}) \times S^1 \times D^2$ such that the resulting cylinders intersect transversely, and such that the homotopy is constant on ∂U . Glue $[-\delta, \delta] \times S^3$ to B^4 by identifying ∂B^4 with $\{-\delta\} \times S^3$ and note that the cylinders $\hat{w}_1|_{[-\delta, \delta] \times S^1}$ and $\hat{w}_2|_{[-\delta, \delta] \times S^1}$ extend the surfaces S_1 and S_2 to Σ_1 and Σ_2 . By construction, Σ_1 and Σ_2 are surfaces in a 4-ball that bound the knots K_1^δ and K_2^δ on the boundary. The difference of the linking number

$$\ell(K_1^\delta, K_2^\delta) - \ell(K_1^{-\delta}, K_2^{-\delta})$$

is now given by

$$\sum_{z \in \Sigma_1 \cap \Sigma_2} \epsilon_z - \sum_{z \in S_1 \cap S_2} \epsilon_z = \sum_{z \in (\Sigma_1 \setminus S_1) \cap (\Sigma_2 \setminus S_2)} \epsilon_z,$$

where $\epsilon_z \in \{\pm 1\}$ depends on the orientation of the intersections. By construction, this exactly counts the intersections with sign that appear after a local homotopy of $\hat{w}_1$ and $\hat{w}_2$ near the breaking orbit. $\square$

The number $\lambda(w_1, w_2)$ is the local intersection number of $\hat{w}_1$ and $\hat{w}_2$ near the breaking orbit $\hat{y}$. That is, after a homotopy in a small neighborhood of $\hat{y}$ that makes the cylinders transverse, $\lambda(w_1, w_2)$ equals the number of intersections with signs in this neighborhood.

We now apply this to the breaking of Floer cylinders in Floer theory. Suppose that we have a homotopy of Hamiltonians $H : \mathbb{R} \times S^1 \times S \rightarrow \mathbb{R}$ such that $H(s, t, x)$ is independent of s for $|s|$ large enough. We want to consider pseudoholomorphic cylinders $u : S^1 \times \mathbb{R} \rightarrow S$ that satisfy the equation

$$\partial_s u + J_{s,t}(\partial_t u - X_{H_{s,t}} \circ u) = 0$$

and such that as $s \rightarrow \pm\infty$ we have $u(s, t) \rightarrow x^\pm(t)$, which consequently are periodic orbits of the Hamiltonian flow induced by $H_{\pm\infty}(t, x) = \lim_{s \rightarrow \pm\infty} H(s, t, x)$. Fix an orientation-preserving diffeomorphism $\varphi : (-1, 1) \rightarrow \mathbb{R}$. The cylinders $u \circ \varphi$ extend to a cylinder $\tilde{u} : [-1, 1] \times S^1 \rightarrow S$. In this case, two cylinders $\tilde{u}_1$ and $\tilde{u}_2$ are admissible if and only if the positive limiting orbits of u_1 and u_2 are pointwise distinct, and likewise the negative limits. We analogously define $\hat{u}$ to be the parameterization of the graph of $\tilde{u}$

$$\hat{u} : [-1, 1] \times S^1 \rightarrow [-1, 1] \times S^1 \times S, \quad (s, t) \mapsto (s, t, \tilde{u}(s, t))$$

and we say that u_1, u_2 are admissible if $\tilde{u}_1, \tilde{u}_2$ are so. Moreover, for admissible u_1, u_2 we define

$$\iota(u_1, u_2) = \iota(\tilde{u}_1, \tilde{u}_2).$$

Proposition 2.2. *Let u_1, u_2 be admissible cylinders satisfying the Floer equation*

$$\partial_s u_i + J_{s,t}(\partial_t u_i - X_{H_{s,t}} \circ u_i) = 0.$$

Then $\iota(u_1, u_2) \geq 0$ and if $\iota(u_1, u_2) = 0$ the maps $\hat{u}_1$ and $\hat{u}_2$ do not intersect.

Proof. We use the Gromov trick and give an almost complex structure on $\mathbb{R}_s \times S_t^1 \times S$ such that the graphs $\hat{u}_i$ are unperturbed holomorphic cylinders. Namely, define

$$\tilde{J}(\partial_s) = \partial_t + X_{H_{s,t}}, \quad \tilde{J}(\partial_t) = -\partial_s - J_{s,t}X_{H_{s,t}}, \quad \tilde{J}|_{TS} = J_{s,t},$$

where we identify $X_{H_{s,t}}(z)$ with its image under the differential of the inclusion

$$\iota_{s,t} : S \hookrightarrow \mathbb{R} \times S^1 \times S, \quad z \mapsto (s, t, z).$$

We then have

$$\partial_s \hat{u}_i + \tilde{J} \partial_t \hat{u}_i = 0.$$

The result now follows from standard theory of intersection of pseudoholomorphic curves in almost complex 4-manifolds (see [MS04]). $\square$

2.2. Gluing and Intersection Numbers. We now study how the intersection number behaves under gluing of pseudoholomorphic cylinders. We briefly recall the standard gluing construction for Floer cylinders (see [AD14, Chapter 9]). More concretely, fix a Hamiltonian $H : \mathbb{R} \times S^1 \times S \rightarrow \mathbb{R}$ and a smooth family of almost complex structures $\{J_{s,t}\}_{s \in \mathbb{R}, t \in S^1}$ and let $\mathcal{M}(x, z)$ denote the space of solutions $u : \mathbb{R} \times S^1 \rightarrow S$ to

$$\partial_s u + J_{s,t}(\partial_t u - X_{H_{s,t}} \circ u) = 0$$

such that $\lim_{s \rightarrow \infty} u(s, t) = z(t)$ and $\lim_{s \rightarrow -\infty} u(s, t) = x(t)$. For a generic choice of $H_{s,t}$ and $J_{s,t}$, the space $\mathcal{M}(x, z)$ is a finite-dimensional manifold carrying the C_{loc}^∞ topology. In our context, a gluing map is a continuous map

$$\psi : (\rho_0, \infty) \rightarrow \mathcal{M}(x, z)$$

such that there exist $u \in \mathcal{M}(x, y)$ and $v \in \mathcal{M}(y, z)$ and smooth functions $f^\pm : \mathbb{R} \rightarrow \mathbb{R}$ with $\lim_{\rho \rightarrow \infty} \psi_\rho(s + f^-(\rho), t) = u(s, t)$ and $\lim_{\rho \rightarrow \infty} \psi_\rho(s + f^+(\rho), t) = v(s, t)$ where we write ψ_ρ for $\psi(\rho)$. We then say that the curves ψ_ρ converge to the broken trajectory (u, v) with breaking orbit y . For us, the map ψ , if it exists, is always constructed as follows. Given $u \in \mathcal{M}(x, y)$ and $v \in \mathcal{M}(y, z)$, we define a *pre-gluing map* $w_\rho : \mathbb{R} \times S^1 \rightarrow S$, where the *gluing parameter* $\rho > 0$ is large enough. This pre-gluing map approximates the broken trajectory (u, v) from x to z which breaks at the orbit y .

First, let $\beta^+ : \mathbb{R} \rightarrow [0, 1]$ be smooth and increasing such that

$$\beta^+(s) = \begin{cases} 0, & s \leq \varepsilon \\ 1, & s \geq 1 \end{cases}$$

and let $\beta^-(s) = \beta^+(-s)$. Then we define

$$w_\rho(s, t) = \begin{cases} u(s + \rho, t) & s \leq -1 \\ \exp_{y(t)} \left(\beta^-(s) \exp_{y(t)}^{-1}(u(s + \rho, t)) + \beta^+(s) \exp_{y(t)}^{-1}(v(s - \rho, t)) \right) & s \in [-1, 1] \\ v(s - \rho, t) & s \geq 1 \end{cases}$$

where $\rho > \rho_0$ and ρ_0 is chosen large enough that the exponential maps are well defined.

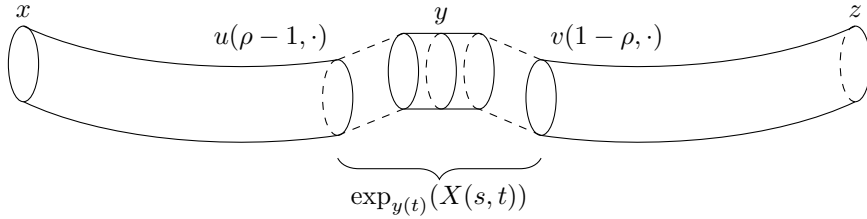


FIGURE 8. Pre-gluing of cylinders.

The map ψ is now given by

$$(4) \quad \psi_\rho(s, t) = \exp_{w_\rho(s, t)}(\xi_\rho(s, t)),$$

where $\xi_\rho(s, t)$ is a well-chosen continuous vector field with $\xi_\rho(s, t) \rightarrow 0$ as $|s| \rightarrow \infty$ and such that $\exp_{w_\rho(s, t)}(\xi_\rho(s, t))$ is indeed an element of $\mathcal{M}(x, z)$ (see, for example, [AD14, Chapter 9]). We now analyze the behavior of intersection numbers under standard Floer gluing.

Lemma 2.3. *Let $\varphi : (-\infty, 0) \rightarrow \mathbb{R}$ be a diffeomorphism such that $\varphi|_{(-\infty, -\varepsilon]}(x) = x$. We identify a broken trajectory (u, v) with the map*

$$u \# v : \mathbb{R} \times S^1 \rightarrow S, \quad (s, t) \mapsto \begin{cases} u(\varphi(s), t), & s < 0 \\ y(t), & s = 0 \\ v(-\varphi(-s), t) & s > 0 \end{cases}$$

where $\lim_{s \rightarrow \infty} u(s, t) = \lim_{s \rightarrow -\infty} v(s, t) = y(t)$. Then $u \# v$ and ψ_ρ are homotopic relative to the limiting orbits for every ρ large enough. In particular, $\widehat{u \# v}$ and $\hat{\psi}_\rho$ are homotopic for every ρ large enough.

Proof. By (4), ψ_ρ is homotopic to w_ρ for ρ large enough. For every $\delta \in (0, 1]$ define

$$\tilde{w}_\rho^\delta(s, t) = \begin{cases} u(\varphi(s) + \rho, t), & s \leq -\delta \\ \exp_{y(t)} \left(\beta^-\left(\frac{s}{\delta}\right) \exp_{y(t)}^{-1}(u(\varphi(s) + \rho, t)) + \beta^+\left(\frac{s}{\delta}\right) \exp_{y(t)}^{-1}(v(-\varphi(-s) - \rho, t)) \right) & s \in [-\delta, \delta] \\ v(-\varphi(-s) - \rho, t) & s \geq \delta \end{cases}$$

which is defined for any $\delta \in (0, 1]$ if ρ is large enough. Since $\varphi|_{(-\infty, -\varepsilon]} = \text{id}_{(-\infty, -\varepsilon]}$ (where ε is from the definition of $\beta^\pm$) we have $\tilde{w}_\rho^1 = w_\rho$ for ρ large enough. Now, for $\delta \rightarrow 0$ the map $\tilde{w}_\rho^\delta$ converges to (u, v) in the C^0 -topology. Hence, for every ρ we obtain a homotopy from (u, v) to $\tilde{w}_\rho^1$. Since w_ρ is homotopic to $\psi_\rho = \exp_{w_\rho}(\xi_\rho)$, we conclude that ψ_ρ is homotopic to $u \# v$ for ρ large enough. $\square$

A direct consequence is the following lemma.

Lemma 2.4 (Stability of Intersection). *Let $u_i \in \mathcal{M}(x_i, y_i)$ and $v_i \in \mathcal{M}(y_i, z_i)$ for $i = 1, 2$ be pairs of admissible curves, that is*

$$x_1(t) \neq x_2(t), \quad y_1(t) \neq y_2(t), \quad z_1(t) \neq z_2(t)$$

for all $t \in S^1$. Let $\psi^{(i)} : (\rho_0, \infty) \rightarrow \mathcal{M}(x_i, z_i)$ for $i = 1, 2$ be gluing maps of u_i, v_i . Then

$$\iota(u_1, u_2) + \iota(v_1, v_2) = \iota(\psi^{(1)}(\rho), \psi^{(2)}(\rho))$$

for every $\rho > 0$ large enough.

Proof. By Lemma 2.3 the cylinders $u_i \# v_i$ and $\psi^{(i)}(\rho)$ are homotopic relative to the boundary when ρ is large enough and hence

$$\iota(u_1 \# v_1, u_2 \# v_2) = \iota(\psi^{(1)}(\rho), \psi^{(2)}(\rho)).$$

Furthermore,

$$\iota(u_1, u_2) + \iota(v_1, v_2) = \iota(u_1 \# v_1, u_2 \# v_2)$$

since orbits satisfy $y_1(t) \neq y_2(t)$ for all $t \in S^1$ so that the count of intersection points split. $\square$

We will later encounter pairs of broken trajectories in a degenerate breaking position, as defined in Section 2.1, that break at the same periodic orbit y ; that is, we have two cylinders $u_1 \# v_1$ and $u_2 \# v_2$ that break at y . To use the symmetry of this configuration, we “swap” the latter two cylinders, giving another configuration in a degenerate breaking position (see Figure 9). The following lemma states that this operation does not change the cylinders’ intersection number and is essential to the proofs of the main theorems. For the proof of the lemma we use the expression

$$\iota(u_1 \# v_1, u_2 \# v_2) = \lambda(u_1 \# v_1, u_2 \# v_2) + \sum_{z \in \widehat{u_1 \# v_1} \cap \widehat{u_2 \# v_2} \setminus \hat{y}} \epsilon_z$$

as in Lemma 2.1. We then use the description of λ by linking numbers

$$\lambda(u_1 \# v_1, u_2 \# v_2) = \ell(K_1^\delta, K_2^\delta) - \ell(K_1^{-\delta}, K_2^{-\delta})$$

and use the fact that the linking number is symmetric.

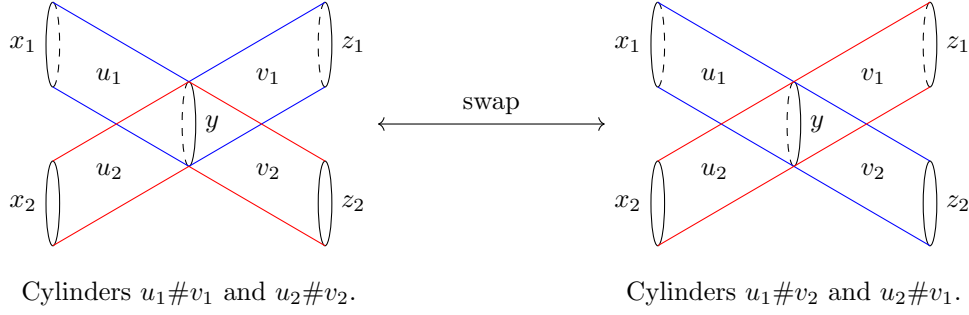


FIGURE 9. The swapping operation.

Lemma 2.5 (Swapping Lemma). *Let $u_i \in \mathcal{M}(x_i, y)$ and $v_i \in \mathcal{M}(y, z_i)$ for $i = 1, \dots, n$ and let $\{x_i\}$ and $\{z_i\}$ be sets of pairwise distinct orbits. Then the maps $\{u_i \# v_i\}_i$ are pairwise admissible and for every σ in the symmetric group of n elements, we have*

$$\sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j) = \sum_{i \neq j} \iota(u_i \# v_{\sigma(i)}, u_j \# v_{\sigma(j)}).$$

Proof. By the asymptotic behavior of the curves, the intersections of $\hat{u}_i, \hat{u}_j$ and $\hat{v}_i, \hat{v}_j$ respectively are disjoint from $\hat{y}$ and do not accumulate at $\hat{y}$ (see, for example, [Sie08, Cor. 2.5], whose hypothesis holds due to unique continuation). After a small homotopy through admissible pairs of curves, small enough that they intersect transversely, positively, and only finitely many times away from $\hat{y}$. Hence, we may assume that the curves $\{\hat{u}_i \# \hat{v}_i\}$ are in degenerate breaking position and Lemma 2.1 we have

$$\iota(u_i \# v_i, u_j \# v_j) = \lambda(u_i \# v_i, u_j \# v_j) + |u_i \cap u_j| + |v_i \cap v_j|.$$

Now, $\lambda(u_i \# v_i, u_j \# v_j) = \ell_{i,j}^+ - \ell_{i,j}^-$ where $\ell_{i,j}^\pm$ is the linking number induced by the knots of $\hat{u}_i, \hat{u}_j$ (for negative sign) and $\hat{v}_i, \hat{v}_j$ (for positive sign) near $\hat{y}$. Thus, we have

$$\begin{aligned} \sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j) &= \sum_{i \neq j} \ell_{i,j}^+ - \ell_{i,j}^- + |u_i \cap u_j| + |v_i \cap v_j| \\ &= \sum_{i \neq j} \ell_{\sigma(i), \sigma(j)}^+ - \ell_{i,j}^- + |u_i \cap u_j| + |v_{\sigma(i)} \cap v_{\sigma(j)}| \\ &= \sum_{i \neq j} \iota(u_i \# v_{\sigma(i)}, u_j \# v_{\sigma(j)}) \end{aligned}$$

□

3. THE BRAIDED FLOER CHAIN COMPLEX AND ITS HOMOLOGY

A (closed pure surface) braid is a set of embeddings $\{s_1, \dots, s_k\}$ with pairwise disjoint images where the strands s_i are given by

$$s_i : S^1 \rightarrow S \times S^1, \quad t \mapsto (x_i(t), t)$$

for loops $x_i : S^1 \rightarrow S$. We say two braids are isotopic if an isotopy of the strands makes their images agree. We refer to an isotopy class of this type as a *braid class with k strands*. For a smooth Hamiltonian $H : S^1 \times S \rightarrow \mathbb{R}$ we define

$$\mathcal{P}(H) = \{x : S^1 \rightarrow S \mid x(t) = \phi_H^t(x(0))\}$$

the space of 1-periodic orbits where we identify $S^1 = \mathbb{R}/\mathbb{Z}$. For every $k \in \mathbb{Z}_{\geq 1}$ and a set X , we denote by $\Delta_X \subset X^k$ the big diagonal

$$\Delta_X = \{(x_1, \dots, x_k) \in X^k \mid x_i = x_j \text{ for some } i \neq j\}.$$

Moreover, let

$$B_k \mathcal{P}(H) = (\mathcal{P}(H)^k - \Delta_{\mathcal{P}(H)})/S_k$$

be the set of *braids of orbits with k strands*, where we quotient by the action of the symmetric group of k elements S_k that permutes components. Any set $x = \{x_1, \dots, x_k\} \in B_k \mathcal{P}(H)$ gives rise to a closed surface braid defined by

$$\mathcal{B}(H, x) = [\{(x_1(t), t), \dots, (x_k(t), t)\}_{t \in S^1}].$$

Here the brackets $[\cdot]$ denote the isotopy class of the closed surface braid. The set of unordered k -tuples of orbits that form a β -braid is denoted by

$$B\mathcal{P}_\beta(H) = \{x \in B_k \mathcal{P}(H) \mid \mathcal{B}(H, x) = \beta\}.$$

The ordered k -tuples of orbits that form a β -braid is defined by

$$OB\mathcal{P}_\beta(H) := \pi^{-1}(B\mathcal{P}_\beta(H)),$$

where $\pi : OB_\beta \mathcal{P}(H) \rightarrow B_\beta \mathcal{P}(H)$ is the map that forgets the ordering. An element in $B\mathcal{P}_\beta(H)$ is called a β -*braid of orbits* of the flow of H . An orbit in a braid of orbits is called a *strand orbit*. The set of strands in a β -braid of orbits is denoted by

$$\mathcal{P}_\beta(H) = \{x_1 \in \mathcal{P}(H) \mid \exists \{x_1, \dots, x_k\} \text{ such that } \mathcal{B}(H, \{x_1, \dots, x_k\}) = \beta\}.$$

The vector space generated by β -braids of orbits is denoted

$$\text{BCF}_\beta(H) = \langle B\mathcal{P}_\beta(H) \rangle_{\mathbb{Z}_2}.$$

As the notation suggests, we call these spaces *braided Floer complexes in the braid class β* . For a non-degenerate Hamiltonian H and the additional generic auxiliary data of an almost complex structure, these vector spaces can be endowed with a differential ∂ that preserves the homotopy class of the braids, justifying the name.

3.1. The Differential and Continuation Maps. In this section we define a homological differential ∂ on the space $\text{BCF}_\beta(H, J)$ and prove that $\partial^2 = 0$. This differential counts Floer-theoretic isotopies that are given by tuples of Floer cylinders of Conley–Zehnder index 1 that have pairwise vanishing intersection numbers.

Unfortunately, the case $S = T^2$ brings some additional technicalities when considering braids β with strands represented by non-contractible loops. This is because the moduli space defined shortly is not compact, since there are no global action bounds on holomorphic curves connecting non-contractible orbits. **In the remainder of Section 3 we will only consider braids on T^2 where every strand is contractible.** However, the author believes that introducing Novikov coefficients resolves this issue, and that the proofs below go through analogously. For simplicity, we will not address this situation.

The Conley–Zehnder index for a (not necessarily contractible) orbit x is defined as follows: For every homotopy class α and fixed reference loop η_α in this class, choose a symplectic trivialization of $\eta_\alpha^* TS$. Choose a cylinder u from η_α to x and push the trivialization along u . This induces a symplectic trivialization on x , which is unique up to homotopy. The trivialization allows us to represent the differential $((d\phi_H^t)_{x(t)})_{t \in [0,1]}$ as a path of symplectic matrices. The index $\mu(x, u)$ is now given by the *Maslov index* of this path. For genus $g \geq 1$ the index is independent of u , and we write $\mu(z)$, while for $g = 0$ the index differs by a multiple of 4 for different choices (since $c_1(S^2) = 2$), and we write $\mu(z)$ for the index in

$\mathbb{Z}_4$. For a cylinder $w : \mathbb{R} \times S^1 \rightarrow S$ connecting periodic orbits $x(t) = \lim_{s \rightarrow -\infty} w(s, t)$ and $y(t) = \lim_{s \rightarrow \infty} w(s, t)$ we define the index

$$\mu(w) = \mu(x, u) - \mu(y, u \# w)$$

which is independent of the capping u of x . Before we get into the technicalities, note the following.

Remark 3.0.1. In general, the homology is not invariant under the choice of Hamiltonian. For a C^2 -small Hamiltonian h , the generated Hamiltonian diffeomorphism has only non-degenerate and constant periodic orbits. Hence, in any non-trivial braid class β , the braided Floer homology $\text{BHF}_\beta(h, J_h)$ is trivial for any choice of generic almost complex structure J_h . However, this need not be the case in general. For example, consider a surface of genus $g > 0$ and a Hamiltonian H on S . Let $\beta_{H, \text{tot}}$ be the braid class given by the collection of all periodic orbits. Hence $\text{BCF}_{\beta_{H, \text{tot}}}(H)$ is one-dimensional. Since the total braid contains the orbit of minimal and maximal Conley–Zehnder index, no incoming or outgoing Floer isotopies can exist from this braid of orbits. Hence, the class is non-trivial in homology and $\text{BHF}_{\beta_{H, \text{tot}}}(H, J_H)$ is one-dimensional. Now, in general $\beta_{H, \text{tot}}$ is non-trivial and hence $\text{BHF}_{\beta_{H, \text{tot}}}(H, J_H) \neq \text{BHF}_{\beta_{H, \text{tot}}}(h, J_h)$.

For every non-degenerate Hamiltonian H a generic choice of almost complex structure J ensures that the moduli space $\mathcal{M}_\ell(x, y)$ of solutions $u : \mathbb{R} \times S^1 \rightarrow S$ to the Floer equation

$$\partial_s u + J(\partial_t u - X_H \circ u) = 0$$

that are of index ℓ and connect periodic orbits x to y is a manifold of dimension ℓ when equipped with the C_{loc}^∞ -topology (see [AD14]). We call such a pair (H, J) *regular*. For all $x, y \in \mathcal{P}(H)^k \setminus \Delta_{\mathcal{P}(H)}$ we define

$$\mathcal{M}_\ell(x, y) = \{u \in \mathcal{M}_\ell(x_1, y_1) \times \cdots \times \mathcal{M}_\ell(x_k, y_k) \mid \iota(u) = 0\}$$

which are the connected components of $\mathcal{M}_\ell(x_1, y_1) \times \cdots \times \mathcal{M}_\ell(x_k, y_k)$ on which the intersection number evaluates to 0. On $\mathcal{M}_\ell(x, y)$, an $\mathbb{R}^k$ -action shifts the s -coordinate in every component. The quotient

$$\widehat{\mathcal{M}}_\ell(x, y) := \mathcal{M}_\ell(x, y) / \mathbb{R}^k$$

is a manifold of dimension $(\ell - 1)k$. In the case $\ell = 1$, all components of $\widehat{\mathcal{M}}_\ell(x, y)$ are 0-dimensional manifolds and compact, as in standard Hamiltonian Floer theory. Hence, $\widehat{\mathcal{M}}_1(x, y)$ is a finite set, which allows us to define the differential of braided Floer complexes by

$$\partial : \text{BCF}_\beta(H, J) \rightarrow \text{BCF}_\beta(H, J), \quad [x] \mapsto \sum_{y \in \text{OB}\mathcal{P}_\beta(H)} |\widehat{\mathcal{M}}_1(x, y)| [y].$$

For convenience we will write $\xi(x, y) = |\widehat{\mathcal{M}}_1(x, y)|$.

Lemma 3.1. *The differential ∂ is well-defined.*

Proof. First, note that for every $\sigma \in S_k$ we have

$$\sum_{y \in \text{OB}\mathcal{P}_\beta(H)} \xi(x, y)[y] = \sum_{y \in \text{OB}\mathcal{P}_\beta(H)} \xi(\sigma(x), \sigma(y))[\sigma(y)] = \sum_{y \in \text{OB}\mathcal{P}_\beta(H)} \xi(\sigma(x), y)[y].$$

Hence, ∂ is independent of the choice of a representative of $[x]$. $\square$

Remark 3.1.1. The homology theory we build differs essentially from the one defined by van den Berg, Ghrist, Vandervorst, and Wójcik in [vdBGVW14]. This is already evident from the fact that their braid Floer homology is independent of H . A concrete difference is the following. In [vdBGVW14] the differential essentially counts single cylinders of index 1 that connect a collection of orbits that form a braid in some fixed (relative) class. In contrast, we count collections of multiple cylinders all of index 1 that connect strands of the braids pairwise.

Let us now prove that ∂ is indeed a differential.

Proposition 3.2. *For every regular pair (H, J) and braid class β we have $\partial^2 = 0$.*

This shows that $\text{BCF}_\beta(H, J)$ is an honest chain complex. We denote its homology by

$$\text{BHF}_\beta(H, J) = \ker(\partial) / \text{im}(\partial).$$

Proof. Note that ∂^2 counts exactly the number of tuples of broken trajectories with intersection number 0

$$\partial^2([x]) = \sum_{y \in \text{OB}\mathcal{P}_\beta(H)} \sum_{z \in \text{OB}\mathcal{P}_\beta(H)} \xi(x, y) \xi(y, z) [z],$$

where $\xi(x, y) \xi(y, z)$ is the number of broken trajectories from x to z breaking at y that have intersection number 0. The proof reduces to showing the following.

Claim: Let k be the number of strands of β . Then for every $x, z \in \text{OB}\mathcal{P}_\beta(H)$ we have

$$\sum_{\sigma \in S_k} \sum_{y \in \text{OB}\mathcal{P}_\beta(H)} \xi(x, y) \xi(y, \sigma(z)) = 0 \pmod{2}.$$

We first finish the proof of the proposition assuming this claim. Namely, we have

$$\begin{aligned} \partial^2([x]) &= \sum_{y \in \text{OB}\mathcal{P}_\beta(H)} \sum_{z \in \text{OB}\mathcal{P}_\beta(H)} \xi(x, y) \xi(y, z) [z] \\ &= \sum_{[z] \in \text{B}\mathcal{P}_\beta(H)} \sum_{\sigma \in S_k} \sum_{y \in \text{OB}\mathcal{P}_\beta(H)} \xi(x, y) \xi(y, \sigma(z)) [z] \\ &= 0. \end{aligned}$$

Let us now prove the claim. For periodic orbits $x, z \in \mathcal{P}(H)$, the space $\widehat{\mathcal{M}}_2(x, z)$ consists of a union of open intervals and copies of S^1 . Let $\widehat{\mathcal{M}}_2^{\text{int}}(x, z)$ denote the union of the connected components that are intervals and hence not yet compact. From standard Floer theory, it follows that the space $\widehat{\mathcal{M}}_2^{\text{int}}(x, z)$ can be compactified by gluing broken trajectories

$$\mathcal{B}(x, z) = \bigcup_{y \in \mathcal{P}(H)} \widehat{\mathcal{M}}_1(x, y) \times \widehat{\mathcal{M}}_1(y, z).$$

For every $x, z \in \text{OB}\mathcal{P}_\beta(H)$ let us denote the space

$$\widehat{\mathcal{M}}_2^{\text{cube}}(x, z) = \{(u_1, \dots, u_k) \in \widehat{\mathcal{M}}_2^{\text{int}}(x_1, z_1) \times \dots \times \widehat{\mathcal{M}}_2^{\text{int}}(x_k, z_k) \mid \iota(u) = 0\}$$

which is the union of connected components of $\widehat{\mathcal{M}}_\ell(x, z)$ that are hypercubes. The space $\widehat{\mathcal{M}}_2^{\text{int}}(x_1, z_1) \times \dots \times \widehat{\mathcal{M}}_2^{\text{int}}(x_k, z_k)$ can be compactified by compactifying each component. We obtain a union of closed hypercubes whose corner points are given by

$$\mathcal{B}(x, z) = \bigcup_{y \in \mathcal{P}(H)^k} (\widehat{\mathcal{M}}_1(x_1, y_1) \times \widehat{\mathcal{M}}_1(y_1, z_1)) \times \dots \times (\widehat{\mathcal{M}}_1(x_k, y_k) \times \widehat{\mathcal{M}}_1(y_k, z_k)).$$

By Lemma 2.3, gluing preserves the algebraic intersection number of broken trajectories. Hence, $\hat{\mathcal{M}}_2^{\text{cube}}(x, z)$ can be compactified and we obtain a disjoint union of closed hypercubes with corner point

$$\mathcal{B}^0(x, z) = \{((u_1, v_1), \dots, (u_k, v_k)) \in \mathcal{B}(x, z) \mid \sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j) = 0\}.$$

Abusing notation, let us denote the compactification again by $\hat{\mathcal{M}}_2^{\text{cube}}(x, z)$, and note that this space is a compact manifold consisting of hypercubes that are given by products of closed intervals. The 0-dimensional corner points are given by $\mathcal{B}^0(x, z)$ and therefore, we find that

$$(5) \quad |\mathcal{B}^0(x, z)| = 0 \pmod{2}.$$

To any element $w \in \mathcal{B}^0(x, z)$, we can associate its breaking orbits $y_w \in \mathcal{P}(H)^k$. An element $w \in \mathcal{B}^0(x, z)$ is called *degenerate* if $y_w \in \Delta_{\mathcal{P}(H)}$, otherwise it is called *regular*. Let us define

$$\mathcal{D}(x, z) = \{w \in \mathcal{B}^0(x, z) \mid y_w \in \Delta_{\mathcal{P}(H)}\}$$

the set of degenerate broken trajectories.

Observation: Let $w \in \mathcal{B}^0(x, z)$ such that $y_w \in \mathcal{P}(H)^k$ is regular. Then we have $y_w \in \text{OB}\mathcal{P}_\beta(H)$ and therefore, $w \in \hat{\mathcal{M}}_1(x, y_w) \times \hat{\mathcal{M}}_1(y_w, z)$.

Proof. For $w = ((u_1, v_1), \dots, (u_k, v_k)) \in \mathcal{B}^0(x, z)$ such that y_w is regular, we have that

$$0 = \sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j) = \sum_{i \neq j} \iota(u_i, u_j) + \iota(v_i, v_j)$$

and hence $\iota(u_i, u_j) = \iota(v_i, v_j) = 0$ for all $i \neq j$. Abusing notation let us write $u_i \in \mathcal{M}(x_i, (y_w)_i)$ for a representative of u_i and note that we now have $u_i(s, t) \neq u_j(s, t)$ for all $(s, t) \in \mathbb{R} \times S^1$ and $i \neq j$. Hence, the cylinders $u_1, \dots, u_k$ give rise to an isotopy of braids associated to x and y_w and therefore $y_w \in \text{OB}\mathcal{P}_\beta(H)$. This concludes the proof of the observation. $\square$

By the observation,

$$\mathcal{B}^0(x, z) = \bigcup_{y \in \text{OB}\mathcal{P}_\beta(H)} \hat{\mathcal{M}}_1(x, y) \times \hat{\mathcal{M}}_1(y, z) \cup \mathcal{D}(x, z)$$

since breaking orbits that lie in $\text{OB}\mathcal{P}_\beta(H)$ are exactly the regular ones. Together with (5) we conclude that

$$0 = \sum_{y \in \text{OB}\mathcal{P}_\beta(H)} \xi(x, y) \xi(y, z) + |\mathcal{D}(x, z)| \pmod{2}.$$

To finish the proof of the claim it now suffices to show that

$$(6) \quad \sum_{\sigma \in S_k} |\mathcal{D}(x, \sigma(z))| = 0 \pmod{2}.$$

For $y \in \Delta_{\mathcal{P}(H)}$ denote

$$\mathcal{D}(x, y, z) = \{w \in \mathcal{D}(x, z) \mid y_w = y\}$$

and note that

$$\mathcal{D}(x, z) = \bigcup_{y \in \Delta_{\mathcal{P}(H)}} \mathcal{D}(x, y, z).$$

The equality (6) follows from a symmetry argument. Namely, for $y \in \Delta_{\mathcal{P}(H)}$ write

$$\text{Fix}(y) = \{\sigma \in S_k \mid \sigma(y) = y\} \subset S_k$$

for the subgroup of elements that fix y . We now define the map

$$T_{\sigma}^{x,y,z} : \mathcal{D}(x, y, z) \rightarrow \mathcal{D}(x, y, \sigma(z)), \quad (u, v) \mapsto (u, \sigma(v)).$$

This is well-defined since after the swapping Lemma 2.5 we have $\iota(u \# v) = \iota(u \# \sigma(v))$. Moreover, it is a bijection with inverse given by $T_{\sigma^{-1}}^{x,y,\sigma(z)}$. Hence,

$$(7) \quad |\mathcal{D}(x, y, z)| = |\mathcal{D}(x, y, \sigma(z))|$$

for every $\sigma \in \text{Fix}(y)$. We now have

$$\begin{aligned} \sum_{\sigma \in S_k} |\mathcal{D}(x, \sigma(z))| &= \sum_{\sigma \in S_k} \sum_{y \in \Delta_{\mathcal{P}(H)}} |\mathcal{D}(x, y, \sigma(z))| \\ &= \sum_{y \in \Delta_{\mathcal{P}(H)}} \sum_{[\mu] \in S_k / \text{Fix}(y)} \sum_{\sigma \in \text{Fix}(y)} |\mathcal{D}(x, y, \sigma(\mu(z)))| \\ &\stackrel{(7)}{=} \sum_{y \in \Delta_{\mathcal{P}(H)}} \sum_{[\mu] \in S_k / \text{Fix}(y)} \sum_{\sigma \in \text{Fix}(y)} |\mathcal{D}(x, y, \mu(z))| \\ &= \sum_{y \in \Delta_{\mathcal{P}(H)}} \sum_{[\mu] \in S_k / \text{Fix}(y)} |\text{Fix}(y)| |\mathcal{D}(x, y, \mu(z))| \end{aligned}$$

where $S_k / \text{Fix}(y) = \{\text{Fix}(y)\mu \mid \mu \in S_k\}$. The subgroup $\text{Fix}(y)$ is a product of permutation groups, so its cardinality $|\text{Fix}(y)|$ is divisible by 2. $\square$

3.2. Continuation Maps. We now define continuation maps on the braided Floer chain complex and show that they are chain maps. We first recall some standard Hamiltonian Floer theory. Given two non-degenerate Hamiltonians $H^{-\infty}$ and H^{∞} and generic choices of smooth loops of almost complex structures $J_{\pm\infty} = (J_{t,\pm\infty})_{t \in S^1}$, the *continuation data* $\Gamma = (H, J)$ connecting $(H_{-\infty}, J_{-\infty})$ and (H_{∞}, J_{∞}) consists of a Hamiltonian $H : \mathbb{R} \times S^1 \times S \rightarrow \mathbb{R}$ and a smooth family of almost complex structures $J = (J_{s,t})_{t \in S^1, s \in \mathbb{R}}$ such that there exists $R > 0$ and

$$H(s, t, x) = \begin{cases} H_{-\infty}(x, t), & s \leq -R \\ H_{\infty}(x, t), & s \geq R \end{cases}$$

and

$$J_{s,t} = \begin{cases} J_{t,-\infty}, & s \leq -R \\ J_{t,\infty}, & s \geq R. \end{cases}$$

For periodic orbits $x \in \mathcal{P}(H^{-\infty})$ and $y \in \mathcal{P}(H_{\infty})$ we denote by $\mathcal{M}_{\ell}^{\Gamma}(x, y)$ the space of solutions $u : S^1 \times \mathbb{R} \rightarrow S$ of the Floer equation

$$\partial_s u + J_{s,t}(\partial_t u - X_{H_{s,t}} \circ u) = 0.$$

Generic continuation data Γ is *regular*, which is a transversality condition ensuring that the subspace $\mathcal{M}_{\ell}^{\Gamma}(x, y) \subset \mathcal{M}^{\Gamma}(x, y)$ of solutions that are cylinders with Maslov index ℓ is a manifold of dimension ℓ equipped with the C_{loc}^{∞} -topology (see [AD14]).

Now given $x \in \mathcal{P}(H_{-\infty})^k \setminus \Delta_{\mathcal{P}(H_{-\infty})}$ and $y \in \mathcal{P}(H_{\infty})^k \setminus \Delta_{\mathcal{P}(H_{\infty})}$ a *continuation isotopy* connecting x and y is given by a tuple $u = (u_1, \dots, u_k)$ of cylinders $u_i \in \mathcal{M}^{\Gamma}(x_i, y_i)$ such that

for all $i \neq j$ and for all $(s, t) \in \mathbb{R} \times S^1$ we have $u_i(s, t) \neq u_j(s, t)$. This condition can be imposed purely algebraically. Namely, consider the intersection number

$$\iota : \mathcal{M}^\Gamma(x_1, y_1) \times \cdots \times \mathcal{M}^\Gamma(x_k, y_k) \rightarrow \mathbb{Z}_{\geq 0}, \quad \iota(u) = \sum_{i \neq j} \iota(u_i, u_j).$$

By positivity of intersection, this number is non-negative, and if $\iota(u) = 0$ we have $\iota(u_i, u_j) = 0$ for all $i \neq j$. This in particular means that there cannot be any $(s, t) \in \mathbb{R} \times S^1$ such that $u_i(s, t) = u_j(s, t)$ since this would amount to a positive intersection. In other words, u is an isotopy if and only if $\iota(u) = 0$. Let us define the moduli space of continuation isotopies

$$\mathcal{M}^\Gamma(x, y) = \{(u_1, \dots, u_k) \in \mathcal{M}^\Gamma(x_1, y_1) \times \cdots \times \mathcal{M}^\Gamma(x_k, y_k) \mid \iota(u) = 0\}$$

and denote the subspace of isotopies consisting of curves with index $\ell \in \mathbb{Z}$ by

$$\mathcal{M}_\ell^\Gamma(x, y) = \{(u_1, \dots, u_k) \in \mathcal{M}_\ell^\Gamma(x_1, y_1) \times \cdots \times \mathcal{M}_\ell^\Gamma(x_k, y_k) \mid \iota(u) = 0\}.$$

Note that any path in the space $\mathcal{M}^\Gamma(x_1, y_1) \times \cdots \times \mathcal{M}^\Gamma(x_k, y_k)$ is, in particular, a homotopy of cylinders relative to limiting orbits. This means that ι is locally constant and $\mathcal{M}^\Gamma(x, y)$ is simply given by the connected component of $\mathcal{M}^\Gamma(x_1, y_1) \times \cdots \times \mathcal{M}^\Gamma(x_k, y_k)$. Hence, for generic continuation data Γ , the moduli space $\mathcal{M}^\Gamma(x, y)$ is a manifold. Similarly, $\mathcal{M}_\ell^\Gamma(x, y)$ is a manifold of dimension ℓk . (Warning: $\mathcal{M}_\ell^\Gamma(x, y)$ is not the part of $\mathcal{M}^\Gamma(x, y)$ with local dimension ℓk .)

For regular pairs (H, J_H) and (G, J_G) and regular continuation data Γ between the pairs let

$$\Phi^\Gamma : \text{BCF}_\beta(H) \rightarrow \text{BCF}_\beta(G), \quad [x] \mapsto \sum_{y \in \text{OB}\mathcal{P}_\beta(G)} |\mathcal{M}_0^\Gamma(x, y)|[y].$$

We from now on write

$$\xi^\Gamma(x, y) = |\mathcal{M}_0^\Gamma(x, y)|.$$

As in Lemma 3.1, this map is well-defined.

Proposition 3.3. *Let (H, J_H) and (G, J_G) be regular pairs and Γ regular continuation data connecting (H, J_H) to (G, J_G) . Then continuation maps are chain maps,*

$$\Phi^\Gamma \circ \partial_{(H, J_H)} = \partial_{(G, J_G)} \circ \Phi^\Gamma.$$

Before proving Proposition 3.3, we first discuss the proof strategy. To conclude that $\Phi^\Gamma \circ \partial_{(H, J_H)} + \partial_{(G, J_G)} \circ \Phi^\Gamma = 0$ we need to show that the number of tuples of broken trajectories $((u_1, v_1), \dots, (u_k, v_k))$ connecting braided orbits x to y and y to z which are of type β is even. First for $x, z \in \text{OB}\mathcal{P}_\beta(H)$ we define a moduli space which is given by a compactification of a subspace of the product

$$\mathcal{M}_1^\Gamma(x_1, z_1) \times \cdots \times \mathcal{M}_1^\Gamma(x_k, z_k)$$

and consists of a union of hypercubes

$$Q = I_1 \times \cdots \times I_k,$$

where I_j is a closed interval in the compactification of $\mathcal{M}_1^\Gamma(x_j, z_j)$. This moduli space is such that all broken trajectories $((u_1, v_1), \dots, (u_k, v_k))$ as above appear as corner points of its hypercubes. Such corner points, corresponding to tuples of broken trajectories, come in two types:

$$\text{Type 1: } (u_i, v_i) \in \mathcal{M}_0^\Gamma(x_i, y_i) \times \hat{\mathcal{M}}_1(y_i, z_i)$$

$$\text{Type 2: } (u_i, v_i) \in \hat{\mathcal{M}}_1(x_i, y_i) \times \mathcal{M}_0^\Gamma(y_i, z_i).$$

A corner point is called *pure* if all components are either of type I or of type II. These pure corner points are exactly the ones that are counted by the map $\Phi^\Gamma \circ \partial_{(H, J_H)} + \partial_{(G, J_G)} \circ \Phi^\Gamma$. To conclude that the number of these corner points is even, we will apply the following combinatorial lemma.

Lemma 3.4. *Let $\rho_i : \{\pm 1\} \rightarrow \{a, b\}$ for $i = 1, \dots, k$ be colorings and view the set $\{\pm 1\}^k$ as the corner points of a hypercube. We call a corner point $c = (c_1, \dots, c_k) \in \{\pm 1\}^k$ pure if $\rho_i(c_i) = a$ for all i or $\rho_i(c_i) = b$ for all i . Then the number of pure corner points is even.*

Proof. We prove the lemma by induction over k . Let n_a and n_b denote the number of a -pure and b -pure corner points, respectively, of a k -dimensional hypercube. For $k = 1$, the lemma holds; there are two corner points, and both are pure. Now suppose the result holds for $k \geq 1$. Let $\tilde{n}_a$ and $\tilde{n}_b$ denote the number of pure a - or b -corner points of the hypercubes obtained by forgetting the last coordinate. Using that $2 = |\rho_{k+1}^{-1}(a)| + |\rho_{k+1}^{-1}(b)|$ we find

$$\begin{aligned} n_a + n_b &= \tilde{n}_a |\rho_{k+1}^{-1}(a)| + \tilde{n}_b |\rho_{k+1}^{-1}(b)| \\ &= \tilde{n}_a (2 - |\rho_{k+1}^{-1}(b)|) + \tilde{n}_b |\rho_{k+1}^{-1}(b)| \\ &= 2\tilde{n}_a + (\tilde{n}_b - \tilde{n}_a) |\rho_{k+1}^{-1}(b)| \end{aligned}$$

which is even since $\tilde{n}_a + \tilde{n}_b$ and therefore also $\tilde{n}_b - \tilde{n}_a$ is even by assumption. $\square$

The boundary components of I_j can be colored by the type of broken trajectories. We will apply Lemma 3.4 to this coloring.

Proof of Proposition 3.3. Note that for $x \in \text{OB}\mathcal{P}_\beta(H)$ the map $\Psi = \Phi^\Gamma \circ \partial_{(H, J_H)} + \partial_{(G, J_G)} \circ \Phi^\Gamma$ is given by

$$\Psi([x]) = \sum_{z \in \text{OB}\mathcal{P}_\beta(G)} \eta(x, z)[z],$$

where $\eta(x, z)$ is the cardinality of the following set consisting of broken trajectories,

$$\mathcal{B}(x, z) = \bigcup_{y \in \text{OB}\mathcal{P}_\beta(G)} \mathcal{M}_0^\Gamma(x, y) \times \widehat{\mathcal{M}}_1(y, z) \cup \bigcup_{y \in \text{OB}\mathcal{P}_\beta(H)} \widehat{\mathcal{M}}_1(x, y) \times \mathcal{M}_0^\Gamma(y, z).$$

To prove the proposition, we need to show that $\Psi = 0$. First note that we have

$$\Psi([x]) = \sum_{z \in \text{OB}\mathcal{P}_\beta(G)} \eta(x, z)[z] = \sum_{[z] \in \mathcal{B}\mathcal{P}_\beta(G)} \sum_{\sigma \in S_k} \eta(x, \sigma(z))[z].$$

In the rest of the proof, we will show that for $x \in \text{OB}\mathcal{P}_\beta(H)$ and $z \in \text{OB}\mathcal{P}_\beta(G)$ we have

$$(8) \quad \sum_{\sigma \in S_k} \eta(x, \sigma(z)) = 0 \pmod{2}$$

which proves the proposition.

For $x \in \mathcal{P}(H)$ and $z \in \mathcal{P}(G)$, the moduli space $\mathcal{M}_1^\Gamma(x, z)$ is a disjoint union of open intervals and copies of S^1 . Let $\mathcal{M}_1^{\Gamma, \text{int}}(x, z)$ denote the subset that is given by the union of all intervals. From standard Floer theory we know that $\mathcal{M}_1^{\Gamma, \text{int}}(x, z)$ can be compactified by gluing broken trajectories

$$\bigcup_{y \in \mathcal{P}(G)} \mathcal{M}_0^\Gamma(x, y) \times \widehat{\mathcal{M}}_1(y, z) \cup \bigcup_{y \in \mathcal{P}(H)} \widehat{\mathcal{M}}_1(x, y) \times \mathcal{M}_0^\Gamma(y, z)$$

as boundary components. It follows that for $x \in \text{OB}\mathcal{P}_\beta(H)$ and $z \in \text{OB}\mathcal{P}_\beta(G)$ the space $\mathcal{M}_1^{\Gamma, \text{int}}(x_1, z_1) \times \cdots \times \mathcal{M}_1^{\Gamma, \text{int}}(x_k, z_k)$ consisting of open hypercubes can be compactified to a disjoint union of closed hypercubes; corner points are given by

$$\mathcal{C}(x, z) = \prod_{i=1}^k \left(\bigcup_{y_i \in \mathcal{P}(G)} \mathcal{M}_0^\Gamma(x_i, y_i) \times \widehat{\mathcal{M}}_1(y_i, z_i) \cup \bigcup_{y_i \in \mathcal{P}(H)} \widehat{\mathcal{M}}_1(x_i, y_i) \times \mathcal{M}_0^\Gamma(y_i, z_i) \right).$$

Let now

$$\mathcal{M}_1^{\Gamma, \text{cube}}(x, z) = \{(u_1, \dots, u_k) \in \mathcal{M}_1^{\Gamma, \text{int}}(x_1, z_1) \times \cdots \times \mathcal{M}_1^{\Gamma, \text{int}}(x_k, z_k) \mid \iota(u) = 0\}$$

which is the subset of $\mathcal{M}_1^\Gamma(x, z)$ given by the disjoint union of all hypercubes. By Lemma 2.3 the intersection number of curves at corner points agrees with the intersection number of curves given by points in the interior of the hypercube. Hence, by taking the subset on which the intersection number ι evaluates to 0, we find that $\mathcal{M}_1^{\Gamma, \text{cube}}(x, z)$ can be compactified to a union of hypercubes with corner points given by

$$\mathcal{C}^0(x, z) = \{((u_1, v_1), \dots, (u_k, v_k)) \in \mathcal{C}(x, z) \mid \sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j) = 0\}.$$

A component of a corner point is given by a broken trajectory and can come in two types

$$\text{Type 1: } (u_i, v_i) \in \mathcal{M}_0^\Gamma(x_i, y_i) \times \widehat{\mathcal{M}}_1(y_i, z_i)$$

$$\text{Type 2: } (u_i, v_i) \in \widehat{\mathcal{M}}_1(x_i, y_i) \times \mathcal{M}_0^\Gamma(y_i, z_i).$$

A corner point $w \in \mathcal{C}^0(x, z)$ is called *pure* if all components of w are of the same type, that is w lies in

$$\left(\prod_{i=1}^k \bigcup_{y_i \in \mathcal{P}(G)} \mathcal{M}_0^\Gamma(x_i, y_i) \times \widehat{\mathcal{M}}_1(y_i, z_i) \right) \cup \left(\prod_{i=1}^k \bigcup_{y_i \in \mathcal{P}(H)} \widehat{\mathcal{M}}_1(x_i, y_i) \times \mathcal{M}_0^\Gamma(y_i, z_i) \right) \cap \mathcal{C}^0(x, z)$$

Note that this set is in bijection to the set

$$\mathcal{C}_{\text{pure}}^0(x, z) = \bigcup_{y \in \mathcal{P}(G)^k} \mathcal{M}_0^\Gamma(x, y) \times \widehat{\mathcal{M}}_1(y, z) \cup \bigcup_{y \in \mathcal{P}(H)^k} \widehat{\mathcal{M}}_1(x, y) \times \mathcal{M}_0^\Gamma(y, z)$$

Now we can apply the combinatorial Lemma 3.4 to deduce that

$$(9) \quad |\mathcal{C}_{\text{pure}}^0(x, z)| = 0 \pmod{2}.$$

Indeed, any $\mathcal{M}_1^{\Gamma, \text{cube}}(x, z)$ is given by a disjoint union of hypercubes, which are products of intervals. The boundary components of these intervals can be colored with respect to their type of broken trajectory. The lemma asserts that the number of corner points with components of the same type is even. However, these are exactly the pure corner points.

Now, for any $w \in \mathcal{C}_{\text{pure}}^0(x, z)$, we denote its breaking orbit by $y_w \in \mathcal{P}(H)^k$ or $y_w \in \mathcal{P}(G)^k$, depending on its type of breaking. A corner point w is called *degenerate* if $y_w \in \Delta_{\mathcal{P}(H)} \cup \Delta_{\mathcal{P}(G)}$ and is called *regular* otherwise. Let

$$\mathcal{D}(x, z) = \{w \in \mathcal{C}_{\text{pure}}^0(x, z) \mid y_w \in \Delta_{\mathcal{P}(H)} \cup \Delta_{\mathcal{P}(G)}\}$$

be the set of degenerate pure cornerpoints.

Observation 1: Let $w \in \mathcal{C}_{\text{pure}}^0(x, z)$ such that y_w is regular. Then $y_w \in \text{OB}\mathcal{P}_\beta(H)$ or $y_w \in \text{OB}\mathcal{P}_\beta(G)$ depending on the type of breaking.

Proof. Assume that $y_w \in \mathcal{P}(H)^k \setminus \Delta_{\mathcal{P}(H)}$, the other case is analogous. Suppose that $w = ((u_1, v_1), \dots, (u_k, v_k))$ so that

$$0 = \sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j) = \sum_{i \neq j} \iota(u_i, u_j) + \iota(v_i, v_j),$$

where we use that $y_w \notin \Delta_{\mathcal{P}(H)}$ for the last equality. By positivity of intersection we can conclude that $\iota(u_i, u_j) = \iota(v_i, v_j) = 0$ for all $i \neq j$. Hence for every $(s, t) \in \mathbb{R} \times S^1$ we have $u_i(s, t) \neq u_j(s, t)$ and the cylinders $u_1, \dots, u_k$ induces an isotopy of braids corresponding to x and y . Since the associated braid of x is of class β , so must the braid of y be as well. This concludes the proof of the observation. $\square$

We now have that

$$\begin{aligned} \mathcal{C}_{\text{pure}}^0(x, z) &= \bigcup_{y \in \mathcal{P}(H)^k \setminus \Delta_{\mathcal{P}(H)}} \mathcal{M}_0^\Gamma(x, y) \times \widehat{\mathcal{M}}_1(y, z) \cup \bigcup_{y \in \mathcal{P}(G)^k \setminus \Delta_{\mathcal{P}(G)}} \widehat{\mathcal{M}}_1(x, y) \times \mathcal{M}^{\Gamma_0}(y, z) \\ &\cup \bigcup_{y \in \Delta_{\mathcal{P}(H)}} \mathcal{M}_0^\Gamma(x, y) \times \widehat{\mathcal{M}}_1(y, z) \cup \bigcup_{y \in \Delta_{\mathcal{P}(G)}} \widehat{\mathcal{M}}_1(x, y) \times \mathcal{M}^{\Gamma_0}(y, z) \\ &= \bigcup_{y \in \text{OB}\mathcal{P}_\beta(H)} \mathcal{M}_0^\Gamma(x, y) \times \widehat{\mathcal{M}}_1(y, z) \cup \bigcup_{y \in \text{OB}\mathcal{P}_\beta(G)} \widehat{\mathcal{M}}_1(x, y) \times \mathcal{M}^{\Gamma_0}(y, z) \cup \mathcal{D}(x, z) \\ &= \mathcal{B}(x, z) \cup \mathcal{D}(x, z) \end{aligned}$$

To show that

$$\sum_{\sigma \in S_k} \eta(x, \sigma(z)) = \sum_{\sigma \in S_k} |\mathcal{B}(x, \sigma(z))| = 0 \pmod{2}$$

it suffices to show that

$$(10) \quad \sum_{\sigma \in S_k} |\mathcal{D}(x, \sigma(z))| = 0 \pmod{2}$$

since after (9) we have $|\mathcal{C}_{\text{pure}}^0(x, z)| = 0 \pmod{2}$. For every $y \in \Delta_{\mathcal{P}(H)} \cup \Delta_{\mathcal{P}(G)}$ let us define

$$\mathcal{D}(x, y, z) = \{w \in \mathcal{D}(x, z) \mid y_w = y\}$$

so that

$$\mathcal{D}(x, z) = \bigcup_{y \in \Delta_{\mathcal{P}(H)} \cup \Delta_{\mathcal{P}(G)}} \mathcal{D}(x, y, z).$$

For every $y \in \Delta_{\mathcal{P}(H)} \cup \Delta_{\mathcal{P}(G)}$ let us define

$$\text{Fix}(y) = \{\sigma \in S_k \mid \sigma(y) = y\}$$

as the subgroup of S_k that fixes y .

Observation 2: For every $\sigma \in \text{Fix}(y)$ we have

$$|\mathcal{D}(x, y, z)| = |\mathcal{D}(x, y, \sigma(z))|.$$

Proof. For every $\sigma \in \text{Fix}(y)$ let us define

$$T_\sigma^{x, y, z} : \mathcal{D}(x, y, z) \rightarrow \mathcal{D}(x, y, \sigma(z)), \quad ((u_1, v_1), \dots, (u_k, v_k)) \mapsto ((u_1, v_{\sigma(1)}), \dots, (u_k, v_{\sigma(k)})).$$

To see that this map is well defined, note that after the swapping Lemma 2.5 we have that

$$0 = \sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j) = \sum_{i \neq j} \iota(u_i \# v_{\sigma(i)}, u_j \# v_{\sigma(j)})$$

and hence $((u_1, v_{\sigma(1)}), \dots, (u_k, v_{\sigma(k)})) \in \mathbb{C}_{\text{pure}}^0(x, \sigma(z))$. Since the breaking orbit y was left unchanged, we have that $((u_1, v_{\sigma(1)}), \dots, (u_k, v_{\sigma(k)})) \in \mathcal{D}(x, y, \sigma(z))$. Now, $T_{\sigma}^{x, y, z}$ is a bijection since $T_{\sigma^{-1}}^{x, y, \sigma(z)}$ is its inverse. This concludes the proof of the observation. $\square$

Now, the subgroup $\text{Fix}(y)$ is isomorphic to a product $S_{n_1} \times \dots \times S_{n_m}$ where n_i are the multiplicities of breaking orbits in y . Since $y \in \Delta_{\mathcal{P}(H)} \cup \Delta_{\mathcal{P}(G)}$ there must be some i such that $n_i \geq 2$. It follows that $|\text{Fix}(y)| = 0 \pmod{2}$. We now have that

$$\begin{aligned} \sum_{\sigma \in S_k} |\mathcal{D}(x, \sigma(z))| &= \sum_{y \in \Delta_{\mathcal{P}(H)} \cup \Delta_{\mathcal{P}(G)}} \sum_{\sigma \in S_k} |\mathcal{D}(x, y, \sigma(z))| \\ &= \sum_{y \in \Delta_{\mathcal{P}(H)} \cup \Delta_{\mathcal{P}(G)}} \sum_{[\mu] \in S_k / \text{Fix}(y)} \sum_{\sigma \in \text{Fix}(y)} |\mathcal{D}(x, y, \sigma(\mu(z)))| \\ &= \sum_{y \in \Delta_{\mathcal{P}(H)} \cup \Delta_{\mathcal{P}(G)}} \sum_{[\mu] \in \text{Fix}(y) S_k} |\text{Fix}(y)| |\mathcal{D}(x, y, \sigma(\mu(z)))| \\ &= 0 \pmod{2} \end{aligned}$$

This concludes the proof of Proposition 3.3. $\square$

Next, let us discuss the composition of continuation maps. consider regular pairs (H^a, J^a) , (H^b, J^b) , and (H^c, J^c) and continuation data $\Gamma' = (F', J')$ from (H^a, J^a) to (H^b, J^b) and continuation data $\Gamma'' = (F'', J'')$ from (H^b, J^b) to (H^c, J^c) . For a real number $\rho > 0$ that is large enough we define continuation data $\Gamma'' \#_{\rho} \Gamma' = (F'' \#_{\rho} F', J'' \#_{\rho} J')$ from (H^a, J^a) to (H^c, J^c) in the following way.

$$F'' \#_{\rho} F'(s, t, x) = \begin{cases} F'(s + \rho, t, x), & s \leq 0 \\ F''(s - \rho, t, x), & s \geq 0 \end{cases},$$

$$(J'' \#_{\rho} J')_s = \begin{cases} J'_{s+\rho}, & s \leq 0 \\ J''_{s-\rho}, & s \geq 0 \end{cases}.$$

Note that since the continuation data is eventually constant in the s parameter, the composition $\Gamma'' \#_{\rho} \Gamma'$ is well-defined for ρ large enough. From standard Floer theory (see for example [AD14]) it follows that (after possibly a arbitrarily small perturbation of the continuation data) for every $x \in \mathcal{P}(H^a)$ any $z \in \mathcal{P}(H^c)$ there is a gluing map

$$\chi : \bigcup_{y \in \mathcal{P}(H^b)} \mathcal{M}_0^{\Gamma'}(x, y) \times \mathcal{M}_0^{\Gamma''}(y, z) \rightarrow \mathcal{M}_0^{\Gamma'' \#_{\rho} \Gamma'}(x, z)$$

which is a bijection. For every $(u, v) \in \mathcal{M}_0^{\Gamma'}(x, y) \times \mathcal{M}_0^{\Gamma''}(y, z)$, the curve $\chi(u, v)$ is obtained by the exponential flow along a pre-gluing map associated to u and v (see [AD14]) and hence falls into the realm we discussed in Section 2.2.

Proposition 3.5. *consider regular pairs (H^a, J^a) , (H^b, J^b) , and (H^c, J^c) and braid class β . For continuation data Γ' from (H^a, J^a) to (H^b, J^b) and continuation data Γ'' from (H^b, J^b) to (H^c, J^c) we have*

$$\Phi^{\Gamma''} \circ \Phi^{\Gamma'} = \Phi^{\Gamma'' \#_{\rho} \Gamma'} : \text{BCF}_{\beta}(H^a) \rightarrow \text{BCF}_{\beta}(H^c)$$

for $\rho > 0$ large enough.

Proof. For every $x \in \text{OB}\mathcal{P}_\beta(H)$ we have

$$\Phi^{\Gamma''} \circ \Phi^{\Gamma'}([x]) = \sum_{y \in \text{OB}\mathcal{P}_\beta(H^b)} \sum_{z \in \text{OB}\mathcal{P}_\beta(H^c)} \xi^{\Gamma'}(x, y) \xi^{\Gamma''}(y, z)[z]$$

and

$$\Phi^{\Gamma'' \#_\rho \Gamma'}([x]) = \sum_{z \in \text{OB}\mathcal{P}_\beta(H^c)} \xi^{\Gamma'' \#_\rho \Gamma'}(x, z)[z].$$

We would like to prove $\Phi^{\Gamma''} \circ \Phi^{\Gamma'} = \Phi^{\Gamma'' \#_\rho \Gamma'}$ which boils down to showing the following equality. For every $x \in \text{OB}\mathcal{P}_\beta(H^a)$ and $z \in \text{OB}\mathcal{P}_\beta(H^c)$ we have

$$(11) \quad \sum_{\sigma \in S_k} \sum_{y \in \text{OB}\mathcal{P}_\beta(G)} \xi^{\Gamma'}(x, y) \xi^{\Gamma''}(y, \sigma(z)) = \sum_{\sigma \in S_k} \xi^{\Gamma'' \#_\rho \Gamma'}(x, \sigma(z)) \pmod{2}.$$

Assuming this, let us finish the proof of Proposition 3.5. For every $x \in \text{OB}\mathcal{P}_\beta(H^a)$

$$\begin{aligned} \Phi^{\Gamma''} \circ \Phi^{\Gamma'}([x]) &= \Phi^{\Gamma''} \left(\sum_{y \in \text{OB}\mathcal{P}_\beta(H^b)} \xi^{\Gamma'}(x, y)[y] \right) \\ &= \sum_{z \in \text{OB}\mathcal{P}_\beta(H^c)} \sum_{y \in \text{OB}\mathcal{P}_\beta(H^b)} \xi^{\Gamma'}(x, y) \xi^{\Gamma''}(y, z)[z] \\ &= \sum_{[z] \in \text{B}\mathcal{P}_\beta(H^c)} \sum_{\sigma \in S_k} \sum_{y \in \text{OB}\mathcal{P}_\beta(H^b)} \xi^{\Gamma'}(x, y) \xi^{\Gamma''}(y, \sigma(z))[z] \\ &= \sum_{[z] \in \text{B}\mathcal{P}_\beta(H^c)} \sum_{\sigma \in S_k} \xi^{\Gamma'' \#_\rho \Gamma'}(x, \sigma(z))[z] \\ &= \sum_{z \in \text{OB}\mathcal{P}_\beta(H^c)} \xi^{\Gamma'' \#_\rho \Gamma'}(x, z)[z] \\ &= \Phi^{\Gamma'' \#_\rho \Gamma'}([x]) \end{aligned}$$

We now prove the equality (11). For every $x \in \mathcal{P}(H^a)$ and $z \in \mathcal{P}(H^c)$ there is a gluing map

$$\chi : \bigcup_{y \in \mathcal{P}(H^b)} \mathcal{M}_0^{\Gamma'}(x, y) \times \mathcal{M}_0^{\Gamma''}(y, z) \rightarrow \mathcal{M}_0^{\Gamma'' \#_\rho \Gamma'}(x, z)$$

that is constructed by gluing solutions to the Floer equation as in the setting of Section 2.2. For every $x \in \text{OB}\mathcal{P}_\beta(H^a)$ and $z \in \text{OB}\mathcal{P}_\beta(H^c)$ this induces a bijective map

$$\chi : \bigcup_{y \in \mathcal{P}(H^b)^k} \mathcal{M}_0^{\Gamma'}(x, y) \times \mathcal{M}_0^{\Gamma''}(y, z) \rightarrow \mathcal{M}_0^{\Gamma'' \#_\rho \Gamma'}(x, z),$$

$$((u_1, \dots, u_k), (v_1, \dots, v_k)) \mapsto (\chi(u_1, v_1), \dots, \chi(u_k, v_k)).$$

For every $y \in \mathcal{P}(H^b)^k$ and $(u, v) = ((u_1, \dots, u_k), (v_1, \dots, v_k)) \in \mathcal{M}^{\Gamma'}(x, y) \times \mathcal{M}^{\Gamma''}(y, z)$ we have after stability of intersections (Lemma 2.3) that

$$\iota(u \# v) = \iota(\chi(u, v))$$

since the map χ is given by pairwise gluing the cylinders of the tuples u and v . Moreover, if $y \in \mathcal{P}(H^b)^k - \Delta_{\mathcal{P}(H^b)}$, where $\Delta_{\mathcal{P}(H^b)}$ is the big diagonal of $\mathcal{P}(H^b)^k$, then we additionally have

$$\iota(u) + \iota(v) = \iota(u \# v) = \iota(\chi(u, v)) = 0.$$

By positivity of intersections we have that $\iota(u) = \iota(v) = 0$ and we must have that $y \in \text{OBP}_\beta(H^b)$ since the tuples of cylinders u give rise to an isotopy. Let now,

$$\mathcal{D}(x, z) = \bigcup_{y \in \Delta_{\mathcal{P}(H^b)}} \mathcal{M}_0^{\Gamma'}(x, y) \times \mathcal{M}_0^{\Gamma''}(y, z).$$

For every $w \in \mathcal{M}_0^{\Gamma' \# \Gamma''}(x, z)$ we find a unique $(u, v) \in \bigcup_{y \in \mathcal{P}(H^b)^k} \mathcal{M}^{\Gamma'}(x, y) \times \mathcal{M}^{\Gamma''}(y, z)$ such that $\chi(u, v) = w$ and with tuple of breaking orbits $y \in \mathcal{P}(H^b)^k$. Let y_w denote this tuple of breaking orbits. Now, y_w is called *degenerate* if $y_w \in \Delta_{\mathcal{P}(H^b)}$, otherwise it is called *regular*. Above we saw that when y_w is regular, we have $y_w \in \text{OBP}_\beta(H^b)$. With this we can see that

$$\sum_{y \in \text{OBP}_{\beta, f}(H^b)} \xi^{\Gamma'}(x, y) \xi^{\Gamma''}(y, z)$$

is exactly the cardinality of

$$\bigcup_{y \in \mathcal{P}(H^b)^k} \mathcal{M}_0^{\Gamma'}(x, y) \times \mathcal{M}_0^{\Gamma''}(y, z) \setminus \mathcal{D}(x, z) = \bigcup_{y \in \text{OBP}_\beta(H^b)} \mathcal{M}_0^{\Gamma'}(x, y) \times \mathcal{M}_0^{\Gamma''}(y, z).$$

It follows that we have

$$\sum_{y \in \text{OBP}_\beta(H^b)} \xi^{\Gamma'}(x, y) \xi^{\Gamma''}(y, z) + |\mathcal{D}(x, z)| = \xi^{\Gamma' \# \Gamma''}(x, z).$$

To conclude the proof we show that $\sum_{\sigma \in S_k} |\mathcal{D}(x, \sigma(z))|$ is divisible by 2. First, for every $y \in \Delta_{\mathcal{P}(H^b)}$ let us define

$$\mathcal{D}(x, y, z) = \{(u, v) \in \mathcal{M}_0^{\Gamma'}(x, y) \times \mathcal{M}_0^{\Gamma''}(y, z) \mid \iota(\chi(u, v)) = 0\}$$

and note that

$$\mathcal{D}(x, z) = \bigcup_{y \in \Delta_{\mathcal{P}(H^b)}} \mathcal{D}(x, y, z)$$

is given by a disjoint union. For every $y \in \Delta_{\mathcal{P}(H^b)}$ let

$$\text{Fix}(y) = \{\sigma \in S_k \mid \sigma(y) = y\}.$$

Now it follows from the swapping Lemma 2.5 that for every $(u, v) \in \mathcal{D}(x, y, z)$ and $\sigma \in S_k$ we have

$$\iota(u \# v) = \iota(u \# \sigma(v)).$$

In particular, if $(u, v) \in \mathcal{D}(x, y, z)$ then for every $\sigma \in \text{Fix}(y)$ we have $(u, \sigma(v)) \in \mathcal{D}(x, y, z)$. Hence, for every $y \in \Delta_{\mathcal{P}(H^b)}$ and $\sigma \in \text{Fix}(y)$ the map

$$T_\sigma^{x, y, z} : \mathcal{D}(x, y, z) \rightarrow \mathcal{D}(x, y, \sigma(z)), \quad (u, v) \mapsto (u, \sigma(v))$$

is well-defined. Moreover, it is a bijection with inverse $T_{\sigma^{-1}}^{x, y, \sigma(z)}$. It follows that

$$|\mathcal{D}(x, y, z)| = |\mathcal{D}(x, y, \sigma(z))|$$

for every $\sigma \in \text{Fix}(y)$. We calculate

$$\begin{aligned} \sum_{\sigma \in S_k} |\mathcal{D}(x, \sigma(z))| &= \sum_{\sigma \in S_k} \sum_{y \in \Delta_{\mathcal{P}(H^b)}} |\mathcal{D}(x, y, \sigma(z))| \\ &= \sum_{y \in \Delta_{\mathcal{P}(H^b)}} \sum_{[\mu] \in S_k / \text{Fix}(y)} \sum_{\sigma \in \text{Fix}(y)} |\mathcal{D}(x, y, \sigma(\mu(z)))| \\ &= \sum_{y \in \Delta_{\mathcal{P}(H^b)}} \sum_{[\mu] \in S_k / \text{Fix}(y)} |\text{Fix}(y)| |\mathcal{D}(x, y, \mu(z))|, \end{aligned}$$

where $S_k/\text{Fix}(y) = \{\text{Fix}(y)\mu \mid \mu \in S_k\}$. Now, $\text{Fix}(y) \cong S_{n_1} \times \cdots \times S_{n_m}$ where n_i are the multiplicities of the loops appearing in y . Since $y \in \Delta_{\mathcal{P}(H^b)}$ has at least one loop appearing at least twice, we have $n_i \geq 2$ for some i , so $|\text{Fix}(y)|$ is divisible by 2. Hence, $\sum_{\sigma \in S_k} |\mathcal{D}(x, \sigma(z))|$ is divisible by 2. This concludes the proof of Proposition 3.5 $\square$

3.3. Braided Floer Homology of the Trivial Braid. On a fixed symplectic surface (S, ω) of genus g , let β_0 denote the trivial braid with $2g+2$ strands. We now prove Theorem C by using continuation maps. We show this in the following way. For a C^2 -small Hamiltonian h , it follows from standard arguments that for a suitable almost complex structure J_h we have

$$\dim \text{BHF}_{\beta_0}(h, J_h) = 1.$$

For such a Hamiltonian h we then show that for a regular pair (H, J_H) there is an injective continuation map

$$\Phi : \text{BHF}_{\beta_0}(h, J_h) \hookrightarrow \text{BHF}_{\beta_0}(H, J_H).$$

We show the injectivity of this map by showing that there is a left inverse. Namely, we first prove in Proposition 3.6 that for certain regular pairs (h, J_h) and continuation data Γ from (h, J_h) to itself, the continuation data is the identity on a chain level

$$\Phi^\Gamma = \text{id}_{\text{BCF}_{\beta_0}(h)}.$$

We then choose Γ' from (h, J_h) to (H, J_H) and Γ'' from (H, J_H) to (h, J_h) and have that

$$\text{id}_{\text{BHF}_{\beta_0}(h, J_h)} = \Phi^{\Gamma''} \#_{\rho} \Gamma' = \Phi^{\Gamma''} \circ \Phi^{\Gamma'}.$$

Proposition 3.6. *Suppose H is a Hamiltonian on a closed surface of genus g such that the following holds.*

- i *There are exactly $2g+2$ 1-periodic orbits, and they are all constant.*
- ii *For all periodic orbits $x, y \in \mathcal{P}(H)$ the set $\hat{\mathcal{M}}_1(x, y)$ has even cardinality.*
- iii *For all $x, y \in \mathcal{P}(H)$, all $u \in \mathcal{M}_1(x, y)$ are time-independent, that is, $u(s, t) = u(s, 0)$ for all $t \in S^1$.*

Suppose that (H, J) is a regular pair and Γ is regular continuation data starting and ending at (H, J) . Then

$$\Phi^\Gamma = \text{id}_{\text{BCF}_{\beta_0}(H, J)}.$$

In the proof of Proposition 3.6 the following topological lemma will be essential.

Lemma 3.7. *Let $u, w, v_1, v_2 : [-1, 1] \times S^1 \rightarrow S$ such that the following holds (see Figure 10).*

- i *v_1 and v_2 are independent of $t \in S^1$*

$$v_1(s, t) = v_1(s, 0) \text{ and } v_2(s, t) = v_2(s, 0).$$
- ii *The ends of v_1 and v_2 agree*

$$v_1(-1, 0) = v_2(-1, 0) \text{ and } v_1(1, 0) = v_2(1, 0).$$
- iii *The ends of u and w are independent of $t \in S^1$*

$$u(\pm 1, t) = u(\pm 1, 0) \text{ and } w(\pm 1, t) = w(\pm 1, 0).$$
- iv *$u(1, 0) = v_1(-1, 0) = v_2(-1, 0)$ so that $u \# v_1$ and $u \# v_2$ are well-defined.*
- v *$u(-1, 0) \neq w(-1, 0), v_1(1, 0) \neq w(1, 0)$ (and $v_2(1, 0) \neq w(1, 0)$) so that the intersection number $\iota(u \# v_1, w)$ and $\iota(u \# v_2, w)$ are well-defined.*

Then

$$\iota(u \# v_1, w) = \iota(u \# v_2, w).$$

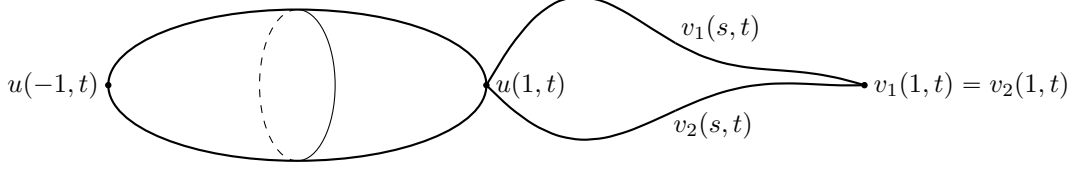


FIGURE 10. The concatenation $u \# v_i$ from Lemma 3.7.

Proof. For $z \in S$ let

$$\text{const}_z : [-1, 1] \times S^1 \rightarrow S, \quad (s, t) \mapsto z$$

be the constant cylinder. Note that w is homotopic to $w \# \text{const}_{w(1,0)}$ relative endings. Hence we have

$$\iota(u \# v_i, w) = \iota(u \# v_i, w \# \text{const}_{w(1,0)}).$$

For dimension reasons, after an arbitrarily small perturbation of u, v_1, v_2 , we may assume the lemma's assumptions hold, and we also have $v_i(s, t) \neq w(1, 0)$ for every s, t (and, in particular, $u(1, 0) \neq w(1, 0)$). After another perturbation of u relative to its endpoints, we may assume that the graphs of $u \# v_i$ and $w \# \text{const}_{w(1,0)}$ are transverse. The intersection number $\iota(u \# v_i, w \# \text{const}_{w(1,0)})$ is now simply a count of the intersections of the graphs of the curves. Since $v_i(s, t) \neq w(1, t)$ for every s, t , the part associated to v_i does not contribute to the count, and we have

$$\iota(u \# v_1, w \# \text{const}_{w(1,0)}) = \iota(u \# v_2, w \# \text{const}_{w(1,0)}).$$

□

To prove Proposition 3.6, let us first recall some facts from standard Hamiltonian Floer theory. Let $\Gamma_0 = (F_0, J_{F_0})$ and $\Gamma_1 = (F_1, J_{F_1})$ be regular continuation data connecting $H_{-\infty}$ to H_{∞} and a generic interpolation $(\Gamma_s)_{s \in [0,1]} = (F_s, J_{F_s})_{s \in [0,1]}$ of continuation data such that Γ_s is constant near the boundary. For every $x \in \mathcal{P}(H^{-\infty})$ and $z \in \mathcal{P}(H^{\infty})$ the moduli space

$$\mathcal{M}^{\{\Gamma_s\}}(x, z) = \{(u, \lambda) \mid u \in \mathcal{M}_0^{\Gamma_\lambda}(x, z)\}$$

is a smooth 1-dimensional manifold with boundary. The boundary is given by

$$\mathcal{M}_0^{\Gamma_0}(x, z) \cup \mathcal{M}_0^{\Gamma_1}(x, z).$$

It can be compactified by gluing the boundary of the form of broken trajectories

$$\Pi^{\{\Gamma_\lambda\}}(x, z) = \bigcup_{y \in \mathcal{P}(H^{\infty})} \mathcal{M}_{-1}^{\{\Gamma_\lambda\}}(x, y) \times \hat{\mathcal{M}}_1(y, z) \cup \bigcup_{y \in \mathcal{P}(H^{-\infty})} \hat{\mathcal{M}}_1(x, y) \times \mathcal{M}_{-1}^{\{\Gamma_\lambda\}}(y, z).$$

The identification of this space as the boundary of the moduli space is done via a gluing map of the form as we discussed in Section 2.2 (see [AD14]).

Proof of Proposition 3.6. Let $\Gamma_0 = (H, J_H)$ be the constant continuation data and $\Gamma_1 = \Gamma$, and let $(\Gamma_s)_{s \in [0,1]}$ be a generic choice of interpolation of continuation data between Γ_0 and Γ_1 such that for any $x, z \in \mathcal{P}(H)$, $\mathcal{M}^{\{\Gamma_s\}}(x, z)$ is a smooth manifold with boundary that consists of a disjoint union of intervals and copies of S^1 . Let $\mathcal{M}^{\{\Gamma_\lambda\}, \text{int}}(x, z)$ denote the subset of $\mathcal{M}^{\{\Gamma_\lambda\}}(x, z)$ given by the union of all open intervals. We can color the boundary of

the compactification of $\mathcal{M}^{\{\Gamma_s\}, \text{int}}(x, z)$ with three colors by the property that the boundary component either lies in $\mathcal{M}^{\Gamma_0}(x, z)$, in $\mathcal{M}^{\Gamma_1}(x, z)$ or $\Pi^{\{\Gamma_s\}}(x, z)$. Now for $x, z \in \text{OBP}_{\beta_0}(H)$ let

$$\mathcal{M}^{\{\Gamma_s\}, \text{cube}}(x, z) = \{((u_1, \lambda_1), \dots, (u_{2g+2}, \lambda_{2g+2})) \in \prod_{i=1}^{2g+2} \mathcal{M}^{\{\Gamma_s\}, \text{int}}(x_i, z_i) \mid \sum_{i \neq j} \iota(u_i, u_j) = 0\}.$$

This space is a manifold given by the union of hypercubes that are products of intervals. Let $Q_1, \dots, Q_\ell$ denote the resulting hypercubes. For every hypercube $Q_j = I_{1,j} \times \dots \times I_{k,j}$ we color the boundary components of the intervals $I_{i,j}$ as follows

$$(12) \quad \rho_{i,j} : \partial I_{i,j} \rightarrow \{a, b, c\}, \quad p \mapsto \begin{cases} a, & p \in \mathcal{M}^{\Gamma_0}(x_i, z_i) \\ b, & p \in \mathcal{M}^{\Gamma_1}(x_i, z_i) \\ c, & p \in \Pi^{\{\Gamma_s\}}(x_i, z_i) \end{cases}.$$

Note that by assumption there are $2g+2$ periodic orbits and if $x \in \text{OBP}_{\beta_0}$ then every orbit appears in the tuple x . Moreover, for any $z \in \text{OBP}_{\beta_0}(H)$ there exists $\sigma \in S_{2g+2}$ such that $z = \sigma(x)$.

Observation: If $z = x$ there is exactly one a -pure corner point and there is none if $z \neq x$.

Proof. Since Γ_0 is the constant continuation data, we have that $\mathcal{M}^{\Gamma_0}(x_i, z_i)$ is empty if $x_i \neq z_i$ and consists of the constant cylinder at x_i if $x_i = z_i$. If there is an a -pure corner point, it must consist of the $2g+2$ constant cylinders at the $2g+2$ constant periodic orbits, which forces $x_i = z_i$. On the other hand, if $x = z$, then the $2g+2$ constant cylinders must appear as a corner point since their pairwise intersection number is trivial. This proves the observation. $\square$

Note that

$$\Phi^{\Gamma_1}([x]) = \sum_{\sigma \in S_k} \xi^{\Gamma_1}(x, \sigma(x))[x],$$

where $\xi^{\Gamma_1}(x, z)$ is the number of b -pure corner points for fixed braid of orbits x and z . We say a boundary component is d -colored if it is either b - or c -colored. Applying Lemma 3.4, we can conclude that the number of a -pure corner points plus the number of d -pure corner points is even. We now study the cardinality of the latter corner points. Let us denote

$$\Pi^{\{\Gamma_s\}}(x, y, z) := \mathcal{M}_{-1}^{\{\Gamma_s\}}(x, y) \times \widehat{\mathcal{M}}_1(y, z) \cup \widehat{\mathcal{M}}_1(x, y) \times \mathcal{M}_{-1}^{\{\Gamma_s\}}(y, z),$$

so that

$$\Pi^{\{\Gamma_s\}}(x, z) = \bigcup_{y \in \mathcal{P}(H)} \Pi^{\{\Gamma_s\}}(x, y, z),$$

since in our specific case $H^\infty = H^{-\infty} = H$. For every $y \in (\mathcal{P}(H) \cup \{\emptyset\})^k$ we define

$$\mathcal{C}(y) = \left\{ (w_1, \dots, w_{2g+2}) \left| \begin{array}{ll} w_i \in \begin{cases} \mathcal{M}^{\Gamma_1}(x_i, z_i), & y_i = \emptyset \\ \mathcal{M}_{-1}^{\{\Gamma_s\}}(x_i, y_i) \times \widehat{\mathcal{M}}_1(y_i, z_i) \cup \widehat{\mathcal{M}}_1(x_i, y_i) \times \mathcal{M}_{-1}^{\{\Gamma_s\}}(y_i, z_i), & y_i \neq \emptyset \end{cases} & \\ \sum_{i \neq j} \iota(w_i, w_j) = 0 \end{array} \right. \right\},$$

which is the set of corner points where every component is b - or c -colored consisting of trajectories breaking at the tuple y (where a component is not broken if $y_i = \emptyset$). Note that

$$|\mathcal{C}(\emptyset, \dots, \emptyset)| = \text{number of } b\text{-pure corner points.}$$

Claim: For every $y \in (\mathcal{P}(H) \cup \{\emptyset\})^k$ such that $y \neq (\emptyset, \dots, \emptyset)$ we have

$$|\mathcal{C}(y)| \equiv 0 \pmod{2}.$$

Assuming this claim, let us conclude the result. We first conclude that the parity of the number of b -pure corner points equals that of the number of d -pure corner points. Indeed, we have

$$\begin{aligned} \text{number of } d\text{-pure corner points} &= \sum_{y \in (\mathcal{P}(H) \cup \{\emptyset\})^k} |\mathcal{C}(y)| \\ &= |\mathcal{C}(\emptyset, \dots, \emptyset)| + \sum_{\substack{y \in (\mathcal{P}(H) \cup \{\emptyset\})^k \\ y \neq (\emptyset, \dots, \emptyset)}} |\mathcal{C}(y)| \\ &= \text{number of } b\text{-pure corner points} \pmod{2}, \end{aligned}$$

where we use the claim in the last line. Now, if $z = x$, we know from the observation that the number of a -pure corner points is odd; hence the number of d -pure corner points, and therefore also the number of b -pure corner points, is odd. It follows that $\xi^{\Gamma_1}(x, x)$ is odd. If $z \neq x$, we similarly conclude that the number of b -pure corner points is even, and hence $\xi^{\Gamma_1}(x, z)$ is even. We can conclude that $\Phi^{\Gamma_1}([x]) = [x]$.

Let us now prove the claim. After relabeling, we may and do assume that $y_1 \neq \emptyset$. If $\mathcal{C}(y)$ is empty there is nothing to show. Otherwise, there exists $w_1 = (u_1, v_1)$ where $u_1 \in \mathcal{M}_{-1}^{\{\Gamma_\lambda\}}(x_1, y_1) \times \hat{\mathcal{M}}_1(y_1, z_1)$ or $w_1 = (v_1, u_1) \in \hat{\mathcal{M}}_1(x_1, y_1) \times \mathcal{M}_{-1}^{\{\Gamma_\lambda\}}(y_1, z_1)$. We will assume that the first holds; the latter case is analogous. By assumption the cardinality of $\hat{\mathcal{M}}_1(y_1, z_1)$ is even. Hence, the set admits a free $\mathbb{Z}_2$ -action

$$s : \hat{\mathcal{M}}_1(y_1, z_1) \times \mathbb{Z}_2 \rightarrow \hat{\mathcal{M}}_1(y_1, z_1).$$

This induces the following $\mathbb{Z}_2$ -action on $\mathcal{C}(y)$

$$\bar{s} : \mathcal{C}(y) \times \mathbb{Z}_2 \rightarrow \mathcal{C}(y), \quad ((u_1, v_1), w_2, \dots, w_{2g+2}) \mapsto ((u_1, s(v_1)), w_2, \dots, w_{2g+2}).$$

By Lemma 3.7 we have that

$$0 = \iota(u_1 \# v_1, w_i) = \iota(u_1 \# s(v_1), w_i)$$

and hence $((u_1, s(v_1)), w_2, \dots, w_{2g+2})$ is an element of $\mathcal{C}(y)$. Since s is free, the action $\bar{s}$ is free, and hence every orbit of $\bar{s}$ consists of two elements. Thus $|\mathcal{C}(y)|$ is even, which concludes the proof of the claim and the proposition. $\square$

Lemma 3.8. *There is a non-degenerate Hamiltonian H satisfying the conditions of Proposition 3.6*

Proof. Choose a Morse function H on S such that there are exactly $2g + 2$ critical points. By considering H/n for n large enough, we may assume that the periodic orbits $\mathcal{P}(H)$ are non-degenerate, constant, and correspond to the critical points of H . For a generic almost complex structure, we may further assume that any $u \in \mathcal{M}_1(x, y)$ satisfies $u(s, t) = u(s, 0)$ for all $(s, t) \in \mathbb{R} \times S^1$ (see [AD14]). Now, since the standard Hamiltonian Floer complex and homology with coefficients in $\mathbb{Z}_2$ satisfy

$$\dim CF(H, J) = \dim HF(H, J)$$

we can conclude that the differential must be trivial and the cardinality of moduli spaces $\hat{\mathcal{M}}_1(x, y)$ must be divisible by 2. $\square$

We can now prove Theorem C.

Proof of Theorem C. By Lemma 3.8, there is a non-degenerate Hamiltonian H and almost complex structure J_H that satisfy the assumptions of Proposition 3.6. For every non-degenerate Hamiltonian diffeomorphism ϕ , let G be a generating Hamiltonian and J_G a generic choice of almost complex structure. Choose continuation data Γ' from (H, J_H) to (G, J_G) and Γ'' from (G, J_G) to (H, J_H) . Let β_0 be the trivial braid with $2g + 2$ strands. Then by Proposition 3.5 we have

$$\Phi^{\Gamma''} \circ \Phi^{\Gamma'} = \Phi^{\Gamma'' \# \Gamma'}$$

since any periodic orbit of G is non-degenerate. Here, the continuation maps relate the braided Floer complexes in the braid class β_0 . Since $\Gamma'' \# \Gamma'$ is self continuation data at (H, J_H) by Proposition 3.6 we have that

$$\Phi^{\Gamma'' \# \Gamma'} = id_{\text{BCF}_{\beta_0}(H, J)}.$$

By Proposition 3.3, we have that the continuation maps are chain maps and hence $\Phi^{\Gamma'' \# \Gamma'}$ induces the identity in homology. In particular, we have that

$$\Phi^{\Gamma'} : \text{BHF}_{\beta_0}(H, J_H) \hookrightarrow \text{BHF}_{\beta_0}(G, J_G)$$

is injective and hence, $\dim \text{BHF}_{\beta_0}(G, J_G) \geq 1$. $\square$

4. BRAID STABILITY AND BOUNDED CONTINUATION MAPS

A *framed braid* is a braid $b = \{s_1, \dots, s_k\}$ together with a framing

$$f : \{s_1, \dots, s_k\} \rightarrow \mathbb{Z} \quad \text{or} \quad f : \{s_1, \dots, s_k\} \rightarrow \mathbb{Z}_4.$$

We say two framed braids are isotopic if they are isotopic as braids via an isotopy that respects the framing. For a framed closed surface braid class (β, f) we define the *framed braided Floer space of class (β, f)* of a non-degenerate Hamiltonian H as

$$\text{BCF}_{\beta, f}(H) = \langle x = \{x_1, \dots, x_k\} \subset \mathcal{P}(H) \mid \mathcal{B}(H, x) = \beta, f_x = f \rangle_{\mathbb{Z}_2},$$

where $f_x : \{x_1, \dots, x_k\} \rightarrow \mathbb{Z}$ when the genus $g \geq 1$ and $f_x : \{x_1, \dots, x_k\} \rightarrow \mathbb{Z}_4$ when the genus $g = 0$ is given by $f_x(x_i) = \mu(x_i)$. In this section, we prove the braid stability statement from Theorem B: For every non-degenerate Hamiltonian H and $x \subset \mathcal{P}(H)$ we define a quantity $\mathcal{S}(H, x) \in \mathbb{R} \cup \{\infty\}$ such that for every non-degenerate Hamiltonian G satisfying $kE(H - G) < \mathcal{S}(H, x)$ we have

$$\dim \text{BCF}_{\beta, f}(G) \geq 1,$$

where $\beta = \mathcal{B}(H, x)$, $f = f_x$, and k is the number of strands. We prove this by using ε -bounded continuation map

$$\Phi_\varepsilon : \text{BCF}_{\beta, f}(H) \rightarrow \text{BCF}_{\beta, f}(G)$$

for $\varepsilon \geq 0$ which counts Floer-theoretic braid isotopies consisting of cylinders $u_1, \dots, u_k$ each connecting orbits in a braid of class (β, f) of H and G whose action difference is absolutely bounded by ε .

4.1. Bounded Continuation Maps. For two Hamiltonians $H, G : S^1 \times S \rightarrow \mathbb{R}$ and a cylinder $u : \mathbb{R} \times S^1 \rightarrow S$ such that $\lim_{s \rightarrow -\infty} u(s, t) = x(t)$ and $\lim_{s \rightarrow \infty} u(s, t) = y(t)$ we will consider the action difference of two capped periodic orbits

$$\mathcal{A}_H([x, C]) - \mathcal{A}_G([y, C \# u]),$$

where C is a capping of x , that is, a cylinder from x to a fixed reference loop in its homotopy class. Note that this expression is independent of the choice of capping. Moreover, for a smooth family of ω -compatible almost complex structures $(J_{s,t})_{(s,t) \in \mathbb{R} \times S^1}$ we define the *energy of the cylinder u* by

$$E(u) = \int_{\mathbb{R}} \int_0^1 \omega(\partial_s u, J_{s,t} \partial_t u) dt ds.$$

In the case when $(J_{s,t}) = (J_t)$ is independent of s and u is a Floer-cylinder solving

$$\partial_s u + J_t(\partial_t u - X_{H_t} \circ u) = 0$$

a standard calculation shows that $E(u)$ equals the action difference of the limiting orbits

$$E(u) = \mathcal{A}_H([x, C_{x_i}]) - \mathcal{A}_G([y, C_{x_i} \# u])$$

which, in particular, is non-negative. Let us introduce a measure of the energy of the continuation data $\Gamma = (H_s, J_s)$. That is, let

$$E^+(\Gamma) = \int_{-\infty}^{\infty} \int_0^1 \max_{x \in S} \partial_s H(s, t, x) dt ds$$

and note that we have

$$E^+(\Gamma'' \#_{\rho} \Gamma') = E^+(\Gamma'') + E^+(\Gamma')$$

for $\rho > 0$ large enough. For every $\varepsilon \geq 0$ we refine the moduli space $\mathcal{M}_{\ell}^{\Gamma}(x, y)$ and define

$$\mathcal{M}_{\ell, \leq \varepsilon}^{\Gamma}(x, y) = \left\{ u \in \mathcal{M}_{\ell}^{\Gamma}(x, y) \left| \left| \sum_{i=1}^k \mathcal{A}_H([x_i, C]) - \mathcal{A}_G([y_i, C \# u_i]) \right| \leq \varepsilon \right. \right\},$$

where C_{x_i} is an arbitrary choice of capping of x_i . Since the expression $\mathcal{A}_H([x_i, C_{x_i}]) - \mathcal{A}_G([y_i, C_{x_i} \# u_i])$ is invariant under homotopy of u_i relative limiting orbits, it is locally constant on $\mathcal{M}_{\ell}^{\Gamma}(x, y)$. Hence, $\mathcal{M}_{\ell, \leq \varepsilon}^{\Gamma}(x, y)$ is the union of connected components for which any element u and any component of it satisfies $\left| \sum_{i=1}^k \mathcal{A}_H([x_i, C]) - \mathcal{A}_G([y_i, C \# u_i]) \right| \leq \varepsilon$. A standard computation (which is also done in the proof of Proposition 4.2 in more generality) shows that for all $u_i \in \mathcal{M}_{\ell}^{\Gamma}(x_i, z_i)$

$$-E^+(\Gamma) \leq \mathcal{A}_H([x, C_{x_i}]) - \mathcal{A}_G([z_i, C_{x_i} \# u_i])$$

and

$$E(u_i) \leq \mathcal{A}_H([x, C_{x_i}]) - \mathcal{A}_G([z_i, C_{x_i} \# u_i]) + E^+(\Gamma).$$

It follows that there is the following energy bound

$$E(u_i) \leq kE^+(\Gamma) + \varepsilon.$$

With this energy bound, as in standard Hamiltonian Floer theory, the space $\mathcal{M}_{0, \leq \varepsilon}^{\Gamma}(x, y)$ is compact and therefore consists of a finite number of points. Importantly, in contrast to Section 3, we here also consider the case of non-contractible orbits on T^2 . Compactness of the moduli spaces in these cases is guaranteed by the prescribed action bound. This leads us to the definition of *ε -bounded continuation maps of braided Floer spaces in the*

class (β, f) . Namely, regular pairs (H, J_H) and (G, J_G) and regular continuation data Γ connecting them we define

$$\Phi_\varepsilon^\Gamma : \text{BCF}_{\beta, f}(H) \rightarrow \text{BCF}_{\beta, f}(G), \quad [x] \rightarrow \sum_{y \in \text{OB}\mathcal{P}_{\beta, f}(G)} |\mathcal{M}_{0, \leq \varepsilon}^\Gamma(x, y)|[y],$$

where $[\cdot]$ stands for the equivalence class of unordered braids. For better readability, let us from now on write

$$\xi_\varepsilon^\Gamma(x, y) = |\mathcal{M}_{0, \leq \varepsilon}^\Gamma(x, y)|.$$

Lemma 4.1. *The map Φ_ε^Γ is well-defined.*

Proof. We need to check that the map does not depend on the representative x of the class $[x]$. Indeed, if $\sigma \in S_k$ is a permutation of k objects, we have

$$\sum_{y \in \text{OB}\mathcal{P}_{\beta, f}(G)} \xi_\varepsilon^\Gamma(x, y)[y] = \sum_{\sigma(y) \in \text{OB}\mathcal{P}_{\beta, f}(G)} \xi_\varepsilon^\Gamma(\sigma(x), \sigma(y))[\sigma(y)] = \sum_{y \in \text{OB}\mathcal{P}_{\beta, f}(G)} \xi_\varepsilon^\Gamma(\sigma(x), y)[y],$$

where $\sigma(z) = (z_{\sigma(1)}, \dots, z_{\sigma(k)})$. $\square$

4.2. The Quantity $\mathcal{S}(H, x)$. We now define the quantity $\mathcal{S}(H, x)$ for a non-degenerate H and $x \subset \mathcal{P}(H)$. Let β be a fixed closed surface braid class. For a smooth family of ω -compatible almost complex structures $(J_t)_{t \in S^1}$ and $x, y \in \text{OB}\mathcal{P}_\beta(H)$ we define

$$\mathcal{M}(x, y) = \{u \in \mathcal{M}(x_1, y_1) \times \dots \times \mathcal{M}(x_k, y_k) \mid \iota(u) = 0\}$$

the space of *Floer isotopies*. The energy of a Floer isotopy is defined as the sum of the energy of the Floer cylinders

$$E(u) = \sum_{i=1}^k E(u_i),$$

and we write

$$\mu_{\max}(u) = \max_i \mu(u_i).$$

The stability radius is defined as

$$\mathcal{S}(H, x) = \frac{1}{k} \sup_J \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\},$$

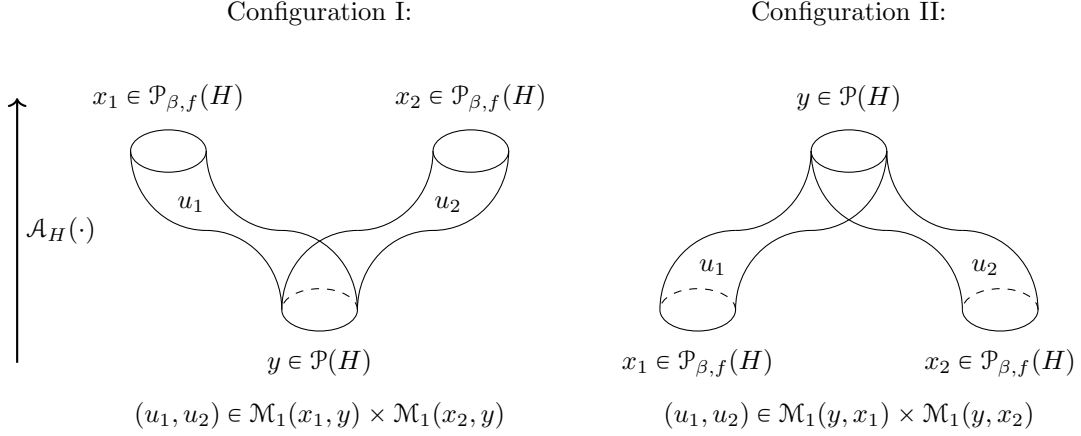
where the supremum runs over all J such that (H, J) is regular and k is the number of strands $k = |x|$. The quantities $\mathcal{E}(H, J, x)$, $\mathcal{E}^{\text{int}}(H, J, x)$, and $\mathcal{E}^2(H, J, x)$ are the minimal energy $E(u)$ of Floer cylinders in certain constellations of Floer cylinders. First, the quantity $\mathcal{E}(H, J, x)$ is the minimal energy of Floer isotopies $u \in \mathcal{M}(x, y)$ or $u \in \mathcal{M}(y, x)$ for $y \in \text{OB}\mathcal{P}_\beta(H)$ with $\mu_{\max}(u) = 1$

$$\mathcal{E}(H, J, x) = \min\{E(u) \mid y \in \text{OB}\mathcal{P}_\beta(H), u \in \mathcal{M}(x, y) \cup \mathcal{M}(y, x), \mu_{\max}(u) = 1\}.$$

This is the most relevant quantity of the three appearing in the definition of $\mathcal{S}(H, x)$, as the other two can be neglected when choosing the right framing. Namely, the quantity $\mathcal{E}^{\text{int}}(H, J, x)$ is the minimal energy of an index 1 cylinder connecting orbits x_i to x_j

$$\mathcal{E}^{\text{int}}(H, J, x) = \min\{E(u) \mid u \in \mathcal{M}(x_i, x_j), \mu(u) = 1\}$$

which is by convention equal to ∞ when $\mu(x_i) \neq \mu(x_j) + 1$ for all $i \neq j$. The quantity $\mathcal{E}^2(H, J, x)$ is the most nuanced of the three and accounts for the configuration (see Fig. 11) of pairs of Floer trajectories that start (or end) in distinct orbits in a single braid of class β and end (or start) in a single orbit $y \in \mathcal{P}(H)$ (warning: y need not be a strand of a braid of periodic orbits of type (β, f)). We define

FIGURE 11. Configurations of Floer cylinders relevant for $\mathcal{E}^2(H, J, x_1, x_2)$.

$$\mathcal{E}^2(H, J, x) = \inf \left\{ \max\{E(u_1), E(u_2)\} \mid \begin{array}{l} y \in \mathcal{P}(H), 1 \leq i < j \leq k, \\ (u_1, u_2) \in \mathcal{M}_1(x_i, y) \times \mathcal{M}_1(x_j, y) \cup \mathcal{M}_1(y, x_i) \times \mathcal{M}_1(y, x_j) \end{array} \right\}$$

which considers constellations as in Figure 11. Note that when $\mu(x_i) \neq \mu(x_j)$ for $i \neq j$, then the infimum runs over the empty set and $\mathcal{E}^2(H, J, x) = \infty$ by convention. Hence in the case where f_x is injective:

$$\mathcal{S}(H, x) = \frac{1}{k} \sup_J \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x)\}$$

as promised in the Introduction 1. For every $x \in \text{OB}\mathcal{P}_\beta(H)$ with framing $f_x = f$ let

$$\pi_{[x]} : \text{BCF}_{\beta,f}(H) \rightarrow \langle [x] \rangle_{\mathbb{Z}_2}$$

be the projection to the factor $[x]$. We prove the following result, which is essential for braid stability.

Proposition 4.2. *Let (H, J) be a regular pair and Γ continuation data connecting (H, J) to itself. Let $\varepsilon \geq 0$ and suppose there exists $x \in \text{OB}\mathcal{P}_\beta(H)$ with $f = f_x$ and such that $0 \leq E^+(\Gamma)$ and*

$$kE^+(\Gamma) + \varepsilon < \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\},$$

where k is the number of strands. Then we have

$$\pi_{[x]} \circ \Phi_\varepsilon^\Gamma([x]) = [x].$$

The proof shows the following. When $kE^+(\Gamma) + \varepsilon < \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\}$ holds for all $x \in \mathcal{P}(H)$ such that $\beta = \mathcal{B}(H, x)$ and $f_x = f$ then

$$\Phi_\varepsilon^\Gamma = \text{id}_{\text{BCF}_{\beta,f}(H)}.$$

Before proving this proposition, we first do some preparation. First, recall some standard facts from Hamiltonian Floer theory. Given regular pairs (H, J_H) and (G, J_G) and continuation data Γ_0 and Γ_1 connecting them, one can choose a generic smooth interpolation of

continuation data $(\Gamma_\lambda)_{\lambda \in [0,1]}$ where $\Gamma_\lambda = \Gamma_0$ for λ close to 0 and $\Gamma_\lambda = \Gamma_1$ for λ close to 1. For two $x \in \mathcal{P}(H)$ and $z \in \mathcal{P}(G)$ and $\ell \in \mathbb{Z}$ the space

$$\mathcal{M}_\ell^{\{\Gamma_\lambda\}}(x, z) = \{(\lambda, u) \mid \lambda \in [0, 1], u \in \mathcal{M}_\ell^{\Gamma_\lambda}(x, z)\}$$

is a smooth finite dimensional manifold with boundary of dimension $\ell + 1$. Importantly, if $\ell = -1$, the manifold is compact and consists of finitely many points. When $\ell = 0$, the possibly non-compact manifold is 1-dimensional with boundary given by

$$\mathcal{M}_0^{\Gamma_0}(x, z) \cup \mathcal{M}_0^{\Gamma_1}(x, z).$$

To compactify this space, one adds a boundary of the form of broken trajectories. Namely, consider

$$\Pi^{\{\Gamma_\lambda\}}(x, z) = \bigcup_{y \in \mathcal{P}(H)} \widehat{\mathcal{M}}_1(x, y) \times \mathcal{M}_{-1}^{\{\Gamma_\lambda\}}(y, z) \cup \bigcup_{y \in \mathcal{P}(H)} \mathcal{M}_{-1}^{\{\Gamma_\lambda\}}(x, y) \times \widehat{\mathcal{M}}_1(y, z).$$

Any broken trajectory $(u, v) \in \Pi^{\{\Gamma_\lambda\}}(x, z)$ is identified as the boundary of $\mathcal{M}^{\{\Gamma_\lambda\}}(x, z)$ by a gluing process that can be performed with a method as described in Section 2.2. For this brief recapitulation from standard Floer theory, see [AD14]. Now, let us return to braided orbits. For pairs (H, J_H) and (G, J_G) and $x \in \text{OBP}_{\beta, f}(H), y \in \text{OBP}_{\beta, f}(G)$ we define

$$\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z) = \left\{ ((u_1, \lambda_1), \dots, (u_k, \lambda_k)) \left| \begin{array}{l} (u_i, \lambda_i) \in \mathcal{M}_0^{\{\Gamma_\lambda\}}(x_i, z_i), \quad \sum_{i \neq j} \iota(u_i, u_j) = 0, \\ \left| \sum_{i=1}^k \mathcal{A}_H([x_i, C_{x_i}]) - \mathcal{A}_G([z_i, C_{x_i} \# u_i]) \right| \leq \varepsilon \end{array} \right. \right\}.$$

Any path in $\mathcal{M}_0^{\{\Gamma_\lambda\}}(x_1, y_1) \times \dots \times \mathcal{M}_0^{\{\Gamma_\lambda\}}(x_k, y_k)$ gives rise to a smooth homotopy of the cylinders $u_1, \dots, u_k$ relative limiting orbits. Since the intersection number is invariant under homotopy, the space $\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z)$ is given by the connected component of $\mathcal{M}_0^{\{\Gamma_\lambda\}}(x_1, y_1) \times \dots \times \mathcal{M}_0^{\{\Gamma_\lambda\}}(x_k, y_k)$ on which the intersection number evaluates to 0 and we have

$$\left| \sum_{i=1}^k \mathcal{A}_H([x_i, C_{x_i}]) - \mathcal{A}_G([z_i, C_{x_i} \# u_i]) \right| \leq \varepsilon.$$

These connected components are products of 1-dimensional manifolds.

To prove Proposition 4.2 we first choose a suitable homotopy of continuation data from $\Gamma_1 := \Gamma$ to the constant one Γ_0 . For every $z \in \text{OBP}_{\beta, f}(H)$ with $[z] = [x]$ (which means $z = \sigma(x)$) any $u \in \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x, z) \cup \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x, z)$ arises as the corner point of a hypercube which is a connected component of $\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z)$. We show that under the conditions in Proposition 4.2, the connected components with such corner points are compact and hence are hypercubes, that is, products of closed intervals in $\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x_i, z_i)$. corner points of these hypercubes are elements in

$$(\mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_1, z_1) \cup \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_1, z_1)) \times \dots \times (\mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_k, z_k) \cup \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_k, z_k)).$$

Such a corner point is called *pure* if it lies in

$$\mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_1, z_1) \times \dots \times \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_k, z_k) \cup \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_1, z_1) \times \dots \times \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_k, z_k)$$

and they are called *mixed* otherwise. The maps $\Phi_0^{\Gamma_0}$ and $\Phi_0^{\Gamma_1}$ count exactly the pure corner points hypercubes in $\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z)$. To show $\Phi_0^{\Gamma_0} + \Phi_0^{\Gamma_1} = 0$, we use the fact that the number

of pure corner points is even, as stated in Lemma 3.4. For a compact hypercube Q in $\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z)$, we have that

$$Q = I_1 \times \cdots \times I_k,$$

where $I_i \subset \mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x_i, z_i)$ is a closed interval. The boundary components of these intervals come in two colors

Type 0: $(u_i, 0)$ where $u_i \in \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_i, z_i)$,

Type 1: $(u_i, 1)$ where $u_i \in \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_i, z_i)$.

Applying Lemma 3.4 to the induced coloring, we find that the number of pure corner points is even. Let us now prove Proposition 4.2.

Proof of Proposition 4.2. Let $z \in \mathcal{OBP}_\beta(H)$ with $[z] = [x]$. Let us write $\Gamma_1 = \Gamma = (F_1, J_1)$ and let us define a homotopy to the constant continuation data $\Gamma_0 = (H, J) = (F_0, J_0)$ given by $\Gamma_\lambda = (F_\lambda, J_\lambda)$ for $\lambda \in [0, 1]$ where

$$F_\lambda(s, t, x) = H(t, x) + f(\lambda)(F_1(s, t, x) - H(t, x))$$

and $f : \mathbb{R} \rightarrow [0, 1]$ is a smooth increasing function such that $f(s) = 0$ for $s < \frac{1}{3}$ and $f(s) = 1$ for $s > \frac{2}{3}$. After a choice of almost complex structure and a C^∞ -small compactly supported perturbation, we can assume that Γ_1 and $\{\Gamma_\lambda\}_{[0,1]}$ are regular. However, for the following energy estimates, we can neglect this perturbation. A standard computation shows that for every $\lambda \in [0, 1]$ and solution $w \in \mathcal{M}_\ell^{\Gamma_\lambda}(x, z)$ we have that

$$\mathcal{A}_H([z, C\#w]) - \mathcal{A}_H([x, C]) \leq E^+(\Gamma).$$

Indeed we have

$$\begin{aligned} 0 &\leq \int_{-\infty}^{\infty} \int_0^1 \omega(\partial_s w, J_\lambda \partial_s w) dt ds \\ &= \int_{-\infty}^{\infty} \int_0^1 \omega(\partial_s w, \partial_t u - X_{F_\lambda} \circ w) dt ds \\ (13) \quad &= \int_{-\infty}^{\infty} \int_0^1 \omega(\partial_s w, \partial_t w) + \omega(X_{F_\lambda} \circ w, \partial_s w) dt ds \\ &= \int_{-\infty}^{\infty} \int_0^1 \omega(\partial_s w, \partial_t w) dt ds - \int_{-\infty}^{\infty} \int_0^1 dF_\lambda \partial_s w dt ds. \end{aligned}$$

and

$$\begin{aligned} \int_{-\infty}^{\infty} \int_0^1 dF_\lambda \partial_s w dt ds &= \int_{-\infty}^{\infty} \int_0^1 \partial_s (F_\lambda \circ w) dt ds - \int_{-\infty}^{\infty} \int_0^1 \partial_s F_\lambda \circ w dt ds \\ &= \int_0^1 H(t, z(t)) dt - \int_0^1 H(t, x(t)) dt - \int_{-\infty}^{\infty} \int_0^1 f(\lambda) \partial_s F_1(s, t, w(s, t)) dt ds \\ &\geq \int_0^1 H(t, z(t)) dt - \int_0^1 H(t, x(t)) dt - f(\lambda) E^+(\Gamma). \end{aligned}$$

In total, we have

$$\begin{aligned} \mathcal{A}_H([z, C\#w]) - \mathcal{A}_H([x, C]) &= - \int_{-\infty}^{\infty} \int_0^1 \omega(\partial_s w, \partial_t w) dt ds - \int_0^1 H(t, x(t)) - H(t, z(t)) dt \\ &\leq f(\lambda) E^+(\Gamma) \leq E^+(\Gamma). \end{aligned}$$

A broken trajectory $(u, v) \in \Pi^{\{\Gamma_\lambda\}}(x, z)$ can come in two forms, either $u \in \widehat{\mathcal{M}}(x, y)$ or $v \in \widehat{\mathcal{M}}(y, z)$. Let us consider the first case; the second case is analogous. Abusing the notation, let u again denote the representative in $\mathcal{M}(x, y)$. The existence of a broken trajectory ensures that there is a sequence (λ_n) such that $\lambda_n \rightarrow \lambda_*$, a sequence (s_n) such that $s_n \rightarrow \infty$, and a sequence $w_n \in \mathcal{M}^{\lambda_n}(x, z)$ such that $w_n(s - s_n, t)$ converges to u and $w_n(s + s_n, t)$ converges to v in C_{loc}^∞ -topology. For every such sequence w_n we have that for n large enough w_n is homotopic to $u \# v$ relative to limiting orbits. We find

$$\begin{aligned} E(u) &= \mathcal{A}_H([x, C]) - \mathcal{A}_H([y, C \# u]) \\ &= (\mathcal{A}_H([x, C]) - \mathcal{A}_H([z, C \# w_n])) - (\mathcal{A}_H([y, C \# u]) - \mathcal{A}_H([z, C \# u \# v])) \\ &\leq \mathcal{A}_H([x, C]) - \mathcal{A}_H([z, C \# w_n]) + E^+(\Gamma). \end{aligned}$$

This inequality will be the tool to rule out the breaking of orbits.

The goal is to show that $\Phi_\varepsilon^{\Gamma_0}([x]) + \Phi_\varepsilon^{\Gamma_1}([x]) = 0$ which concludes the result since $\Phi_\varepsilon^{\Gamma_0}$ is the identity. We do this by first observing that $\Phi_\varepsilon^{\Gamma_0}([x]) + \Phi_\varepsilon^{\Gamma_1}([x])$ counts certain corner points of certain hypercubes in the manifold $\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z)$ where $z \in \text{OBP}_{\beta, f}(H)$ with $[z] = [x]$. We then show that this count is even.

The space $\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z)$ is a product of 1-dimensional moduli spaces, all of which can be compactified by gluing broken trajectories. The compactification of these moduli spaces induces a compactification of $\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z)$, given by a disjoint union of products of compact 1-dimensional manifolds. We consider the subset given by the union of all hypercubes, which are given by a product of intervals, and denote this subspace by

$$\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}, \text{cube}}(x, z) \subset \mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z).$$

The 0-dimensional corner points of such a hypercube is are element of

$$\prod_{i=1}^k \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_i, z_i) \cup \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_i, z_i) \cup \Pi^{\{\Gamma_\lambda\}}(x_i, z_i).$$

Note that all elements in

$$\prod_{i=1}^k \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_i, z_i) \cup \prod_{i=1}^k \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_i, z_i)$$

appear as corner points of a hypercube in $\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}, \text{cube}}(x, z)$. A corner point is called *pure* if all components either lie in $\mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_i, z_i)$ (in which case we say it is 0-pure), all in $\mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_i, z_i)$ (in which case we say 1-pure), or all in $\Pi^{\{\Gamma_\lambda\}}(x_i, z_i)$ (in which case we say purely broken). To show that $\Phi_\varepsilon^{\Gamma_0} + \Phi_\varepsilon^{\Gamma_1}([x]) = 0$ it suffices to show that the number of 0- and 1-pure corner points of hypercubes in $\mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z)$ is even. To do so we show that if a hypercube $Q \subset \mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}, \text{cube}}(x, z)$ has a 0- or 1-pure corner points, then it trivially has an even number of 0- or 1-pure corner points, or there no component of a corner point is a broken trajectory, that is all corner points lie in

$$\prod_{i=1}^k \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_i, z_i) \times \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_i, z_i)$$

In the latter case a broken trajectory must not appear in any component of a corner point. We show this in the case where a 1-pure corner point exists. The other case is analogous.

So, let $Q \subset \mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z)$ be a hypercube that has a 1-pure corner point. Suppose that some corner point contains broken trajectories. The hypercube is given by a product of closed intervals

$$Q = I_1 \times \cdots \times I_k,$$

where points in the interior of I_j correspond to pairs (s_λ, λ) with $s_\lambda \in \mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x_j, z_j)$ and the boundary correspond to either a broken trajectory or an element in $\mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_j, z_j) \cup \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_j, z_j)$ while at. In contrast, one boundary corresponds to a point in $\mathcal{M}_{1, \leq \varepsilon}^{\Gamma_0}(x_j, z_j)$. If both boundary components of some interval I_j corresponds to an element in $\mathcal{M}^{\Gamma_1}(x_j, z_j)$ there is an even number of 1-pure corer points in the hypercube. This does not affect the count modulo 2, hence we may assume that there are none. If the other boundary component corresponds to a broken trajectory (u_j, v_j) or (v_j, u_j) , we can associate with it a $\lambda_j \in (0, 1)$ that is such that

$$(u_j, v_j) \in \mathcal{M}_{-1}^{\Gamma_{\lambda_j}}(x_j, y_j) \times \widehat{\mathcal{M}}_1(y_j, z_j) \text{ or } (v_j, u_j) \in \widehat{\mathcal{M}}_1(x_j, y_j) \times \mathcal{M}_{-1}^{\Gamma_{\lambda_j}}(y_j, z_j).$$

Let $\lambda_\star = \max \lambda_j$ where the maximum runs over λ_j associated to a broken trajectory at a boundary of some I_j (in the case of 0-pure corner points we would take the minimum here). We find that for all j , there exists a broken trajectory with $\lambda_j = \lambda_\star$ corresponding to a bounded point of I_j or simply $u_j \in \mathcal{M}^{\Gamma_{\lambda_\star}}(x_j, z_j)$ corresponding to an interior point of I_j . In the first case, we write y_j for the breaking orbit of the broken trajectory. After relabeling, we have the following tuple at our hand

$$u = (u_1, \dots, u_p, (u_{p+1}, v_{p+1}), \dots, (u_q, v_q), (v_{q+1}, u_{q+1}), \dots, (v_k, u_k)),$$

where $u_j \in \mathcal{M}^{\Gamma_{\lambda_\star}}(x_j, z_j)$ for $j = 1, \dots, p$, $(u_j, v_j) \in \mathcal{M}^{\Gamma_{\lambda_\star}}(x_j, y_j) \times \widehat{\mathcal{M}}(y_j, z_j)$ for $j = p + 1, \dots, q$, and $(v_j, u_j) \in \widehat{\mathcal{M}}(x_j, y_j) \times \mathcal{M}^{\Gamma_{\lambda_\star}}(y_j, z_j)$ for $j = q + 1, \dots, k$. Abusing notation, we shall from now on write v_j for a representative in $\mathcal{M}_1(x_i, z_j)$ or in $\mathcal{M}_1(y_i, z_i)$. Now, for every loop $\gamma : S^1 \rightarrow S$ let

$$Z_\gamma : \mathbb{R} \times S^1 \rightarrow S, \quad (s, t) = \gamma(t)$$

be the constant cylinder. Let us define

$$w_i = \begin{cases} Z_{x_i} \# u_i \# Z_{z_i}, & \text{for } i = 1, \dots, p \\ Z_{x_i} \# u_i \# v_i, & \text{for } i = p + 1, \dots, q \\ v_i \# u_i \# Z_{z_i}, & \text{for } i = q + 1, \dots, k \end{cases}$$

and denote

$$w_{i,1} = \begin{cases} Z_{x_i}, & i = 1, \dots, q \\ v_i, & i = q + 1, \dots, k \end{cases}, \quad w_{i,2} = u_i, \quad w_{i,3} = \begin{cases} Z_{z_i}, & i = 1, \dots, p \text{ or } i = q + 1, \dots, k \\ v_i, & i = p + 1, \dots, q \end{cases}.$$

Consider the tuple of cylinders given by

$$w = (w_1, \dots, w_k)$$

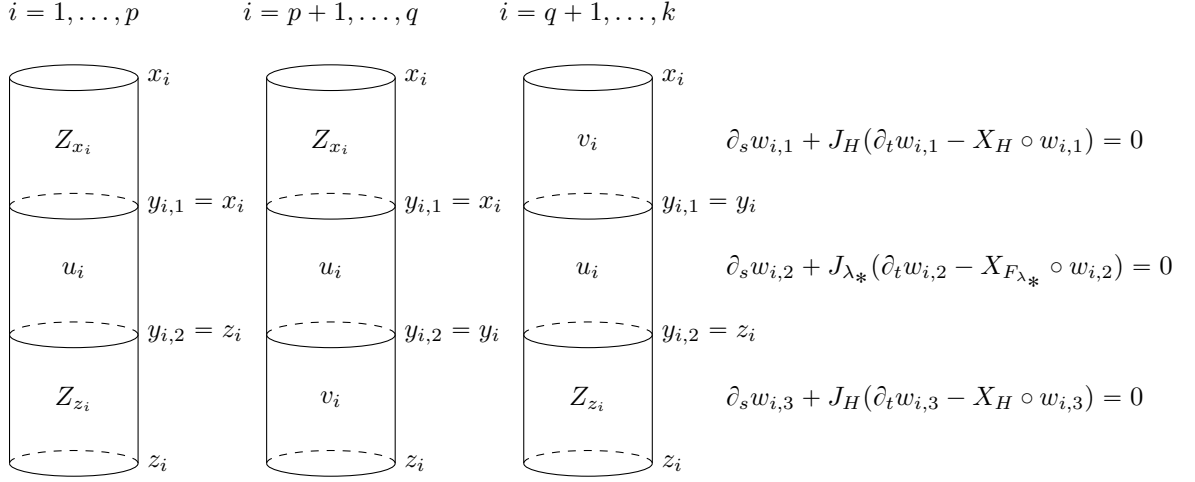
which is homotopic to u and given by extending by constancy in some of the components. Any cylinder in this tuple is divided into three parts $w_{i,1}$, $w_{i,2}$, and $w_{i,3}$ where the first and the last part satisfy

$$\partial_s w_{i,j} + J_H(\partial_t w_{i,j} - X_H \circ w_{i,j}) = 0$$

for $j = 1, 3$ and the middle part satisfies

$$\partial_s w_{i,2} + J_{\lambda_\star}(\partial_t w_{i,2} - X_{F_{\lambda_\star}} \circ w_{i,2}) = 0.$$

Let $y_{j,1}$ and $y_{j,2}$ denote the periodic orbits connecting the first two parts and the latter

FIGURE 12. The cylinders w_i .

two parts, respectively, of the j -th component. That is

$$y_{j,1} = \begin{cases} x_j, & \text{for } j = 1, \dots, q \\ y_j, & \text{for } j = q + 1, \dots, k \end{cases}$$

and

$$y_{j,2} = \begin{cases} z_j, & \text{for } j = 1, \dots, p \text{ or } j = q + 1, \dots, k \\ y_j, & \text{for } j = p + 1, \dots, q \end{cases}.$$

Observation 1: For every hypercube $Q \subset \mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}}(x, z)$ that has a 1-pure corner point c we have

$$\sum_{i=1}^k E(w_{i,1}) + E(w_{i,3}) < \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\}.$$

In particular, for any $v_i = w_{i,j}$ for $j = 1$ or $j = 3$ we have

$$E(v_i) \leq \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\}$$

Proof. We have

$$\mathcal{A}_H([x_i, C_{x_i}]) - \mathcal{A}_H([z_i, C_{x_i} \# s_\lambda]) = E(w_{i,1}) + e_i + E(w_{i,3}),$$

where $e_i = \mathcal{A}_H([y_{i,1}, C_{y_{i,1}}]) - \mathcal{A}_H([y_{i,2}, C_{y_{i,1}} \# w_{i,2}]) \geq -E^+(\Gamma)$. Using that

$$\sum_{i=1}^k \mathcal{A}_H([x_i, C_{x_i}]) - \mathcal{A}_H([z_i, C_{x_i} \# s_\lambda]) \leq \varepsilon$$

we find

$$\begin{aligned}
\sum_{i=1}^k E(w_{i,1}) + E(w_{i,3}) &\leq \sum_{i=1}^k E(w_{i,1}) + E(w_{i,3}) + \sum_{i=1}^k e_i + kE^+(\Gamma) \\
&\leq \left(\sum_{i=1}^k \mathcal{A}_H([x_i, C_{x_i}]) - \mathcal{A}_H([z_i, C_{x_i} \# s_\lambda]) \right) + kE^+(\Gamma) \\
&\leq \varepsilon + kE^+(\Gamma) \\
&< \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\}
\end{aligned}$$

This concludes the proof of the observation. $\square$

The quantities, $\mathcal{E}^{\text{int}}(H, J, x)$ and $\mathcal{E}^2(H, J)$, essentially only play a role in the following observation.

Observation 2: For $i = 1, 2$ the set $\{y_{j,i}\}_{j=1,\dots,k}$ consist of distinct orbits.

Proof. We prove this for $i = 1$; the other case is analogous. (For the other case, recall that we consider $z \in \text{OBP}_\beta(H)$ with $[x] = [z]$. This means that there is $\sigma \in S_k$ such that $z = \sigma(x)$.) Suppose that $y_{j,1} = y_{j',1}$ for $j \neq j'$. For every index ℓ we either have $y_{\ell,1} = x_\ell$ or $y_{\ell,1} = y_\ell$ is a breaking orbit. There are four cases to consider, and we show that each leads to a contradiction.

Case 1: $x_j = y_{j,1}$ and $y_{j',1} = x_{j'}$.

This is the easiest case since we have that $x_j \neq x_{j'}$ by definition.

Case 2: $y_{j,1} = y_j$ and $y_{j',1} = x_{j'}$.

Since $y_{j,1} = y_{j',1}$ by assumption, we have $y_j = x_{j'}$ and in this case $v_j = w_{i,j}$ for $j = 1, 3$ satisfies

$$\partial_s v_i + J_H(\partial_t v_i - X_H \circ v_i) = 0.$$

The cylinder v_j connects x_j and $x_{j'}$ which are both orbits in x . However, in Observation 1 we saw that we have $E(v_i) < \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\} \leq \mathcal{E}^{\text{int}}(H, J, x)$. By the definition of $\mathcal{E}^{\text{int}}(H, J, x)$, this is not possible, and hence we conclude that $y_j \neq x_{j'}$ for every j, j' .

Case 3: $y_{j,1} = x_j$ and $y_{j',1} = y_{j'}$.

This is Case 2 with indices swapped.

Case 4: $y_{j,1} = y_j$ and $y_{j',1} = y_{j'}$.

We find that for one of the curves $v \in \{v_j, v_{j'}\}$ we have

$$\mathcal{E}^2(H, J, x) \leq \max\{E(v_j), E(v_{j'})\} = E(v).$$

This again contradicts Observation 1 where we found that $E(v) = \max\{E(v_j), E(v_{j'})\} < \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\}$.

This concludes the proof of the observation. $\square$

After this observation, we have

$$0 = \iota(u) = \iota(w) = \sum_{i \neq j} \iota(w_{i,1}, w_{j,1}) + \iota(w_{i,2}, w_{j,2}) + \iota(w_{i,3}, w_{j,3}).$$

By positivity of intersection (Lemma 2.2), we can now conclude that for all $i \neq j$

$$\iota(w_{i,1}, w_{j,1}) = \iota(w_{i,2}, w_{j,2}) = \iota(w_{i,3}, w_{j,3}) = 0$$

and hence for all $i \neq j$ and all $(s, t) \in \mathbb{R} \times S^1$ we have

$$w_{i,r}(s, t) \neq w_{j,r}(s, t),$$

for $r = 1, 2, 3$. Hence, the curves induce isotopies between the braids $B(H, \{x_1, \dots, x_k\})$, $B(H, \{y_{1,1}, \dots, y_{k,1}\})$, $B(H, \{y_{1,2}, \dots, y_{k,2}\})$, and $B(H, \{z_1, \dots, z_k\})$ and we have that

$$(w_{1,1}, \dots, w_{k,1}) \in \mathcal{M}(x, (y_{1,1}, \dots, y_{k,1})), \quad (w_{1,3}, \dots, w_{k,3}) \in \mathcal{M}(y_{1,2}, \dots, y_{k,2}, x).$$

By assumption we have that $\mu_{\max}(w_{1,1}, \dots, w_{k,1}) = 1$ or $\mu_{\max}(w_{1,3}, \dots, w_{k,3}) = 1$. Without loss of generality, we may assume that this holds in the first case. After the observation we know that

$$E(w_{1,1}, \dots, w_{k,1}) = \sum_{i=1}^k E(w_{i,1}) < \mathcal{E}(H, J, x).$$

However, this contradicts the definition of $\mathcal{E}(H, J, x)$ and hence the assumption that broken trajectories appear as components of admissible corner hypercubes. This means that the corner points must lie in

$$\prod_{i=1}^k \mathcal{M}^{\Gamma_0}(x_i, z_i) \cup \mathcal{M}^{\Gamma_1}(x_i, z_i).$$

Now for a hypercube $Q \subset \mathcal{M}_{0, \leq \varepsilon}^{\{\Gamma_\lambda\}, \text{cube}}(x, z)$

$$Q = I_1 \times \dots \times I_k$$

with a 0- or 1-pure corner point where $I_j \subset \mathcal{M}_0^{\{\Gamma_\lambda\}}(x_i, z_i)$ are connected components which are closed intervals, we define colorings of the boundary $\rho_i : \partial I_i \rightarrow \{0, 1\}$ by the type of boundary component

Type 0: $(u_i, 0)$ where $u_i \in \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x_i, z_i)$,

Type 1: $(u_i, 1)$ where $u_i \in \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x_i, z_i)$.

By Lemma 3.4, we have that the number of 0- and 1-pure corner points of such hypercubes is even. These corner points correspond to the set

$$\mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x, z) \cup \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x, z).$$

Therefore we find

$$\pi_{[x]}(\Phi_\varepsilon^{\Gamma_1}([x]) + \Phi_\varepsilon^{\Gamma_0}([x])) = \sum_{\substack{z \in OBP_{\beta, f}(H) \\ [z] = [x]}} |\mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x, z) \cup \mathcal{M}_{0, \leq \varepsilon}^{\Gamma_1}(x, z)| [z] = 0 \pmod{2}.$$

Since Γ_0 is the constant continuation data, it follows from standard arguments (see [AD14, Proposition 11.1.14])

$$|\mathcal{M}_{0, \leq \varepsilon}^{\Gamma_0}(x, z)| = \begin{cases} 0, & x \neq z \\ 1, & x = z \end{cases}$$

and hence $\Phi_\varepsilon^{\Gamma_0} = id_{BCF_{\beta, f}(H)}$. This means that

$$\pi_{[x]} \circ \Phi_\varepsilon^{\Gamma_1}([x]) = [x]$$

which concludes the proof of Proposition 4.2. $\square$

Using this result, we can now prove the braid stability statement of Theorem B.

4.3. Braid Stability. This section proves the quantitative braid stability result in Theorem B. We briefly outline the proof strategy. Given non-degenerate Hamiltonians H, G , and $x \in \mathcal{P}(H)$ with associated braid class β we choose suitable almost complex structures and continuation data Γ'' from G to H and Γ' from H to G such that

$$kE^+(\Gamma'' \#_\rho \Gamma') = kE(H - G) < \mathcal{S}(H, x).$$

The continuation data $\Gamma'' \#_\rho \Gamma'$ connects H to itself, and we can apply Proposition 4.2. We conclude that there is an odd number of tuples of continuation cylinders w from x to some $z \in \text{OB}\mathcal{P}_\beta(H)$ with $[x] = [z]$. For such a tuple $w = (w_1, \dots, w_k)$ the continuation cylinders w_i are obtained by gluing the continuation cylinders u_i and v_i with respect to the data Γ'' and Γ' . The tuples $(u_1, \dots, u_k)$ end in breaking orbits $y_1, \dots, y_k \in \mathcal{P}(G)$ which may come with multiplicities. Applying a symmetry argument, we show that the number of such tuples, where the breaking orbits come with multiplicities, is even. This implies that there is at least one tuple of $(u_1, \dots, u_k)$ ending in pairwise distinct orbits. Using that $\iota(u_i, u_j) = 0$ we conclude that the tuple $(u_1, \dots, u_k)$ induces an isotopy of the braids $B(H, x)$ and $B(G, y)$, which means that $\dim \text{BCF}_{\beta, f}(G) \geq 1$.

Proof of Theorem B. Choose an almost complex structure J such that (H, J) is regular pair and $kE(H - G) < \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\}$. We define self-continuation data $\Gamma = \Gamma'' \#_\rho \Gamma'$ at (H, J) and will apply Proposition 4.2 to it. Let $f : \mathbb{R} \rightarrow [0, 1]$ be smooth increasing such that $f(s) = 0$ for $s < -1$ and $f(s) = 1$ for $s > 1$. Let H' and H'' be interpolations $\mathbb{R} \times S^1 \times S \rightarrow \mathbb{R}$ between H and G defined as follows

$$H'(s, t, x) = H(t, x) + f(s)(G(t, x) - H(t, x)),$$

$$H''(s, t, x) = G(t, x) + f(s)(H(t, x) - G(t, x)).$$

Note that

$$\begin{aligned} \int_{-\infty}^{s_0} \int_0^1 \max_{p \in S} \partial_s H'(s, t, p) dt ds &= -f(s_0) \int_0^1 \min_{x \in S} (H(t, x) - G(t, x)) dt \\ \int_{-\infty}^{s_0} \int_0^1 \max_{p \in S} \partial_s H''(s, t, p) dt ds &= f(s_0) \int_0^1 \max_{x \in S} (H(t, x) - G(t, x)) dt. \end{aligned}$$

We have

$$E^+(\Gamma') = \int_0^1 \max_{x \in S} (H(t, x) - G(t, x)) dt \quad \text{and} \quad E^+(\Gamma'') = - \int_0^1 \min_{x \in S} (H(t, x) - G(t, x)) dt.$$

In particular, $kE^+(\Gamma'' \#_\rho \Gamma') = kE(H - G)$ which is bounded above by $k\mathcal{S}(H, x)$ and hence also by $\min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\}$. By choosing suitable interpolations of almost complex structures we have continuation data $\Gamma' = (H', J')$ and $\Gamma'' = (H'', J'')$ such that after a C^∞ -small perturbation we have that Γ', Γ'' , and $\Gamma'' \#_\rho \Gamma'$ is regular for ρ large enough and such that $kE^+(\Gamma'' \#_\rho \Gamma') < \min\{\mathcal{E}(H, J, x), \mathcal{E}^{\text{int}}(H, J, x), \mathcal{E}^2(H, J, x)\}$. Now, applying Proposition 4.2 we find $\pi_{[x]} \circ \Phi_0^{\Gamma'' \#_\rho \Gamma'}([x]) = [x]$. This means that there are an odd number of cylinders in the following set

$$\mathcal{T}(x) = \bigcup_{\substack{z \in \text{OB}\mathcal{P}_\beta \\ [z] = [x]}} \mathcal{M}_{0, \leq 0}^{\Gamma'' \#_\rho \Gamma'}(x, z).$$

We now use a standard result from Floer theory on gluing continuation data. Namely, there is a bijection

$$\chi : \bigcup_{y \in \mathcal{P}(G)} \mathcal{M}_0^{\Gamma'}(x_i, y) \times \mathcal{M}_0^{\Gamma''}(y, x_i) \rightarrow \mathcal{M}_0^{\Gamma'' \# \rho \Gamma'}(x_i, x_i), \quad (u, v) \mapsto \chi(u, v)$$

which is obtained by gluing the cylinders in a fashion described in Section 2.2. In particular, Lemma 2.3 applies which means that for pairs $(u_i, v_i) \in \bigcup_{y \in \mathcal{P}(G)} \mathcal{M}_0^{\Gamma'}(x_i, y) \times \mathcal{M}_0^{\Gamma''}(y, x_i)$ we have

$$\sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j) = \sum_{i \neq j} \iota(\chi(u_i, v_i), \chi(u_j, v_j)).$$

Let

$$B\mathcal{T}(x) = \{(\chi^{-1}(w_1), \dots, \chi^{-1}(w_k)) \mid w = (w_1, \dots, w_k) \in \mathcal{T}(x)\}$$

be the set of tuples of broken trajectories corresponding to a tuple of trajectories $w \in \mathcal{T}(x)$. For every $w \in \mathcal{T}(x)$ let $y^w = (y_1^w, \dots, y_k^w) \in \mathcal{P}(G)^k$ be the tuple of breaking orbits of the broken trajectory $(\chi^{-1}(w_1), \dots, \chi^{-1}(w_k)) \in B\mathcal{T}(x)$. We say a trajectory $w \in \mathcal{T}(x)$ is *degenerate* if the associated tuple of breaking orbits y^w has multiplicities in its entries, that is, if $y^w \in \Delta_{\mathcal{P}(G)}$.

Observation: It is sufficient to prove that there is a regular $w \in \mathcal{T}(x)$.

Proof. Assume that $w \in \mathcal{T}(x)$ is regular. Then consider the associated broken trajectory

$$((u_1, v_1), \dots, (u_k, v_k)) = (\chi^{-1}(w_1), \dots, \chi^{-1}(w_k)) \in B\mathcal{T}(x).$$

We noticed before that

$$0 = \sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j).$$

Since w is regular, the breaking orbits $y_1^w, \dots, y_k^w$ are pairwise distinct which means that the intersection numbers $\iota(u_i, u_j)$ and $\iota(v_i, v_j)$ are well-defined for $i \neq j$ and that

$$0 = \sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j) = \sum_{i \neq j} \iota(u_i, u_j) + \iota(v_i, v_j).$$

After positivity of intersection, this means that

$$0 = \iota(u_i, u_j) = \iota(v_i, v_j)$$

for all $i \neq j$. This in particular means that for all $(s, t) \in \mathbb{R} \times S^1$ we have $u_i(s, t) \neq u_j(s, t)$ and therefore, the cylinders $(u_1, \dots, u_k)$ induce an isotopy of the braids $B(H, x)$ and $B(G, y^w)$. Since the cylinders u_i connect orbits of the same index, the framing is preserved. Thus, there is a braid of orbits of G with framed braid class (β, f) , so $\dim \text{BCF}_{\beta, f}(G) \geq 1$. This concludes the proof of the observation. $\square$

We now show that there is a regular trajectory in $\mathcal{T}(x)$. Let

$$D\mathcal{T}(x) = \{w \in \mathcal{T}(x) \mid y^w \in \Delta_{\mathcal{P}(G)}\}$$

be the set of degenerate trajectories. Since $\mathcal{T}(x)$ splits as a disjoint union into the set of regular and degenerate trajectories, and since we already established that the cardinality of $\mathcal{T}(x)$ is odd, it suffices to show that

$$|D\mathcal{T}(x)| = 0 \pmod{2}.$$

We remark here that if the framing is injective, we are done since in this case $\mu(x_i) = \mu(y_i^w)$ are pairwise distinct and hence, the orbits y_i^w are always pairwise distinct, which means that

$D\mathcal{T}(x) = \emptyset$. In the general case, we argue by symmetry. For every $y = (y_1, \dots, y_k) \in \Delta_{\mathcal{P}(G)}$ and $z \in OB\mathcal{P}_\beta(H)$ such that $[x] = [z]$ let

$$DB\mathcal{T}(x, y, z) = \left\{ ((u_1, v_1), \dots, (u_k, v_k)) \in B\mathcal{T}(x) \left| \begin{array}{l} \text{for } w = (\chi(u_1, v_1), \dots, \chi(u_k, v_k)), \\ y^w = y, w \in \mathcal{M}_0^{\Gamma'' \# \rho \Gamma'}(x, z) \end{array} \right. \right\},$$

the set of broken trajectories with breaking orbits y corresponding to degenerate trajectories connecting x to z . Note that the cardinality of

$$D\mathcal{T}(x) \quad \text{and} \quad \bigcup_{y \in \Delta_{\mathcal{P}(G)}} \bigcup_{\substack{z \in OB\mathcal{P}_\beta(H) \\ [x]=[z]}} DB\mathcal{T}(x, y, z)$$

agrees. For every $y \in \Delta_{\mathcal{P}(G)}$ we show that the cardinality of $\bigcup_{\substack{z \in OB\mathcal{P}_\beta(H) \\ [x]=[z]}} DB\mathcal{T}(x, y, z)$ is even which concludes the result. Let us denote by

$$\text{Fix}(y) = \{\sigma \in S_k \mid \sigma(y) = y\},$$

the set of permutations that leave y invariant. For every $\sigma \in \text{Fix}(y)$ we have

$$(14) \quad |DB\mathcal{T}(x, y, z)| = |DB\mathcal{T}(x, y, \sigma(z))|.$$

Indeed, the following map is a bijection

$$S^{x, y, z, \sigma} : DB\mathcal{T}(x, y, z) \rightarrow DB\mathcal{T}(x, y, \sigma(z)), \quad ((u_1, v_1), \dots, (u_k, v_k)) \mapsto ((u_1, v_{\sigma(1)}), \dots, (u_k, v_{\sigma(k)})).$$

We argue that this map is well-defined. First, note that $((u_1, v_{\sigma(1)}), \dots, (u_k, v_{\sigma(k)})) \in \mathcal{M}_0^{\Gamma'}(x, y) \times \mathcal{M}_0^{\Gamma''}(y, \sigma(z))$ since we have $\sigma(y) = y$. By Lemma 2.5 we have

$$0 = \sum_{i \neq j} \iota(u_i \# v_i, u_j \# v_j) = \sum_{i \neq j} \iota(u_i \# v_{\sigma(i)}, u_j \# v_{\sigma(j)}).$$

By Lemma 2.3 we then also have

$$0 = \sum_{i \neq j} \iota(\chi(u_i, v_{\sigma(i)}), \chi(u_j, v_{\sigma(j)})).$$

Moreover,

$$\begin{aligned} 0 &= \left| \sum_{i=1}^k \mathcal{A}_H([x_i, C_{x_i}]) - \mathcal{A}_H([z_i, C_{x_i} \# u_i \# v_i]) \right| \\ &= \left| \sum_{i=1}^k \mathcal{A}_H([x_i, C_{x_i}]) - \mathcal{A}_H([z_{\sigma(i)}, C_{x_i} \# u_i \# v_{\sigma(i)}]) \right| \end{aligned}$$

and hence $(\chi(u_1, v_{\sigma(1)}), \dots, \chi(u_k, v_{\sigma(k)})) \in \mathcal{T}(x)$. By definition, we find that $((u_1, v_{\sigma(1)}), \dots, (u_k, v_{\sigma(k)})) \in DB\mathcal{T}(x, y, \sigma(z))$. To see that the map $S^{x, y, z, \sigma}$ is a bijection, note that an inverse is given by $S^{x, y, \sigma(z), \sigma^{-1}}$, which is defined since $\text{Fix}(y)$ is a group and $\sigma^{-1} \in \text{Fix}(y)$. This shows (14).

We now have

$$\begin{aligned}
|D\mathcal{T}(x)| &= \sum_{y \in \Delta_{\mathcal{P}(G)}} \sum_{\substack{z \in OB\mathcal{P}_{\beta}(H) \\ [x]=[z]}} |DB\mathcal{T}(x, y, z)| \\
&= \sum_{y \in \Delta_{\mathcal{P}(G)}} \sum_{\sigma \in S_k} |DB\mathcal{T}(x, y, \sigma(x))| \\
&= \sum_{y \in \Delta_{\mathcal{P}(G)}} \sum_{[\nu] \in S_k/\text{Fix}(y)} \sum_{\sigma \in \text{Fix}(y)} |DB\mathcal{T}(x, y, \sigma(\nu(x)))| \\
&= \sum_{y \in \Delta_{\mathcal{P}(G)}} \sum_{[\nu] \in S_k/\text{Fix}(y)} |\text{Fix}(y)| |DB\mathcal{T}(x, y, \nu(x))| \\
&= 0 \pmod{2},
\end{aligned}$$

where we use that $[x] = [z]$ if and only if there is $\sigma \in S_k$ such that $\sigma(z) = x$ and that for every $y \in \Delta_{\mathcal{P}(G)}$ the group $\text{Fix}(y)$ is a non-trivial product of permutation groups and therefore has even cardinality. $\square$

5. AN APPLICATION: CONTRACTIBLE ORBITS AND ENTROPY OF EGGBEATERS ON T^2

The goal of this section is prove Theorem A. We outline the strategy.

- i In Section 5.1 we define a notion of entropy $h(\beta)$ of a closed braid β that satisfies the following. For a Hamiltonian H and periodic orbits $\{x_1, \dots, x_k\} \subset \mathcal{P}(H)$ at fixed points $z_1, \dots, z_k$ of ϕ_H^1 we have

$$h(\mathcal{B}(H, x_1, \dots, x_k)) \leq h_{\text{top}}(\phi_H^1).$$

- ii In Section 5.2 we study closed braids on symplectic tori and define a family of closed braids $\beta_{m,n}$ with two contractible strands such that

$$\frac{1}{5} \log(|mn|) \leq h(\beta_{m,n})$$

and hence $h(\beta_{m,n}) \rightarrow \infty$ as $m, n \rightarrow \infty$.

- iii In Section 5.3 we first define a sequence of Hamiltonians H_k whose flows are eggbeater maps on the torus.
- iv We then analyze the contractible orbits of the flow of H_{2k} and fix a pair of contractible orbits $x_1^{(k)}, x_2^{(k)} \in \mathcal{P}(H_{2k})$ whose associated framings

$$f_k : x_1^{(k)} \mapsto 2, \quad x_2^{(k)} \mapsto -2$$

are injective and such that $\mathcal{B}(H_{2k}, x_1^{(k)}, x_2^{(k)}) = \beta_{2k+1, 2k+1}$.

- v Using topological obstructions we restrict the set of pairs $\{y_1, y_2\} \subset \mathcal{P}(H_{2k})$ that can be connected to $\{x_1, x_2\}$ by an isotopy. Using this restriction on Floer isotopies, a computation then shows the following for the stability radius of the braids of orbits $\{x_1^{(k)}, x_2^{(k)}\}$

$$\mathcal{S}(H_{2k}, x_1^{(k)}, x_2^{(k)}) \rightarrow \infty \quad (k \rightarrow \infty).$$

Here, the fact that we use braids is essential since it refines the set of actions to which we need to compare the action of $x_1^{(k)}, x_2^{(k)}$.

- vi Applying the braid stability result Theorem B we then conclude that in a large Hofer neighborhood of ϕ_{2k} any Hamiltonian diffeomorphism is generated by a Hamiltonian G whose flow has a braid of orbits of type $\beta_{2k+1, 2k+1}$. This gives a lower bound on the topological entropy of ψ and proves Theorem A.

5.1. Preliminaries: Surface Braids and Entropy. For k distinct points $y_1, \dots, y_k \in S$ the *braid group* with k strands is defined by equipping the set of isotopy classes of a set of k (indistinguishable) strands

$$s_i : [0, 1] \rightarrow S \times [0, 1], \quad t \mapsto (x_i(t), t)$$

for paths $x_i : [0, 1] \rightarrow S$ such that $\{x_1(0), \dots, x_k(0)\} = \{x_1(1), \dots, x_k(1)\} = \{y_1, \dots, y_k\}$, with the group multiplication given by juxtaposition of braids (for more details see [FM12]). We denote this group by $\text{BG}_k(y_1, \dots, y_k)$. The *pure braid group*

$$\text{PB}_k(y_1, \dots, y_k) \subset \text{BG}_k(y_1, \dots, y_k)$$

is defined to be the subgroup given by the elements where x_i are loops, that is $x_i(0) = x_i(1)$ for all i . For every element in the pure braid group, the closure gives a well-defined braid class invariant under conjugation in the total braid group. More precisely, there is the following correspondence

$$\{\text{Cl}_{\text{BG}_k}(b) \mid b \in \text{PB}_k(y_1, \dots, y_k)\} \xleftarrow[\text{closure}]{1:1} \{\text{classes of closed surface braids with } k \text{ strands}\},$$

where $\text{Cl}_{\text{BG}_k}(b)$ is the conjugacy class of b in the $\text{BG}_k(y_1, \dots, y_k)$.

Moreover, there is a group homomorphism

$$\text{Push} : \text{BG}_k(y_1, \dots, y_k) \rightarrow \text{MCG}(S \setminus \{y_1, \dots, y_k\}),$$

where $\text{MCG}(S \setminus \{y_1, \dots, y_k\})$ is the mapping class group of the k -times punctured surface $S \setminus \{y_1, \dots, y_k\}$ (see [FM12]). It is characterized by the property that for the flow ϕ^t with fixed points $y_1, \dots, y_k$ of ϕ^1 the mapping class represented by $\phi^1|_{S \setminus \{y_1, \dots, y_k\}}$ is the image of $((\phi^t(y_1), t), \dots, (\phi^t(y_k), t))$.

For a compact topological space X , there are several ways to define the topological entropy of a homeomorphism $f : X \rightarrow X$. When the topology of X is metrizable, one can define it as follows. First, choose a metric d inducing the topology and for every $n \in \mathbb{Z}_{\geq 1}$ define a new metric by

$$d_n(x, y) = \max_{i=0, \dots, n-1} d(f^i(x), f^i(y)).$$

We say a set P is (n, ε) -spanning if for all $x \in X$ there is some $p \in P$ with $d_n(x, p) < \varepsilon$. Let $r(f, n, \varepsilon)$ be the minimal cardinality of a set P that is (n, ε) -spanning; by compactness this number is finite. We then define

$$h_{\text{top}}(f) = \lim_{\varepsilon \rightarrow 0} \limsup_{n \rightarrow \infty} \frac{1}{n} \log(r(f, n, \varepsilon)).$$

This agrees with the general definition of topological entropy when X is not metrizable and is independent of the choice of metric d .

Suppose that $F \subset X$ is a finite set. To a mapping class $[f] \in \text{MCG}(X \setminus F)$ one can associate a notion of entropy, which is a measure of the growth rate of words under the action of f_* on $\pi_1(X \setminus F)$. This algebraic definition of entropy is a lower bound on the topological entropy of the mapping class. Let A be a set of generators of $\pi_1(X \setminus F)$. For every $w \in \pi_1(X \setminus F)$ denote by $L_A(w)$ the minimal length of the word w in the letters $A \cup A^{-1}$. Then define

$$h([f]) = \sup_{w \in \pi_1(X \setminus F)} \limsup_{n \rightarrow \infty} \frac{1}{n} \log(L_A(f_*^n(w))).$$

It is not hard to see that it suffices only to consider $w \in A$ and that the expression is independent of the choice of generators A . Given a homeomorphism $f : X \rightarrow X$ such that

$f(F) = F$, for a finite set F , one can conclude that

$$h([f] \mid_{X \setminus F}) \leq h_{\text{top}}(f),$$

by compactifying the space $S \setminus F$ by adding boundary components at each puncture consisting of copies of S^1 (see, for example, [Bow06]). Passing to conjugacy classes, the push map associates closed surface braids to conjugacy classes of mapping classes. Since the topological entropy of the mapping class is invariant under conjugation, the push map induces a notion of entropy for closed surface braids β

$$h(\beta) := h(\text{Push}(\beta)).$$

Importantly, if ϕ^t is a flow and F is a set of fixed points of ϕ^1 , the topological entropy of the closed braid $\mathcal{B}(\phi^t, F)$, which is the isotopy class of $\{(\phi^t(x))\}_{x \in F}$, gives a lower bound on the entropy of ϕ^1 , namely

$$h(\mathcal{B}(\phi^t, F)) \leq h_{\text{top}}(\phi^1).$$

A class of dynamically interesting homeomorphisms is *pseudo-Anosov* maps. For such maps, there exist two transverse measured foliations and a *dilatation* $\lambda > 1$ such that f stretches one foliation by a factor λ and the other by $1/\lambda$. A mapping class $[f]$ is pseudo-Anosov if there is a pseudo-Anosov representative f . In this case, this representative minimizes the topological entropy in the mapping class, and we have the following (see [FS79])

$$h_{\text{top}}(f) = h([f]) = \log(\lambda).$$

We say λ is the dilatation of the mapping class. There are many ways to estimate λ ; in the case when $[f]$ is the image of a point-push map along a loop γ , we can estimate λ by the self-intersection number defined by

$$i(\gamma) = \frac{1}{2} \min_{\mu} |\{(t_1, t_2) \in S^1 \mid t_1 \neq t_2 \text{ and } \mu(t_1) = \mu(t_2)\}|,$$

where the minimum runs over all loops μ freely homotopic to γ (see [Dow11]). A loop γ on a surface $S_{g,n}$ of genus g with n punctures is called *filling* if any loop freely homotopic to γ intersects any simple closed essential loop on $S_{g,n}$ (where essential means non-contractible and not homotopic to a simple loop around a puncture). By Kra [Kra81], the associated point-push mapping class $[f_\gamma]$ has positive entropy if γ is filling. Analyzing the self-intersection number of γ , one can give a quantitative estimate (see [Dow11]). Let p be a marked point on $S_{g,n}$ and γ a loop starting at p . By [Dow11, Theorem 1.5] if $3g + n > 3$ and γ represents a primitive element in $\pi_1(S_{g,n}, p)$ then for the point push map $\text{Push}(\gamma) \in \text{MCG}(S_{g,n} \setminus \{p\})$ the dilatation λ of the mapping class satisfies

$$\lambda \geq \sqrt[5]{i(\gamma) + 1}.$$

We will work on the once punctured torus $S_{1,1}$ where the inequality $3g + n > 3$ holds.

5.2. Torus Braids. For two base points $z_1, z_2 \in T^2$ the *pure braid group on the torus with two strands* is given by $\text{PB}_2(T) = \pi_1(X_2(T), (z_1, z_2))$ where

$$X_2(T) = \{(x, y) \in T^2 \times T^2 \mid x \neq y\}$$

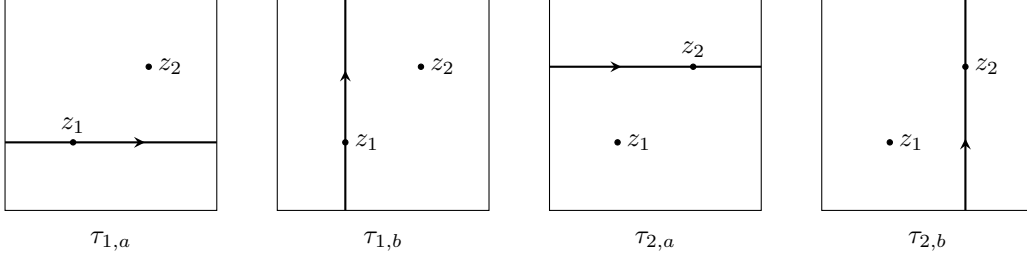
is the configuration space of two points. A representation of this group is given by

$$\text{PB}_2(T) = \langle \tau_{1,a}, \tau_{1,b}, \tau_{2,a}, \tau_{2,b} \mid \text{Relations} \rangle$$

with complicated relations that we omit here (we point the interested reader to the reference [Sco08]). Here the generators $\tau_{i,a}$ and $\tau_{i,b}$ thought of the braids that correspond to

$$(15) \quad \begin{aligned} \tau_{1,a} &= (z_1 + (t, 0), z_2), & \tau_{1,b} &= (z_1 + (0, t), z_2) \\ \tau_{2,a} &= (z_1, z_2 + (t, 0)), & \tau_{2,b} &= (z_1, z_2 + (0, t)) \end{aligned}$$

which are pairs of loops for $t \in [0, 1]$.



Note that there is a homeomorphism

$$\psi : X_2(T) \rightarrow (T^2 \setminus \{(0, 0)\}) \times T^2, \quad (x, y) \mapsto (x - y, y)$$

which gives rise to an isomorphism

$$\psi_* : \text{PB}_2(T) \rightarrow F_2 \times \mathbb{Z}^2,$$

where $F_2 = \langle a, b \rangle$ is the free group with two generators (Attention: we use the convention that words in F_2 and in the braid group are read from left to right). We use that the group F_2 is naturally isomorphic to $\pi_1(T^2 \setminus \{(0, 0)\}, z_1 - z_2)$ via an isomorphism that maps the loops

$$z_1 - z_2 + (t, 0) \mapsto a, \quad z_1 - z_2 + (0, t) \mapsto b.$$

With the representation as in (15) we have an isomorphism

$$\langle \tau_{1,a}, \tau_{1,b}, \tau_{2,a}, \tau_{2,b} \mid \text{Relations} \rangle \rightarrow F_2 \times \mathbb{Z}^2,$$

where

$$(16) \quad \tau_{1,a} \mapsto (a, 0), \quad \tau_{1,b} \mapsto (b, 0),$$

$$(17) \quad \tau_{2,a} \mapsto (a^{-1}, a), \quad \tau_{2,b} \mapsto (b^{-1}, b),$$

where $\mathbb{Z}^2 = \langle a, b \rangle^{\text{ab}}$. Now consider the abelinization

$$\pi : F_2 \times \mathbb{Z}^2 \rightarrow \mathbb{Z}^2 \times \mathbb{Z}^2$$

Lemma 5.1. *A braid has contractible strands if and only if for the corresponding element $(w, c) \in F_2 \times \mathbb{Z}^2$ we have*

$$\pi(w, c) = 0.$$

Proof. Firstly, the second strand is contractible if and only if $c = 0$. We now find a representation of the braid as a product of letters $\tau_{1,a}^{\pm 1}$ and $\tau_{1,b}^{\pm 1}$. The abelianization of w in $\mathbb{Z}^2$ is exactly the homology class of the first strand, which is contractible if and only if this class is trivial. $\square$

A path $\gamma : [0, 1] \rightarrow T^2 \setminus \{(0, 0)\}$ is *admissible* if it intersects $\{0\} \times \mathbb{R}/\mathbb{Z}$ and $\mathbb{R} \times \{0\}/\mathbb{Z}^2$ transversely. We now define a function

$$\text{spell} : \{\gamma : [0, 1] \rightarrow T^2 \setminus \{(0, 0)\} \mid \gamma \text{ admissible}\} \rightarrow F_2.$$

Consider the universal cover

$$p : \mathbb{R}^2 \rightarrow \mathbb{R}^2/\mathbb{Z}^2$$

and a lift $\tilde{\gamma}$ of an admissible path γ . The word $\text{spell}(\gamma)$ is obtained in the following way. For every intersection of $\tilde{\gamma}$ with a horizontal grid line $\mathbb{R} \times \{k\}$ or vertical grid line $\{k\} \times \mathbb{R}$ for some $k \in \mathbb{Z}$, we write a letter in the obvious order given by $\tilde{\gamma}$. Namely, write a if $\tilde{\gamma}$ intersects some vertical grid line from left to right and a^{-1} if it intersects from right to left. Similarly, write b if $\tilde{\gamma}$ intersects a horizontal grid line from bottom to top and write b^{-1} if it intersects from top to bottom. The resulting word is defined to be $\text{spell}(\gamma)$. Clearly for a loop γ the word $\text{spell}(\gamma)$ is a possibly unreduced representative of the word in F_2 that corresponds to the class $[\gamma] \in \pi_1(T^2 \setminus \{(0, 0)\})$. For two loops $x_1, x_2 : S^1 \rightarrow T^2$ such that $x_1(t) \neq x_2(t)$ for all $t \in S^1$ we define

$$\text{spell}(x_1, x_2) = \text{spell}(x_1 - x_2).$$

Note that in general $\text{spell}(x_1, x_2) \neq \text{spell}(x_2, x_1)$.

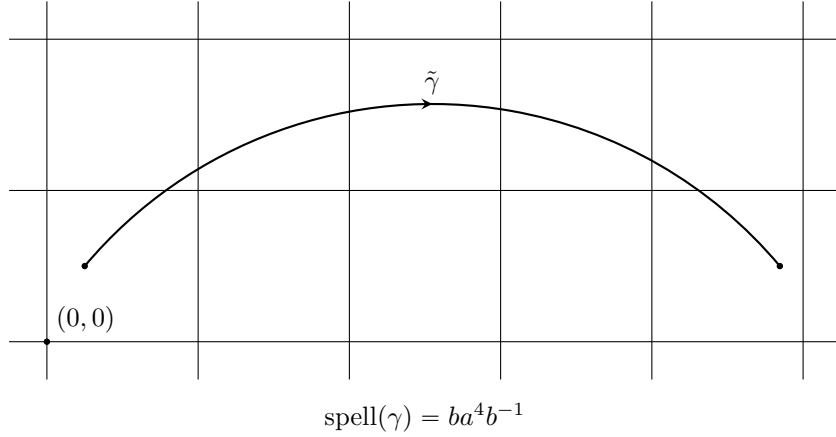


FIGURE 13. The spelling of a path γ .

Lemma 5.2. *Let (x_1, x_2) and (y_1, y_2) be loops in $X_2(T^2)$ such that each loop x_i and y_i is contractible in T^2 . If the loops are homotopic in $X_2(T^2)$ (that is the associated braids are isotopic) then $\text{spell}(x_1, x_2)$ and $\text{spell}(y_1, y_2)$ are conjugate.*

Proof. First, suppose that $(x_1(0), x_2(0)) = (y_1(0), y_2(0))$. In this case, the lemma follows from the fact that the closures of two braids are isotopic if and only if they are conjugate to each other in the braid group at $(x_1(0), x_2(0))$. Indeed, $(\text{spell}(x_1, x_2), 0)$ and $(\text{spell}(y_1, y_2), 0)$ in $F_2 \times \mathbb{Z}^2$ are the corresponding braids in the braid group, and they are conjugate if and only if the first components are conjugate. In general, let (α_1, α_2) be a path from $(x_1(0), x_2(0))$ to $(y_1(0), y_2(0))$. Note that $(\alpha_1 \# y_1 \# \alpha_1^{-1}, \alpha_2 \# y_2 \# \alpha_2^{-1})$ is a braid at $(x_1(0), x_2(0))$ whose closure is isotopic to the one of (y_1, y_2) . Clearly, the word $\text{spell}(\alpha_1 \# y_1 \# \alpha_1^{-1}, \alpha_2 \# y_2 \# \alpha_2^{-1})$ is conjugate to $\text{spell}(y_1, y_2)$. The general case now follows from the case $(x_1(0), x_2(0)) = (y_1(0), y_2(0))$. $\square$

We now discuss the entropy of a family of closed braids that will come up in the construction of eggbeater maps. For integers $n, m \in \mathbb{Z} \setminus \{0\}$ let $\beta_{m,n}$ denote the closed torus braid that is obtained by the closure of a braid given by the element $(a^m b^n a^{-m} b^{-n}, 0)$. Note that this definition is independent of the choice of base points (x_1, x_2) , since by Lemma 5.2 the closures are isotopic if the spelling of the difference loops is conjugate.

Lemma 5.3. *For every $m, n \in \mathbb{Z}$ such that $|n|, |m| \geq 2$ we have*

$$h(\beta_{m,n}) \geq \frac{1}{5} \log(|mn|).$$

Proof. Let $x_2 = (0, 0)$ and $x_1 = (1/2, 1/2)$ in $T^2 = \mathbb{R}^2/\mathbb{Z}^2$. The braid

$$b_{m,n} = \tau_{1,a}^m \tau_{1,b}^n \tau_{1,a}^{-m} \tau_{1,b}^{-n}$$

corresponds to the word $(a^m b^n a^{-m} b^{-n}, 0)$ and hence the closure of the braid is given by $\beta_{m,n}$. The associated mapping class of the braid is given by a point push map of the point x_1 along the loop $\gamma_{m,n} = B^{-n} \# A^{-m} \# B^n \# A^m$ where $A(t) = (1/2 + t, 1/2)$ and $B(t) = (1/2, 1/2 + t)$ for $t \in [0, 1]$. We consider the loop $\gamma_{m,n}$ as a loop in $\pi_1(T^2 \setminus \{x_2\}, x_1)$ where it is primitive (i.e. not a power of another element) and filling. We now have that (see [CL16] for an algorithm to compute the self-intersection number or see [CR04] for a direct calculation)

$$\iota(\gamma_{m,n}) = 4|mn| - 2|m| - 2|n|.$$

After [Dow11] for the dilatation $\lambda_{m,n}$ of the mapping class of $\varphi_{m,n} = \text{Push}(\gamma_{m,n})$ in $\text{MCG}(T^2 \setminus \{x_1, x_2\})$ the following estimate holds

$$\lambda_{m,n} \geq \sqrt[5]{\iota(\gamma_{m,n}) + 1} = \sqrt[5]{4|mn| - 2|m| - 2|n| + 1}.$$

Therefore,

$$h(\beta_{m,n}) = h(\varphi_{m,n}) \geq \log(\lambda_{m,n}) \geq \frac{1}{5} \log(4|mn| - 2|m| - 2|n| + 1) \geq \frac{1}{5} \log(|nm|),$$

where we use [FS79] for the estimate with the dilatation and use that $|n|, |m| \geq 2$ in the last inequality. $\square$

5.3. Proof of Theorem A: Eggbeater Maps, Braids of Orbits, and Action Estimates. In this section, inspired by Khanevsky's eggbeater map [Kha22] on T^2 and eggbeater maps on higher-genus surfaces [PS16, AGKK⁺19], we construct Hamiltonian diffeomorphisms ϕ_k generated by Hamiltonians H_k on $T^2 = \mathbb{R}^2/\mathbb{Z}^2$ equipped with the standard symplectic form $\omega = dx \wedge dy$. If not stated otherwise, we identify a point $z \in T^2$ with the unique tuple $(x, y) \in [0, 1)^2$. We then choose a sequence of pairs of periodic orbits $\{x_1^{(k)}, x_2^{(k)}\} \subset \mathcal{P}(H_{2k})$ with $\mathcal{B}(H_{2k}, x_1^{(k)}, x_2^{(k)}) = \beta_{2k+1, 2k+1}$ such that $\mathcal{S}(H_{2k}, x_1^{(k)}, x_2^{(k)}) \rightarrow \infty$. We use these maps to prove Theorem A by applying the braid stability result from Theorem B.

Let us now define the Hamiltonians H_k . Consider the function

$$s : [0, 1] \rightarrow \mathbb{R}, \quad t \mapsto \begin{cases} -4t, & t \in [0, 1/4] \\ 4t - 2, & t \in [1/4, 3/4] \\ -4t + 4, & t \in [3/4, 1] \end{cases}.$$

For convenience of notation we often will consider s as a function on S^1 as it is periodic. For an integer k define

$$s_k(t) = (k + \varepsilon)s(t),$$

where $\varepsilon > 0$ is chosen (depending on k) to be small enough. Smooth s_k near $t = 1/4, 3/4$ by a C^0 -small perturbation such that

$$\int_0^1 s_k(t) dt = 0,$$

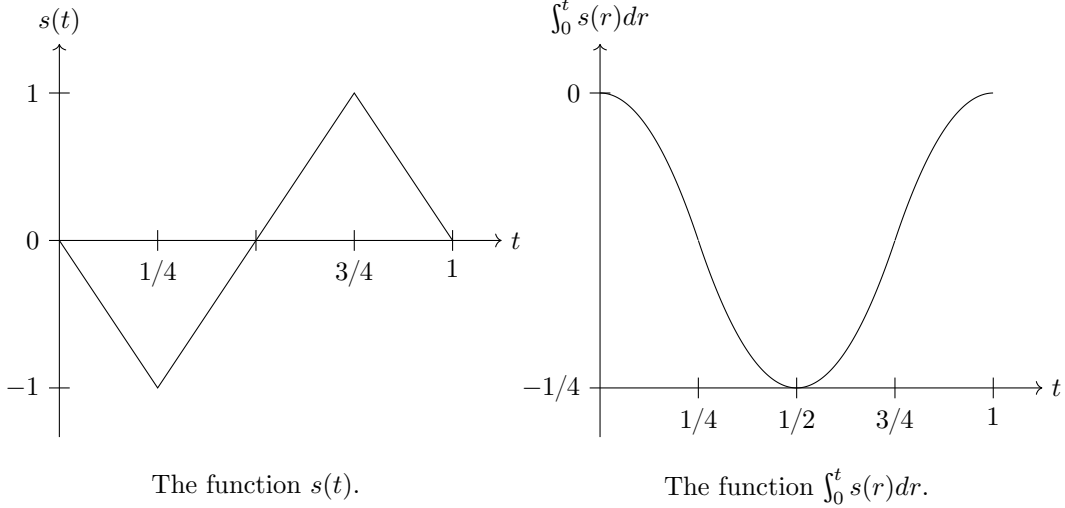
the symmetry

$$s_k(t + 1/2) = -s_k(t)$$

still holds, and that the smoothening is supported on $\{t \in [0, 1] \mid |s_k(t)| > k\}$. For convenience, we will view the function s_k as a 1-periodic function on $\mathbb{R}$. Next, define the functions

$$h_k(t) = \int_0^t s_k(r) dr$$

and let $H_{k,a}(x, y) = h_k(y)$, $H_{k,b}(x, y) = h_k(x)$ be two autonomous Hamiltonians on T^2 .



Denote by $\phi_{k,a}^t$ and $\phi_{k,b}^t$ the flows induced by these Hamiltonians. They are given by

$$\phi_{k,a}^t(x, y) = (x - ts_k(y), y) \quad \text{and} \quad \phi_{k,b}^t(x, y) = (x, y + ts_k(x)).$$

For $\varepsilon > 0$ small enough, consider

$$\phi_k = \phi_{k,b}^1 \circ \phi_{k,a}^1 \circ \phi_{k,b}^1 \circ \phi_{k,a}^1$$

generated by $H_k = H_{k,b} \# H_{k,a} \# H_{k,b} \# H_{k,a}$, where $\#$ denotes the concatenation in the time variable. We will show that ϕ_k is an example of a sequence promised in Theorem A. For every homeomorphism ϕ of T^2 let $\tilde{\phi}$ denote the lift to the universal cover $\mathbb{R}^2$. For every fixed point $z \in [0, 1]^2$ of ϕ_k we call the points

$$z_1 = \tilde{\phi}_{k,a}(z), \quad z_2 = \tilde{\phi}_{k,b}(\tilde{\phi}_{k,a}(z)), \quad z_3 = \tilde{\phi}_{k,a}(\tilde{\phi}_{k,b}(\tilde{\phi}_{k,a}(z))), \quad \text{and} \quad z_4 = \tilde{\phi}_k(z)$$

intermediate points. Note that $z = (x, y)$ is a fixed point of ϕ_k whose orbit is contractible if and only if $\tilde{\phi}_k(z) = z$. In this case there are two real numbers $R_a, R_b \in \mathbb{R}$ such that the intermediate points are given by

$$z_1 = (x + R_a, y), \quad z_2 = (x + R_a, y + R_b), \quad z_3 = (x, y + R_b) \quad \text{and} \quad z_4 = (x, y).$$

Indeed, we have $R_a = -s_k(y)$ and $R_b = -s_k(x)$ as follows from the proof of the following lemma.

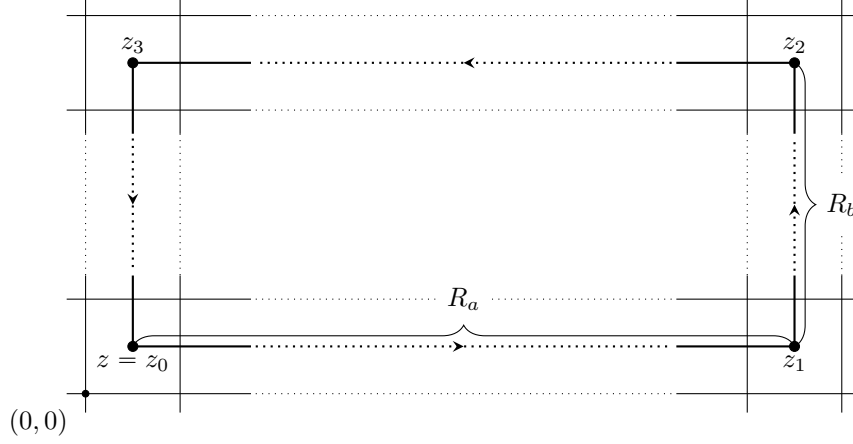


FIGURE 14. Dynamics of the intermediate points on $\mathbb{R}^2$.

Lemma 5.4. *A point $(x, y) \in T^2$ is a fixed point with contractible orbit if and only if*

$$s_k(y) = -s_k(y - s_k(x)) \quad \text{and} \quad s_k(x) = -s_k(x - s_k(y)).$$

Proof. The intermediate points of any point (x_0, y_0) are given by

$$(x_1, y_1) = (x_0 - s_k(y_0), y_0), \quad (x_2, y_2) = (x_1, y_1 + s_k(x_1)), \quad (x_3, y_3) = (x_2 - s_k(y_2), y_2).$$

Moreover, (x_0, y_0) is a fixed point with a contractible orbit if and only if

$$(x_0, y_0) = (x_4, y_4) = (x_3, y_3 + s_k(x_3)).$$

Importantly, we consider the equality in $\mathbb{R}^2$ and not merely on T^2 . We find that $(x_2, y_2) = (x_0 - s_k(y_0), y_0 + s_k(x_0 - s_k(y_0)))$ and, similarly, $(x_0, y_0) = (x_4, y_4) = (x_2 - s_k(y_2), y_2 + s_k(x_2 - s_k(y_2)))$. Adding one equation to the other we find

$$(18) \quad s_k(y_2) = -s_k(y_0) \quad \text{and} \quad s_k(x_2 - s_k(y_2)) = -s_k(x_0 - s_k(y_0))$$

and since $x_0 = x_4 = x_3 = x_2 - s_k(y_2)$ we find

$$(19) \quad s_k(x_0) = -s_k(x_0 - s_k(y_0)).$$

Using (19) we now find

$$s_k(y_2) = s_k(y_1 + s_k(x_1)) = s_k(y_0 + s_k(x_0 - s_k(y_0))) = s_k(y_0 - s_k(x_0)),$$

where we use $y_0 = y_1$. Hence, combining this with (18) we find

$$s_k(y_0) = -s_k(y_0 - s_k(x_0)).$$

On the other hand, assume that $s_k(x_0) = -s_k(x_0 - s_k(y_0))$ and $s_k(y_0) = -s_k(y_0 - s_k(x_0))$. Then we have $s_k(x_0) = -s_k(x_2)$ since $x_2 = x_1 = x_0 - s_k(y_0)$. Moreover, we then have

$s_k(y_0) = -s_k(y_2)$ since $y_0 - s_k(x_0) = y_1 + s_k(x_2) = y_1 + s_k(x_1) = y_2$. We now find

$$\begin{aligned}
z_4 &= (x_4, y_4) \\
&= (x_3, y_3 + s_k(x_3)) \\
&= (x_2 - s_k(y_2), y_2 + s_k(x_2 - s_k(y_2))) \\
&= (x_1 - s_k(y_2), y_1 + s_k(x_1) + s_k(x_1 + s_k(y_0))) \\
&= (x_1 - s_k(y_2), y_1 + s_k(x_2) + s_k(x_0)) \\
&= (x_0 - s_k(y_0) - s_k(y_2), y_0 + s_k(x_2) + s_k(x_0)) \\
&= (x_0, y_0)
\end{aligned}$$

which means that (x_0, y_0) is a fixed point with contractible orbit. $\square$

Lemma 5.5. *For $k \in \mathbb{Z}_{\geq 1}$ large enough and $\varepsilon > 0$ small enough, all fixed points $z = (x, y)$ with contractible orbits are non-degenerate and satisfy $|s_k(x)| < k$ and $|s_k(y)| < k$.*

Proof. We first show that any fixed point $(x, y) \in T^2$ with a contractible orbit has the property that for all intermediate points (x_i, y_i) the coordinates x_i, y_i lie in the part where s_k is linear. Assume that this is not the case. Without loss of generality, we assume this fails, i.e. $x = x_0$ or $y = y_0$. Indeed, if this is not the case, it must hold for x_2 or x_3 (recall that $x_0 = x_3$, $x_1 = x_2$, $y_0 = y_1$, and $y_2 = y_3$). In this case, we consider the fixed point $\tilde{z} = \phi_{k,b}^1 \circ \phi_{k,a}^1(z)$ which is given by (x_2, y_2) .

This means that $|s_k(y)| \in [k, k + 3\varepsilon/2]$ or $|s_k(x + s_k(y))| \in [k, k + 3\varepsilon/2]$ where we assume that the perturbation to a smooth approximation s of the original function is small enough and done in a small enough neighborhood of non-smooth points.

First assume that $|s_k(y)| \in [k, k + 3\varepsilon/2]$. Since s_k is periodic, we have $s_k(x - s_k(y)) \in s_k(x \pm [0, 3\varepsilon/2])$. By Lemma 5.4 we must have $s_k(x) = -s_k(x - s_k(y))$ and hence we find $0 < c \leq 3\varepsilon/2$ such that

$$|s_k(x)| = \frac{1}{2}|s_k(x) - s_k(x - s_k(y))| = \frac{1}{2}|s_k(x) - s_k(x \pm c)|.$$

Choosing ε small enough, we have that $c > 0$ is arbitrarily small. By continuity, for every $\delta > 0$ we have that $|s_k(x)| = \frac{1}{2}|s_k(x) - s_k(x \pm c)| < \delta$ when $\varepsilon > 0$, and therefore also c , is small enough. But this contradicts $s_k(y) = -s_k(y - s_k(x))$ by choosing δ small enough. Indeed, we similarly have

$$|s_k(y)| = \frac{1}{2}|s_k(y) - s_k(y - s_k(x))|$$

which tends to 0 when ε tends to 0 since in this case $|s_k(x)|$ tends to 0. This contradicts the assumption that $|s_k(y)| \geq k$. An analogous argument holds in the case $|s_k(x - s_k(y))| \in [k, k + 3\varepsilon/2]$ by using $|s_k(x - s_k(y))| = |s_k(x)|$.

Hence, we may assume that x_i, y_i lie in the part of $[0, 1]$ where s_k and s_k respectively are linear. A computation shows we have

$$\begin{aligned}
d_z \phi_k &= \begin{pmatrix} 1 & 0 \\ 4\epsilon_4 \lambda & 1 \end{pmatrix} \begin{pmatrix} 1 & 4\epsilon_3 \lambda \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 4\epsilon_2 \lambda & 1 \end{pmatrix} \begin{pmatrix} 1 & 4\epsilon_1 \lambda \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 + 16\epsilon_2 \epsilon_3 \lambda^2 & \dots \\ \dots & 1 + 16(\epsilon_1 \epsilon_2 + \epsilon_3 \epsilon_4 + \epsilon_4 \epsilon_1) \lambda^2 + 256\epsilon_1 \epsilon_2 \epsilon_3 \epsilon_4 \lambda^4 \end{pmatrix},
\end{aligned}$$

where $\lambda = k + \varepsilon$ and $\epsilon_i \in \{\pm 1\}$ depends on z and is given by

$$\epsilon_1 = -\text{sign}(\dot{s}_k(y_0)), \quad \epsilon_2 = \text{sign}(\dot{s}_k(x_1)), \quad \epsilon_3 = -\text{sign}(\dot{s}_k(y_2)), \quad \epsilon_4 = \text{sign}(\dot{s}_k(x_3)).$$

Since $\det(d_z \phi_k) = 1$, 1 is an eigenvalue of $d_z \phi_k$ if and only if $\text{tr}(d_z \phi_k) = 2$. However, we have

$$\text{tr}(d_z \phi_k) = 2 + 16(\epsilon_1 \epsilon_2 + \epsilon_2 \epsilon_3 + \epsilon_3 \epsilon_4 + \epsilon_4 \epsilon_1) \lambda^2 + 256 \epsilon_1 \epsilon_2 \epsilon_3 \epsilon_4 \lambda^4 \neq 2$$

for $k \geq 1$ and $\varepsilon > 0$ small enough. Therefore, any fixed point with a contractible orbit is non-degenerate. $\square$

As a path of Hamiltonian diffeomorphisms from the identity to ϕ_k we consider the following isotopy

$$(20) \quad \{\phi_{k,b}^t\} \# \{\phi_{k,a}^t\} \# \{\phi_{k,b}^t\} \# \{\phi_{k,a}^t\}.$$

We now characterize the fixed points with contractible orbits using the following data. Firstly, for every such fixed point z of ϕ_k the *winding numbers* $(m, n) \in \mathbb{Z}^2$ are defined by

$$(21) \quad m = \lfloor x - s_k(y) \rfloor \quad \text{and} \quad n = \lfloor y - s_k(x) \rfloor,$$

where $\lfloor \cdot \rfloor$ denotes the floor function. Since $|s_k|$ is bounded by $k + \varepsilon$, the winding numbers of any fixed points lie in $m, n \in [-k, k]$. Indeed, by Lemma 5.5 we have $|s_k(y)| < k$ and $x \in [0, 1)$ and therefore $x - s_k(y) \in (-k, k + 1)$ which means $|m| \leq k$. Likewise, we find $|n| \leq k$. Next, we define the *branching index* $\epsilon = \epsilon(z) \in \{\pm 1\}^4$ by

$$\epsilon_1 = -\text{sign}(\dot{s}_k(y_0)), \quad \epsilon_2 = \text{sign}(\dot{s}_k(x_1)), \quad \epsilon_3 = -\text{sign}(\dot{s}_k(y_2)), \quad \epsilon_4 = \text{sign}(\dot{s}_k(x_3)),$$

where $z_i = (x_i, y_i)$ are the intermediate points. The branching index captures whether the flow locally acts as a positive or negative shear at the intermediate points.

We now study the dynamics of fixed points $z = (x, y)$ such that $s(x) \neq 0 \neq s(y)$. Let $m, n \in [-k, k]$ and fix a branching index ϵ . Denote by $\bar{z}_i$ the shifted intermediate points in $[0, 1]^2$ given by

$$\bar{z}_0 = z_0 \quad \bar{z}_1 = z_1 - (m, 0), \quad \bar{z}_2 = z_2 - (m, n), \quad \bar{z}_3 = z_3 - (0, n).$$

The point z_0 is a fixed point of ϕ_k with contractible orbit and winding number (m, n) if and only if it solves the following system of equations.

$$(22) \quad \begin{aligned} \bar{z}_1 &= \bar{z}_0 + (-s_k(\bar{y}_0) - m, 0), & \bar{z}_2 &= \bar{z}_1 + (0, s_k(\bar{x}_1) - n) \\ \bar{z}_3 &= \bar{z}_2 + (-s_k(\bar{y}_2) + m, 0), & \bar{z}_0 &= \bar{z}_3 + (0, s_k(\bar{x}_3) + n). \end{aligned}$$

This can be written as

$$\begin{aligned} \bar{z}_1 &= \begin{pmatrix} 1 & 4\epsilon_1 \lambda \\ 0 & 1 \end{pmatrix} \bar{z}_0 + \nu_1, & \bar{z}_2 &= \begin{pmatrix} 1 & 0 \\ 4\epsilon_2 \lambda & 1 \end{pmatrix} \bar{z}_1 + \nu_2, \\ \bar{z}_3 &= \begin{pmatrix} 1 & 4\epsilon_3 \lambda \\ 0 & 1 \end{pmatrix} \bar{z}_2 + \nu_3, & \bar{z}_0 &= \begin{pmatrix} 1 & 0 \\ 4\epsilon_4 \lambda & 1 \end{pmatrix} \bar{z}_3 + \nu_4, \end{aligned}$$

where $\lambda = k + \varepsilon$ and ν_i are shift parameters given by

$$\begin{aligned} \nu_1 &= \begin{cases} (-m, 0), & \epsilon_1 = 1, s(y) < 0 \\ (-m + 2\lambda, 0), & \epsilon_1 = -1 \\ (-m - 4\lambda, 0), & \epsilon_1 = 1, s(y) > 0 \end{cases}, & \nu_2 &= \begin{cases} (0, -n), & \epsilon_2 = -1, s(x) > 0 \\ (0, -n - 2\lambda), & \epsilon_2 = 1 \\ (0, -n + 4\lambda), & \epsilon_2 = -1, s(x) < 0 \end{cases}, \\ \nu_3 &= \begin{cases} (m, 0), & \epsilon_3 = 1, s(y) > 0 \\ (m + 2\lambda, 0), & \epsilon_3 = -1 \\ (m - 4\lambda, 0), & \epsilon_3 = 1, s(y) < 0 \end{cases}, & \nu_4 &= \begin{cases} (0, n), & \epsilon_4 = -1, s(x) < 0 \\ (0, n - 2\lambda), & \epsilon_4 = 1 \\ (0, n + 4\lambda), & \epsilon_4 = -1, s(x) > 0 \end{cases}, \end{aligned}$$

where the cases depend on the direction of flow and slope of the speed function at the intermediate points, and we use that $s_k(x_1) = -s_k(x_0) = -s_k(x)$ and $s_k(y_2) = -s_k(y_0) = -s_k(y)$. One can formulate this with closed forms by

$$\begin{aligned}\nu_1 &= (-m - \lambda[\text{sign}(s(y))(1 + \epsilon_1) + 2\epsilon_1], 0), \nu_2 = (0, -n - \lambda[\text{sign}(s(x))(1 - \epsilon_2) + 2\epsilon_2]), \\ \nu_3 &= (m + \lambda[\text{sign}(s(y))(1 + \epsilon_3) - 2\epsilon_3], 0), \nu_4 = (0, n + \lambda[\text{sign}(s(x))(1 - \epsilon_4) - 2\epsilon_4]).\end{aligned}$$

Any fixed point then satisfies the equation

$$(23) \quad (A - id)z = -\nu,$$

where

$$A = \begin{pmatrix} 1 & 0 \\ 4\epsilon_4\lambda & 1 \end{pmatrix} \begin{pmatrix} 1 & 4\epsilon_3\lambda \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 4\epsilon_2\lambda & 1 \end{pmatrix} \begin{pmatrix} 1 & 4\epsilon_1\lambda \\ 0 & 1 \end{pmatrix}$$

and ν is the accumulated shift parameter

$$\nu = \nu_4 + \begin{pmatrix} 1 & 0 \\ 4\epsilon_4\lambda & 1 \end{pmatrix} \left[\nu_3 + \begin{pmatrix} 1 & 4\epsilon_3\lambda \\ 0 & 1 \end{pmatrix} \left[\nu_2 + \begin{pmatrix} 1 & 0 \\ 4\epsilon_2\lambda & 1 \end{pmatrix} \nu_1 \right] \right].$$

In the proof of Lemma 5.5, we already showed that A does not have 1 as an eigenvalue and hence $A - id$ is invertible. Hence, there is a unique solution z to (23). The solution to this equation is approximated by

$$\begin{aligned}x &= \frac{1}{2} + \frac{\text{sign}(s(x))(1 - \epsilon_4)}{4} - \frac{\epsilon_4 n}{4k} + \frac{1}{k} \left(\frac{(\epsilon_1 + \epsilon_3)\epsilon_4}{16} \left(\frac{m}{k} + \text{sign}(s(y)) \right) + \frac{\epsilon_4 \text{sign}(s(y))}{8} \right) + O\left(\frac{1}{k^2}\right) \\ y &= \frac{1}{2} + \frac{\text{sign}(s(y))(1 + \epsilon_1)}{4} + \frac{\epsilon_1 m}{4k} + \frac{1}{k} \left(\frac{(\epsilon_2 + \epsilon_4)\epsilon_1}{16} \left(\frac{n}{k} + \text{sign}(s(x)) \right) - \frac{\epsilon_1 \text{sign}(s(x))}{8} \right) + O\left(\frac{1}{k^2}\right),\end{aligned}$$

where we use that $\varepsilon < \frac{1}{k}$ is chosen small enough for each k as can be verified with a Python script. The point (x, y) is a fixed point of ϕ_k exactly if it satisfies $(x, y) \in [0, 1]^2$.

Lastly, note that for every fixed point z of ϕ_k we must either have

$$s(x) = s(y) = 0 \quad \text{or} \quad s(x) \neq 0 \neq s(y).$$

Indeed, if $s(x) = 0$ then Lemma 5.4 asserts that $s(y) = -s(y)$ and hence also must be 0 and analogously if the role of x and y are swapped. We formulate the result of this discussion in the following lemma.

Lemma 5.6. *Any fixed point $z = (x, y)$ of ϕ_k with contractible orbits either satisfies $s(x) = s(y) = 0$ or solves (23).*

Let z be a fixed point of ϕ_k and let $\bar{z}_i$ denote its intermediate points projected to the torus. The intermediate points pass from $\bar{z}_i$ to $\bar{z}_{i+1}$ by either an action of $\phi_{k,a}$ or $\phi_{k,b}$. At these points, the differential $d_{\bar{z}_i}(\phi_{k,a})$ or $d_{\bar{z}_i}\phi_{k,b}$ can take two forms depending on which branch of s the corrdinate $\bar{y}_i$ or $\bar{x}_i$ lies determined by the branching index at ϵ_i . Namely, in the standard trivialization of TT^2 we have

$$d_{\bar{z}_i}\phi_{k,a} = \begin{pmatrix} 1 & \epsilon_i 4\lambda \\ 0 & 1 \end{pmatrix}, \quad d_{\bar{z}_i}\phi_{k,b} = \begin{pmatrix} 1 & 0 \\ \epsilon_i 4\lambda & 1 \end{pmatrix}.$$

The following lemma is somewhat computation-heavy, and we will prove it in Appendix 6.

Lemma 5.7. *Let z be a fixed point of ϕ_k with $\epsilon = \epsilon(z) \in \{\pm 1\}^4$. For k large enough we have*

$$\mu(z) = \frac{(\epsilon_1 + \epsilon_3) - (\epsilon_2 + \epsilon_4)}{2}.$$

We will estimate the stability radius of braids by action estimates. For this, the following action formula will be essential.

Lemma 5.8. *For every fixed point $z = (x, y)$ with contractible orbit such that $s(x) \neq 0 \neq s(y)$, with winding numbers (m, n) and branching index $\epsilon = \epsilon(z) \in \{\pm 1\}^4$ the action of its orbit γ is given by*

$$\mathcal{A}_{H_k}(\gamma) = -mn + \frac{k}{4}(\mu(z) - 2) + \frac{\epsilon_1 + \epsilon_3}{8} \left(\frac{m^2}{k} - 2|m| \right) - \frac{\epsilon_2 + \epsilon_4}{8} \left(\frac{n^2}{k} - 2|n| \right) - \frac{|m| + |n|}{2} + O(1).$$

Proof. Let $z_i = (x_i, y_i) \in \mathbb{R}^2$ be the intermediate points of z and let $\bar{z}_i = (\bar{x}_i, \bar{y}_i)$ denote the intermediate points projected to T^2 . We first show the following formula for the action, and then approximate $s(x)$ and $s(y)$

$$(24) \quad \mathcal{A}_{H_k}(\gamma) = -\lambda^2 s(x)s(y) + \frac{\lambda}{8} [2\mu(z) - (\epsilon_1 + \epsilon_3)s(y)^2 + (\epsilon_2 + \epsilon_4)s(x)^2 - 4] + O(1),$$

where $\lambda = (k + \varepsilon)$. The action is given by

$$\mathcal{A}_{H_k}(\gamma) = h_k(y_0) + h_k(x_1) + h_k(y_2) + h_k(x_3) - (x_1 - x_0)(y_2 - y_1),$$

where $(x_1 - x_0)(y_2 - y_1)$ accounts for the area of the capping, and the rest comes from the Hamiltonian part. First, note that the area term is given by

$$(x_1 - x_0)(y_2 - y_1) = s_k(x_1)(-s_k(y_0)) = s_k(x_0)s_k(y_0) = \lambda^2 s(x_0)s(y_0).$$

We have that

$$\bar{y}_0 = \begin{cases} (-\epsilon_1 s(\bar{y}_0) + 1 - \epsilon_1)/4, & \bar{y}_0 \in [0, 1/2] \\ (-\epsilon_1 s(\bar{y}_0) + 3 + \epsilon_1)/4, & \bar{y}_0 \in [1/2, 1] \end{cases}, \bar{x}_1 = \begin{cases} (\epsilon_2 s(\bar{x}_1) + 1 + \epsilon_2)/4, & \bar{x}_1 \in [0, 1/2] \\ (\epsilon_2 s(\bar{x}_1) + 3 - \epsilon_2)/4, & \bar{x}_1 \in [1/2, 1] \end{cases}$$

$$\bar{y}_2 = \begin{cases} (-\epsilon_3 s(\bar{y}_2) + 1 - \epsilon_3)/4, & \bar{y}_2 \in [0, 1/2] \\ (-\epsilon_3 s(\bar{y}_2) + 3 + \epsilon_3)/4, & \bar{y}_2 \in [1/2, 1] \end{cases}, \bar{x}_3 = \begin{cases} (\epsilon_4 s(\bar{x}_3) + 1 + \epsilon_4)/4, & \bar{x}_3 \in [0, 1/2] \\ (\epsilon_4 s(\bar{x}_3) + 3 - \epsilon_4)/4, & \bar{x}_3 \in [1/2, 1] \end{cases}$$

We have that

$$h_k(t) = \begin{cases} -2\lambda t^2, & t \in [0, 1/4] \\ \lambda(2(t - 1/2)^2 - 1/4), & t \in [1/4, 3/4] \\ -2\lambda(t - 1)^2, & t \in [3/4, 1] \end{cases}.$$

Note that

$$\epsilon_1 = \begin{cases} -1, & y_0 \in [1/4, 3/4] \\ 1, & \text{otherwise} \end{cases}, \quad \epsilon_2 = \begin{cases} 1, & x_1 \in [1/4, 3/4] \\ -1, & \text{otherwise} \end{cases},$$

$$\epsilon_3 = \begin{cases} -1, & y_2 \in [1/4, 3/4] \\ 1, & \text{otherwise} \end{cases}, \quad \epsilon_4 = \begin{cases} 1, & x_3 \in [1/4, 3/4] \\ -1, & \text{otherwise} \end{cases}.$$

Going through all cases, we find the closed form for the Hamiltonian terms

$$h_k(\bar{y}_0) = -\lambda(\epsilon_1 s(\bar{y}_0)^2 + 1 - \epsilon_1)/8, \quad h_k(x_1) = \lambda(\epsilon_2 s(\bar{x}_1)^2 - 1 - \epsilon_2)/8$$

$$h_k(\bar{y}_2) = -\lambda(\epsilon_3 s(\bar{y}_2)^2 + 1 - \epsilon_3)/8, \quad h_k(\bar{x}_3) = \lambda(\epsilon_4 s(\bar{x}_3)^2 - 1 - \epsilon_4)/8$$

Using that $s(\bar{y}_2) = -s(\bar{y}_0)$ and $s(\bar{x}_0) = s(\bar{x}_3) = -s(\bar{x}_1)$ and additionally using $\mu(z) = ((\epsilon_1 + \epsilon_3) - (\epsilon_2 + \epsilon_4))/2$ as in Lemma 5.7 shows (24).

We now estimate the values of $s(x)$ and $s(y)$. By Lemma 5.6, we either have $s(x) = s(y) = 0$ or $s(x) \neq 0 \neq s(y)$. In the first case, formula (24) says that

$$\mathcal{A}_{H_k}(\gamma) = \frac{\lambda}{8}(2\mu(z) - 4) + O(1) = \frac{k}{4}(\mu(z) - 2) + O(1),$$

where we use $\lambda = k + \varepsilon$. The winding numbers are $(m, n) = 0$ in this case and hence we arrive at the promised expression in the lemma. Now assume $s(x) \neq 0 \neq s(y)$. The fixed point solves the system of equations (22) and hence also (23), given by

$$(A - id)z = -\nu.$$

Solving this system of equations, with a Python script, one can verify that we have

$$\begin{aligned} x &= \frac{1}{2} + \frac{\text{sign}(s(x))(1 - \epsilon_4)}{4} - \frac{\epsilon_4 n}{4k} + \frac{1}{k} \left(\frac{(\epsilon_1 + \epsilon_3)\epsilon_4}{16} \left(\frac{m}{k} + \text{sign}(s(y)) \right) + \frac{\epsilon_4 \text{sign}(s(y))}{8} \right) + O\left(\frac{1}{k^2}\right) \\ y &= \frac{1}{2} + \frac{\text{sign}(s(y))(1 + \epsilon_1)}{4} + \frac{\epsilon_1 m}{4k} + \frac{1}{k} \left(\frac{(\epsilon_2 + \epsilon_4)\epsilon_1}{16} \left(\frac{n}{k} + \text{sign}(s(x)) \right) - \frac{\epsilon_1 \text{sign}(s(x))}{8} \right) + O\left(\frac{1}{k^2}\right), \end{aligned}$$

where we use that $\varepsilon < \frac{1}{k}$ is chosen small enough for each k . Plugging this back into $s(\cdot)$ we find

$$\begin{aligned} s(x) &= -\frac{n}{k} + \frac{1}{k} \left(\frac{(\epsilon_1 + \epsilon_3)}{4} \left(\frac{m}{k} + \text{sign}(s(y)) \right) + \frac{\text{sign}(s(y))}{2} \right) + O\left(\frac{1}{k^2}\right), \\ s(y) &= -\frac{m}{k} - \frac{1}{k} \left(\frac{(\epsilon_2 + \epsilon_4)}{4} \left(\frac{n}{k} + \text{sign}(s(x)) \right) - \frac{\text{sign}(s(x))}{2} \right) + O\left(\frac{1}{k^2}\right). \end{aligned}$$

Using that $\lambda = k + O\left(\frac{1}{k}\right)$ we calculate

$$-\lambda^2 s(x)s(y) = -mn + \frac{\epsilon_1 + \epsilon_3}{4} \left(\frac{m^2}{k} - |m| \right) - \frac{\epsilon_2 + \epsilon_4}{4} \left(\frac{n^2}{k} - |n| \right) - \frac{|m| + |n|}{2} + O(1),$$

where we use that $\text{sign}(s(y))m = -|m|$ and $\text{sign}(s(x))n = -|n|$. The Hamiltonian term of the action is then given by

$$\frac{\lambda}{8}(2\mu(z) - (\epsilon_1 + \epsilon_3)s(y)^2 + (\epsilon_2 + \epsilon_4)s(x)^2 - 4) = \frac{k}{4}(\mu(z) - 2) - \frac{\epsilon_1 + \epsilon_3}{8} \frac{m^2}{k} + \frac{\epsilon_2 + \epsilon_4}{8} \frac{n^2}{k} + O(1).$$

Using both terms and (24) we find

$$\mathcal{A}_{H_k}(\gamma) = -mn + \frac{k}{4}(\mu(z) - 2) + \frac{\epsilon_1 + \epsilon_3}{8} \left(\frac{m^2}{k} - 2|m| \right) - \frac{\epsilon_2 + \epsilon_4}{8} \left(\frac{n^2}{k} - 2|n| \right) - \frac{|m| + |n|}{2} + O(1).$$

□

We now consider the construction for $2k$ in place of k and show that for a choice of $x_1^{(k)}, x_2^{(k)} \in \mathcal{P}(H_{2k})$ and any suitable choice of almost complex structure J we have

$$\mathcal{E}(H_{2k}, J, x_1^{(k)}, x_2^{(k)}) \rightarrow \infty \quad (k \rightarrow \infty).$$

Namely, let $x_1^{(k)}, x_2^{(k)} \in \mathcal{P}(H_{2k})$ be the contractible loops with winding numbers (k, k) and $(-k, -k)$ and branching indices $\epsilon(x_1) = (1, -1, 1, -1)$ and $\epsilon(x_2) = (-1, 1, -1, 1)$ so that $\mu(x_1^{(k)}) = 2$ and $\mu(x_2^{(k)}) = -2$. Indeed, solving (23) in this case ensures that these fixed points exist. For better readability, we drop the superscript (k) . In the following, we give an obstruction to the existence of Floer isotopies with one end in $\{x_1, x_2\}$ and another pair $\{y_1, y_2\} \subset \mathcal{P}(H_{2k})$ relying on the winding numbers of the orbits y_1 and y_2 ; this argument is purely topological. We then estimate the energy of the Floer isotopies using the action differences of the orbits x_i and y_i for $i = 1, 2$.

Lemma 5.9. *Let k be large enough and $y_1, y_2 \in \mathcal{P}(H_{2k})$ contractible orbits with winding numbers (m_i, n_i) such that there exists a Floer isotopy (u_1, u_2) , where u_i connects $x_i^{(k)}$ and y_i for $i = 1, 2$. Then we have*

$$m_1 - m_2 = n_1 - n_2 = \pm 2k.$$

Proof. We apply Lemma 5.2 to the loops $\{x_1^{(k)}, x_2^{(k)}\}$ and $\{y_1, y_2\}$ and have that an isotopy can only exist if the words $\text{spell}(x_1, x_2)$ and $\text{spell}(y_1, y_2)$ are conjugate in F_2 . The fixed points $z = x_1^{(k)}(0)$ and $w = x_2^{(k)}(0)$ of ϕ_{2k} and their intermediate points satisfy the following dynamics.

$$\begin{aligned} z_0, w_2 &\in [0, 1/2] \times [0, 1/2] + \mathbb{Z}^2, & z_1, w_3 &\in [1/2, 1] \times [0, 1/2] + \mathbb{Z}^2, \\ z_2, w_0 &\in [1/2, 1] \times [1/2, 1] + \mathbb{Z}^2, & z_3, w_1 &\in [0, 1/2] \times [1/2, 1] + \mathbb{Z}^2. \end{aligned}$$

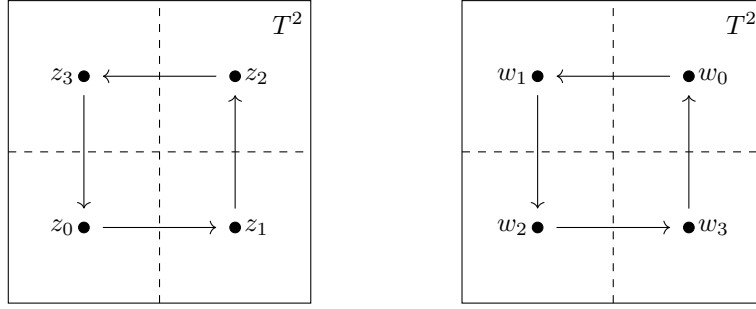


FIGURE 15. The dynamics of intermediate points z_0, z_1, z_2 , and z_3 and w_0, w_1, w_2 , and w_3 .

With this in mind and the fact that the winding numbers are $(\pm k, \pm k)$, we can conclude that

$$\text{spell}(x_1^{(k)}, x_2^{(k)}) = a^{2k+1}b^{2k+1}a^{-2k-1}b^{-2k-1}.$$

Note that for the orbits y_1, y_2 there exists $m, n \in \mathbb{Z}$ such that

$$\text{spell}(y_1, y_2) = a^m b^n a^{-m} b^{-n}.$$

This word is only conjugate to $a^{2k+1}b^{2k+1}a^{-2k-1}b^{-2k-1}$ if $m = n = \pm(2k+1)$ since in F_2 words are conjugate if and only if their cyclically reduced words agree up to cyclic permutations. Let us consider the case $m = n = 2k+1$; the other case is analogous. Let now $z = (x^z, y^z) = y_1(0)$ and $w = (x^w, y^w) = y_2(0)$ be the corresponding fixed points. Denote by $z_i = (x_i^z, y_i^z)$ and $w_i = (x_i^w, y_i^w)$ the intermediate points of z and w . Then we have

$$m = [x_1^z - x_1^w] - [x_0^z - x_0^w] \quad \text{and} \quad n = [y_2^z - y_2^w] - [y_0^z - y_0^w].$$

Observation: $x_0^z < \frac{1}{2} < x_0^w$ and $y_0^z < \frac{1}{2} < y_0^w$.

Proof. First assume that $s_{2k}(x_0^z) \geq 0$. We find the following contradiction

$$\begin{aligned} n = 2k + 1 &= \lfloor y_2^z - y_2^w \rfloor - \lfloor y_0^z - y_0^w \rfloor \\ &= \lfloor y_0^z - s_{2k}(x_0^z) - (y_0^w - s_{2k}(x_0^w)) \rfloor - \lfloor y_0^z - y_0^w \rfloor \\ &\leq \lfloor y_0^z - y_0^w + s_{2k}(x_0^w) \rfloor - \lfloor y_0^z - y_0^w \rfloor \\ &\leq \lfloor s_{2k}(x_0^w) \rfloor + 1 \\ &\leq 2k, \end{aligned}$$

where we use Lemma 5.5 for the last inequality. This shows that $x_0^z < \frac{1}{2}$. Assuming that $s_k(x_0^w) \leq 0$, a similar argument shows that $x_0^w > \frac{1}{2}$. The proof that $y_0^z < \frac{1}{2} < y_0^w$ is analogous. This concludes the proof of the observation. $\square$

This means that $m = \lfloor x_1^z - x_1^w \rfloor + 1$ and $n = \lfloor y_2^z - y_2^w \rfloor + 1$. Moreover, the points z and w follow the same dynamics as in Figure (15) which means that

$$\begin{aligned} x_1^z &\in [m_z + 1/2, m_z + 1) \quad \text{and} \quad x_1^w \in [m_w, m_w + 1/2) \\ y_2^z &\in [n_z + 1/2, n_z + 1) \quad \text{and} \quad y_2^w \in [n_w, n_w + 1/2), \end{aligned}$$

where (m_z, n_z) and (m_w, n_w) are the winding numbers of z and w . This means that $x_1^z - x_1^w \in (m_z - m_w, m_z - m_w + 1)$ and $y_2^z - y_2^w \in (n_z - n_w, n_z - n_w + 1)$ and hence $2k + 1 = m = m_z - m_w + 1$ and $2k + 1 = n = n_z - n_w + 1$. This concludes the proof of the Lemma. $\square$

We now estimate the energy of Floer isotopies with ends in $\{x_1, x_2\}$. At this point, the fact that $\mathcal{E}(H, J, x)$ relies on the energy of Floer isotopies rather than on the energy of the single strands is essential since there are choices of orbits $y_1, y_2 \in \mathcal{P}(H_{2k})$ that are not obstructed by Lemma 5.9 where one of the action difference $\mathcal{A}_{H_{2k}}(x_1) - \mathcal{A}_{H_{2k}}(y_1)$ or $\mathcal{A}_{H_{2k}}(y_2) - \mathcal{A}_{H_{2k}}(x_2)$ is small but the other is of order k .

Lemma 5.10. *For k large enough and any generic almost complex structure J we have*

$$\mathcal{E}(H_{2k}, J, x_1^{(k)}, x_2^{(k)}) \geq \frac{1}{8}k + O(1)$$

Proof. Suppose that $y_1, y_2 \in \mathcal{P}(H_{2k})$ are such that there exists a Floer isotopy (u_1, u_2) with $\mu_{\max}(u_1, u_2) = 1$ connecting x_i and y_i for $i = 1, 2$. Purely for topological reasons, by Lemma 5.9, we must have that y_1 and y_2 are contractible loops with winding numbers $(k + p, k + q)$ and $(-k + p, -k + q)$ or $(-k + p, -k + q)$ and $(k + p, k + q)$ where $p, q \in \mathbb{Z}$ are integers with $|p|, |q| \leq k$. Note that an orbit y with fixed index $\mu(y)$ has the same action when the winding number is $(k + p, k + q)$ as when it is $(-k - p, -k - q)$. Since we consider all integer values of $p, q \in [-k, k]$, we may and do only consider the case when the winding numbers of y_1 and y_2 are $(k + p, k + q)$ and $(-k + p, -k + q)$ since we obtain the other case by inverting p and q . There are two cases to consider: $(\mu(y_1), \mu(y_2)) = (1, -2)$ or $(\mu(y_1), \mu(y_2)) = (2, -1)$. Note that we need not consider the case $(\mu(y_1), \mu(y_2)) = (1, -1)$ as there cannot exist a Floer isotopy between $\{x_1, x_2\}$ and $\{y_1, y_2\}$; since the index does not increase along Floer cylinders, we must either have $\mu(x_i) \leq \mu(y_i)$ or $\mu(y_i) \leq \mu(x_i)$ for $i = 1, 2$ respectively. Hence, in any Floer isotopy, u_1 or u_2 is constant.

Case 1: $\mu(y_1) = 2$ and $\mu(y_2) = -1$.

Since u_1 is constant, the first strand remains unchanged, hence $p = q = 0$. We have that

$$E(u_1, u_2) = E(u_2) \geq (\mathcal{A}_{H_{2k}}(y_2) - \mathcal{A}_{H_{2k}}(x_2)) > 0.$$

Note that we only need to consider the orbits y_2 where this difference is positive. Using Lemma 5.8 a short computation shows that

$$\mathcal{A}_{H_{2k}}(x_2) = -k^2 - 2k + \frac{\mu(x_2)}{8}k + O(1).$$

Moreover, we have

$$\begin{aligned} \mathcal{A}_{H_{2k}}(y_2) &= -k^2 + \frac{k}{2}(\mu(y_2) - 2) + \frac{\epsilon_1 + \epsilon_3}{8} \left(\frac{k^2}{2k} - 2k \right) - \frac{\epsilon_2 + \epsilon_4}{8} \left(\frac{k^2}{2k} - 2k \right) - k + O(1) \\ &= -k^2 - 2k + \frac{\mu(y_2)}{8}k + O(1). \end{aligned}$$

Hence, we find the estimate

$$\mathcal{A}_{H_{2k}}(y_2) - \mathcal{A}_{H_{2k}}(x_2) = \frac{\mu(y_2) - \mu(x_2)}{8}k + O(1) = \frac{1}{8}k + O(1).$$

Hence, we find $E(u_1, u_2) \geq \frac{1}{8}k + O(1)$.

Case 2: $\mu(y_1) = 1$ and $\mu(y_2) = -2$.

This case is very similar to the first one; however, now u_2 is constant and we have $p = q = 0$. We again find

$$E(u_1, u_2) = E(u_1) \geq (\mathcal{A}_{H_{2k}}(x_1) - \mathcal{A}_{H_{2k}}(y_1)) > 0$$

and only need to consider those y_1 where the action difference is positive. Again using Lemma 5.8 a short computation shows

$$\mathcal{A}_{H_{2k}}(x_1) = -k^2 - 2k + \frac{\mu(x_1)}{8}k + O(1).$$

Similarly as in the first case we find

$$\mathcal{A}_{H_{2k}}(y_1) = -k^2 - 2k + \frac{\mu(y_1)}{8}k + O(1).$$

We find

$$\mathcal{A}_{H_{2k}}(x_1) - \mathcal{A}_{H_{2k}}(y_1) = \frac{\mu(x_1) - \mu(y_1)}{8}k + O(1) = \frac{1}{8}k + O(1).$$

Hence, we find $E(u_1, u_2) \geq \frac{1}{8}k + O(1)$. □

Putting everything together, we can now prove Theorem A.

Proof of Theorem A. Consider the diffeomorphism ϕ_{2k} and let $x_1^{(k)}, x_2^{(k)} \in \mathcal{P}(H_{2k})$ be the contractible orbits as in Lemma 5.9. In the proof of Lemma 5.9 we saw that

$$\text{spell}(x_1^{(k)}, x_2^{(k)}) = a^{2k+1}b^{2k+1}a^{-2k-1}b^{-2k-1}$$

and hence as discussed in Section 5.2 we have

$$\mathcal{B}(H_{2k}, x_1^{(k)}, x_2^{(k)}) = \beta_{2k+1, 2k+1}.$$

By Lemma 5.3 the entropy of the braid is bounded by

$$h(\beta_{2k+1, 2k+1}) \geq \frac{2}{5} \log(2k+1).$$

In Lemma 5.10 we saw that $\mathcal{E}(H_{2k}, J, x_1^{(k)}, x_2^{(k)}) \rightarrow \infty$ as $k \rightarrow \infty$ for every almost complex structure J . Since the framing of the braid $B(H_{2k}, x_1^{(k)}, x_2^{(k)})$ is

$$f_{2k} : x_1^{(k)} \mapsto \mu(x_1^{(k)}) = 2, \quad x_2^{(k)} \mapsto \mu(x_2^{(k)}) = -2$$

we have that $\mathcal{E}^{\text{int}}(H_{2k}, J, x_1^{(k)}, x_2^{(k)}) = \infty$ since there are no index 1 Floer cylinders connecting the orbits. Using the definition of the stability radius (1) for injective framings we find

$$\frac{1}{16}k + O(1) \leq \mathcal{S}(H_{2k}, x_1^{(k)}, x_2^{(k)}) \rightarrow \infty \quad (k \rightarrow \infty).$$

For a non-degenerate Hamiltonian diffeomorphism ψ such that

$$d_H(\phi_{2k}, \psi) < \mathcal{S}(H_{2k}, x_1^{(k)}, x_2^{(k)}),$$

there exists a Hamiltonian G generating ψ such that $E(H - G) < \mathcal{S}(H_{2k}, x_1^{(k)}, x_2^{(k)})$. After the braid stability Theorem B we have that $\dim \text{BCF}_{\beta_{2k+1, 2k+1}, f_{2k}}(G) \geq 1$ which means that there are periodic orbits $y_1, y_2 \in \mathcal{P}(G)$ such that $\mathcal{B}(G, y_1, y_2) = \beta_{2k+1, 2k+1}$. It follows that

$$\frac{1}{5} \log(2k+1) < h(\beta_{2k+1, 2k+1}) \leq h_{\text{top}}(\psi).$$

Now, more generally, if ψ is possibly a degenerate Hamiltonian diffeomorphism, there are arbitrarily small C^∞ perturbations of ψ yielding a non-degenerate Hamiltonian diffeomorphism $\tilde{\psi}$ such that still $d_H(\phi_{2k}, \tilde{\psi}) < \mathcal{S}(H_{2k}, x_1^{(k)}, x_2^{(k)})$. For $\tilde{\psi}$ the entropy estimate holds. Since topological entropy is C^∞ -upper-semicontinuous [Yom87, New89], the estimate holds for ψ as well. This means that for $\alpha_k = \frac{1}{5} \log(2k+1)$ we have

$$\frac{1}{16}k + O(1) \leq d_H(\text{Ent}_{\leq \alpha_k}(T^2, \omega), \phi_{2k}),$$

for k large enough. □

6. APPENDIX

Next, we prove Lemma 5.7. We use the Maslov index as defined by Robbin and Salamon in [RS93]. This index assigns to a path of Lagrangian subspaces $\Lambda : [0, 1] \rightarrow \mathcal{L}(n) \subset Gr(n, \mathbb{R}^{2n})$ and a reference subspace $V \in \mathcal{L}(n)$ a half-integer which is invariant under homotopy of this path relative to the endpoints. The index counts the intersections of Λ and V as follows. Namely, if we assume that the crossings are regular, for every $t_0 \in [0, 1]$ there is a well-defined quadratic form

$$Q(\Lambda, t_0) : \Lambda(t_0) \rightarrow \mathbb{R}, \quad v \mapsto \frac{d}{dt} \Big|_{t=t_0} \omega(v, w(t)),$$

where we fix a Lagrangian complement W of $\Lambda(t_0)$ and choose $w(t) \in W$ such that $v + w(t) \in \Lambda(t)$ for small t . Let V be a further Lagrangian subspace. A $t \in [0, 1]$ such that $\Lambda(t) \cap V$ is non-trivial is called *crossing time* and define the associated *crossing form* by

$$\Gamma(\Lambda, V, t_0) = Q(\Lambda, t_0) \Big|_{\Lambda(t_0) \cap V}.$$

The Maslov index of a path of Lagrangians is then given by

$$\mu(\Lambda, V) = \frac{1}{2} \text{sgn}(\Gamma(\Lambda, V, 0)) + \sum_{0 < t < 1} \text{sgn}(\Gamma(\Lambda, V, t)) + \frac{1}{2} \text{sgn}(\Gamma(\Lambda, V, 1)).$$

This index is, in particular, useful since it is well-defined even when $\Lambda(1) = V$, which allows us to consider paths of symplectic matrices which end in a symplectic matrix with eigenvalue 1.

The connection to the Conley–Zehnder index is the following. Since we consider Hamiltonian diffeomorphisms in $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, the standard trivialization of the tangent bundle induces a representation of the differentials $\{d(\phi_k^t)_{\phi_k^t(x_0)}\}$ as a path of symplectic matrices.

For a periodic orbit $x \in \mathcal{P}(H)$ we let $\Psi_x : [0, 1] \rightarrow Sp(2, \mathbb{R})$ denote this path. Furthermore, let for every path $A : [0, 1] \rightarrow Sp(2, \mathbb{R})$

$$\text{Graph}(A(t)) = \{(v, A(t)v) \mid v \in \mathbb{R}^2\}$$

denote the path of the graphs. In our case, the Conley–Zehnder index is then given by paths of Lagrangians in $\mathbb{R}^4$

$$\mu_{CZ}(x) = \mu(\text{Graph}(\Psi_x), \Delta_{\mathbb{R}^2}),$$

where we choose $\omega \oplus -\omega$ on $\mathbb{R}^4$. For more details, see [RS93]. Note that for every $(0, 0) \neq (v, v) \in \text{Graph}(A(t)) \cap \Delta_{\mathbb{R}^2}$ we have $A(t)v = v$ and $A(t)$ has the eigenvalue 1. Hence, the index $\mu(\text{Graph}(A(t)), \Delta_{\mathbb{R}^2})$ sums the intersection form over all times t where $A(t)$ has the eigenvalue 1 (with half counts on the boundary). This index is invariant under so-called *stratum homotopies* (see [RS93]). In particular, this means the following. Given a homotopy $A_s(t)$ of paths of matrices for $s, t \in [0, 1]$, such that $A_s(0) = id$ and none of the matrices $A_s(1)$ has 1 as an eigenvalue, then $\mu(\text{Graph}(A_s(t)), \Delta_{\mathbb{R}^2}) = \mu(\text{Graph}(A_0(t)), \Delta_{\mathbb{R}^2})$. We will make use of this shortly.

Proof of Lemma 5.7. Let us for now fix a large positive integer $k \in \mathbb{Z}_{\geq 1}$ and define $\lambda = 4(k + \varepsilon)$. For every fixed point $z = z_0 = (x_0, y_0)$ with intermediate points (x_i, y_i) let

$$\epsilon_1 = -\text{sign}(\dot{s}_k(y_0)), \quad \epsilon_2 = \text{sign}(\dot{s}_k(x_1)), \quad \epsilon_3 = -\text{sign}(\dot{s}_k(y_2)), \quad \epsilon_4 = \text{sign}(\dot{s}_k(x_3)).$$

For such a fixed point, using the standard trivialization of TT^2 , the path $(d\phi_k^t)_{\phi_k^t(z_0)}$ represented as matrices with respect to the standard basis is homotopic through symplectic matrices to the path

$$\begin{aligned} P_\epsilon(t) &= \begin{pmatrix} 1 & 0 \\ \lambda\epsilon_4 t & 1 \end{pmatrix} \begin{pmatrix} 1 & \lambda\epsilon_3 t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \lambda\epsilon_2 t & 1 \end{pmatrix} \begin{pmatrix} 1 & \lambda\epsilon_1 t \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 + t^2\lambda^2\epsilon_2\epsilon_3 & t\lambda(\epsilon_1 + \epsilon_3 + t^2\lambda^2\epsilon_1\epsilon_2\epsilon_3) \\ t\lambda(\epsilon_2 + \epsilon_4 + t^2\lambda^2\epsilon_2\epsilon_3\epsilon_4) & 1 + t^2\lambda^2(\epsilon_1\epsilon_2 + \epsilon_3\epsilon_4 + \epsilon_4\epsilon_1) + t^4\lambda^4\epsilon_1\epsilon_2\epsilon_3\epsilon_4 \end{pmatrix} \end{aligned}$$

We will distinguish cases and compute the index in each case. Unfortunately, for 12 of the 16 choices, the path $P_\epsilon(t)$ gives rise to degenerate intersection forms. Hence, we need to perturb the path. Let us first discuss the four paths where we do not need to perturb $P_\epsilon(t)$. For every $\epsilon \in \{\pm 1\}^4$ we will show that

$$\mu(z) = \frac{(\epsilon_1 + \epsilon_3) - (\epsilon_2 + \epsilon_4)}{2}.$$

The matrix $P_\epsilon(t)$ has 1 as an eigenvalue for $t \in [0, 1]$ if and only if

$$\text{tr}(P_\epsilon(t)) = 2 + t^2\lambda^2(\epsilon_1 + \epsilon_3)(\epsilon_2 + \epsilon_4) + t^4\lambda^4\epsilon_1\epsilon_2\epsilon_3\epsilon_4 = 2$$

which means that $t = 0$ or

$$t = \sqrt{-\frac{(\epsilon_1 + \epsilon_3)(\epsilon_2 + \epsilon_4)}{\lambda^2\epsilon_1\epsilon_2\epsilon_3\epsilon_4}}.$$

We here compute the intersection form of the crossing time $t = 0$ in generality, as it will appear in any of the cases below. Suppose that

$$\dot{P}_\epsilon(0) = \begin{pmatrix} 0 & p \\ q & 0 \end{pmatrix}.$$

Since $P_\epsilon(0) = id$ we have $\text{Graph}(P_\epsilon(0)) = \Delta \subset \mathbb{R}^4$. Let us choose the Lagrangian complement $W = 0 \times \mathbb{R} \times \mathbb{R} \times 0$ of $\Delta_{\mathbb{R}^4}$. For every $t \in \mathbb{R}$ close to 0 and $v = (x, y, x, y) \in \Delta_{\mathbb{R}^4}$ let $w(t) = (0, \eta(t), \xi(t), 0) \in W$ such that $v + w(t) \in \text{Graph}(P_\epsilon(t))$. This means that

$$\begin{pmatrix} x + \xi(t) \\ y \end{pmatrix} = P_\epsilon(t) \begin{pmatrix} x \\ y + \eta(t) \end{pmatrix}.$$

Differentiating at $t = 0$ and using that $\eta(0) = 0$ as well as $P_\epsilon(0) = id$ we find

$$\begin{pmatrix} \dot{\xi}(0) \\ 0 \end{pmatrix} = \dot{P}_\epsilon(0) \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 0 \\ \dot{\eta}(0) \end{pmatrix}.$$

Hence, we find

$$\dot{\xi}(0) = py \quad \text{and} \quad \dot{\eta}(0) = -qx.$$

We now have that

$$\begin{aligned} Q(x, y) &= \frac{d}{dt} \Big|_{t=0} \omega(v, w(t)) \\ &= \dot{\eta}(0)x + \dot{\xi}(0)y \\ &= -qx^2 + py^2. \end{aligned}$$

Case: $\epsilon = \pm(1, 1, 1, 1)$.

Then the only $t \in [0, 1]$ such that 1 is an eigenvalue of $P_\epsilon(t)$ is $t = 0$. We calculate that

$$\dot{P}_\epsilon(0) = \begin{pmatrix} 0 & \pm 2\lambda \\ \pm 2\lambda & 0 \end{pmatrix}$$

and find

$$Q(x, y) = \mp 2\lambda x^2 \pm 2\lambda y^2$$

which is a quadratic form of signature 0. Thus, the fixed point z has index $\mu(z) = 0$, which agrees with the given formula.

Case: $\epsilon = \pm(1, -1, 1, -1)$.

In this case we find that

$$\dot{P}_\epsilon(0) = \begin{pmatrix} 0 & \pm 2\lambda \\ \mp 2\lambda & 0 \end{pmatrix}$$

and a similar computation as above shows that the crossing form at $t = 0$ is given by

$$Q(x, y) = \pm 2\lambda x^2 \pm 2\lambda y^2$$

which is a quadratic form of signature ± 2 . Hence, the crossing time $t = 0$ contributes ± 1 to the index. However, there is another crossing time

$$t_1 = \frac{2}{\lambda},$$

which lies in $[0, 1]$ when k is large enough. We have that

$$P_\epsilon(t) = \begin{pmatrix} 1 - t^2\lambda^2 & \pm 2t\lambda \mp t^3\lambda^3 \\ \mp 2t\lambda \pm t^3\lambda^3 & 1 - 3t^2\lambda^2 + t^4\lambda^4 \end{pmatrix}$$

and hence

$$P_\epsilon(t_1) = \begin{pmatrix} -3 & \mp 4 \\ \pm 4 & 5 \end{pmatrix}.$$

Note that again $W = 0 \times \mathbb{R} \times \mathbb{R} \times 0$ is a Lagrangian complement to $\text{Graph}(P_\epsilon(t_1))$. We find that the eigenvector of $P_\epsilon(t_1)$ with eigenvalue 1 is given by $(\mp 1, 1)$ and the intersection

$\text{Graph}(P_\epsilon(t_1)) \cap \Delta$ is spanned by the vector $v = (\mp 1, 1, \mp 1, 1)$. For $t \in \mathbb{R}$ close to t_1 we let $w(t) = (0, \eta(t), \xi(t), 0)$ such that $v + w(t) \in \text{Graph}(P_\epsilon(t))$. Similar to above, this means that

$$\begin{pmatrix} x + \xi(t) \\ y \end{pmatrix} = P_\epsilon(t) \begin{pmatrix} x \\ y + \eta(t) \end{pmatrix}.$$

Differentiation at t_1 and using $\eta(t_1) = 0$ that

$$\begin{pmatrix} \dot{\xi}(t_1) \\ 0 \end{pmatrix} = \begin{pmatrix} -4\lambda & \mp 10\lambda \\ \pm 10\lambda & 20\lambda \end{pmatrix} \begin{pmatrix} \mp 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -3 & \mp 4 \\ \pm 4 & 5 \end{pmatrix} \begin{pmatrix} 0 \\ \dot{\eta}(t_1) \end{pmatrix}.$$

We find that

$$\dot{\eta}(t_1) = -2\lambda \quad \text{and} \quad \pm \dot{\xi}(t_1) = 2\lambda.$$

Now, we have that

$$\begin{aligned} \Gamma(\text{Graph}(P_\epsilon(t)), \Delta, t_1)(s) &= Q(\text{Graph}(P_\epsilon(t_1)))|_{\text{Graph}(P_\epsilon(t_1)) \cap \Delta}(s) \\ &= s^2 \frac{d}{dt} \Big|_{t=t_1} \omega(v, w(t)) \\ &= s^2 (\mp \dot{\eta}(t_1) + \dot{\xi}(t_1)) \\ &= \pm s^2 4\lambda. \end{aligned}$$

We therefore find that at crossing time $t = t_1$, the crossing form contributes ± 1 to the index. Finally, we find $\mu(z) = \pm 2$, which agrees with the given formula in the statement of the lemma.

Now consider the remaining cases where $\epsilon_1 + \epsilon_3 = 0$ or $\epsilon_2 + \epsilon_4 = 0$. Unfortunately, in these cases, the crossing form of P_ϵ at $t = 0$ is degenerate. Since

$$\text{tr}(P_\epsilon(t)) = 2 + t^4 \lambda^4 \epsilon_1 \epsilon_2 \epsilon_3 \epsilon_4$$

we have that $\text{tr}(P_\epsilon(t)) = 2$ if and only if $t = 0$. Hence, $P_\epsilon(t)$ has 1 as an eigenvalue if and only if $t = 0$, which is the only (degenerate) crossing time of the Lagrangian path associated to $P_\epsilon(t)$. We perturb the path $P_\epsilon(t)$ such that this remains true and $t = 0$ is the only crossing time but is now regular.

Let $\alpha, \beta \in \mathbb{R}$ be small enough in absolute value and to be fixed later. We consider the perturbed path of symplectic matrices

$$P_{\epsilon, \alpha, \beta}(t) = \begin{pmatrix} 1 & 0 \\ \alpha t & 1 \end{pmatrix} P_\epsilon(t) \begin{pmatrix} 1 & \beta t \\ 0 & 1 \end{pmatrix}.$$

Note that

$$\text{tr}(P_{\epsilon, \alpha, \beta}(1)) = \text{tr}(P_\epsilon(1)) + \alpha(P_\epsilon(1))_{12} + \beta(P_\epsilon(1))_{21} + \alpha\beta(P_\epsilon(1))_{11}.$$

Since

$$\text{tr}(P_\epsilon(1)) = 2 + \lambda^2(\epsilon_1 + \epsilon_3)(\epsilon_2 + \epsilon_4) + \lambda^4 \epsilon_1 \epsilon_2 \epsilon_3 \epsilon_4 \neq 2$$

when k is large enough, we have that $\text{tr}(P_{\epsilon, \alpha, \beta}(1)) \neq 2$ when $|\alpha|, |\beta|$ are small enough. Note that $P_{\epsilon, \alpha, \beta}(0) = \text{id}$. It follows that P_ϵ and $P_{\epsilon, \alpha, \beta}$ are stratum homotopic via the homotopy given by

$$I : [0, 1] \times [0, 1] \rightarrow Sp(2, \mathbb{R}), \quad (s, t) \mapsto \begin{pmatrix} 1 & 0 \\ \alpha s t & 1 \end{pmatrix} P_\epsilon(t) \begin{pmatrix} 1 & \beta s t \\ 0 & 1 \end{pmatrix}.$$

A short computation shows

$$\begin{aligned} \text{tr}(P_{\epsilon, \alpha, \beta}(t)) = & 2 + t^4 \lambda^4 \epsilon_1 \epsilon_2 \epsilon_3 \epsilon_4 + \\ & + t^2 [\alpha \beta + \alpha \lambda (\epsilon_1 + \epsilon_3) + \beta \lambda (\epsilon_2 + \epsilon_4)] \\ & + t^4 [\alpha \lambda^3 \epsilon_1 \epsilon_2 \epsilon_3 + \beta \lambda^3 \epsilon_2 \epsilon_3 \epsilon_4 + \alpha \beta \lambda^2 \epsilon_2 \epsilon_3], \end{aligned}$$

where we use that $\epsilon_1 + \epsilon_3 = 0$ or $\epsilon_2 + \epsilon_4 = 0$. Hence, we have $\text{tr}(P_{\epsilon, \alpha, \beta}(t)) = 2$ if and only if $t = 0$ or $t = \sqrt{D_{\epsilon, \alpha, \beta}}$ and $t \in [0, 1]$ where

$$D_{\epsilon, \alpha, \beta} = -\frac{\alpha \beta + \alpha \lambda (\epsilon_1 + \epsilon_3) + \beta \lambda (\epsilon_2 + \epsilon_4)}{\lambda^4 \epsilon_1 \epsilon_2 \epsilon_3 \epsilon_4 + [\alpha \lambda^3 \epsilon_1 \epsilon_2 \epsilon_3 + \beta \lambda^3 \epsilon_2 \epsilon_3 \epsilon_4 + \alpha \beta \lambda^2 \epsilon_2 \epsilon_3]}.$$

Case: $\epsilon_1 + \epsilon_3 = 0$ and $\epsilon_2 + \epsilon_4 \neq 0$.

Choose $\delta > 0$ small enough and set $\alpha = 0$ and $\beta = \delta(\epsilon_2 + \epsilon_4)\epsilon_1\epsilon_2\epsilon_3\epsilon_4$. Note that in this case we have $\epsilon_1\epsilon_3 = -1$ and $\epsilon_2\epsilon_4 = 1$ and hence $\beta = -\delta(\epsilon_2 + \epsilon_4)$. When δ is small enough, $D_{\epsilon, \alpha, \beta} < 0$, so $t = 0$ is the only crossing time. We have that

$$\dot{P}_{\epsilon, \alpha, \beta}(0) = \dot{P}_\epsilon(0) + \begin{pmatrix} 0 & \beta \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\delta(\epsilon_2 + \epsilon_4) \\ \lambda(\epsilon_2 + \epsilon_4) & 0 \end{pmatrix}.$$

As above, we find that the associated intersection form at $t = 0$ is given by

$$Q(x, y) = -(\epsilon_2 + \epsilon_4) [\lambda x^2 + \delta y^2]$$

which has signature $-(\epsilon_2 + \epsilon_4)$ since $\delta, \lambda > 0$. This contributes $-(\epsilon_2 + \epsilon_4)/2$, and since $t = 0$ is the only crossing time, the index is $-(\epsilon_2 + \epsilon_4)/2$, which agrees with the formula.

Case: $\epsilon_1 + \epsilon_3 \neq 0$ and $\epsilon_2 + \epsilon_4 = 0$.

This case is analogous to the last case, but we set $\beta = 0$ and $\alpha = -\delta(\epsilon_1 + \epsilon_3)$ for small enough $\delta > 0$. We find that $D_{\epsilon, \alpha, \beta} < 0$ and hence that $t = 0$ is the only crossing time. We now have

$$\dot{P}_{\epsilon, \alpha, \beta}(0) = \begin{pmatrix} 0 & \lambda(\epsilon_1 + \epsilon_3) \\ -\delta(\epsilon_1 + \epsilon_3) & 0 \end{pmatrix}.$$

The intersection form at $t = 0$ is then given by

$$Q(x, y) = (\epsilon_1 + \epsilon_3) [\delta x^2 + \lambda y^2],$$

which has signature $(\epsilon_1 + \epsilon_3)$ since $\delta, \lambda > 0$. Hence, the index is given by $(\epsilon_1 + \epsilon_3)/2$, which agrees with the formula stated in the lemma.

Case: $\epsilon_1 + \epsilon_3 = \epsilon_2 + \epsilon_4 = 0$.

In this case we choose $\delta > 0$ small enough and set $\alpha = \beta = \delta$. We have that $\epsilon_1\epsilon_2\epsilon_3\epsilon_4 = 1$, and hence, for $\delta > 0$ sufficiently small, we find $D_{\epsilon, \alpha, \beta} < 0$ and that $t = 0$ is the only crossing time. Moreover, we have

$$\dot{P}_{\epsilon, \alpha, \beta}(0) = \begin{pmatrix} 0 & \delta \\ \delta & 0 \end{pmatrix}$$

and the intersection form at $t = 0$ is given by

$$Q(x, y) = -\delta x^2 + \delta y^2$$

which has signature 0. Hence, the index is 0 in this case, which agrees with the formula given in the lemma. $\square$

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